

Idiosyncratic Expenditure Risk and the Dynamics of Household Wealth and Consumption*

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Abstract

Heterogeneous-agent macroeconomics reproduces the cross-sectional and life-cycle distribution of U.S. household wealth but not how households move through it. In PSID data, I identify *idiosyncratic expenditure risk*: a persistent component of household non-durable consumption (such as medical bills, home and vehicle repairs, or schooling and childcare) that current income, wealth, and demographics leave unpredicted and that does not anticipate higher future income. It is large and persistent, accounting for 15% of the total variance of log consumption with a half-life near nine years. In a calibrated heterogeneous-agent overlapping-generations model with discount-factor heterogeneity, expenditure risk enters as a structural primitive: a persistent shock to the marginal utility of consumption, opening a covariance wedge between the shock and consumption growth in the household's Euler equation. Relative to an income-risk economy calibrated to the same wealth targets, adding expenditure risk closes 78% of the model-data gap on the variance of consumption growth, 85% on escapes from low net worth (non-positive net worth), and 39% on falls into it; the same wedge makes high marginal propensities to consume transient and loosens their tie to current wealth.

Keywords: Expenditure risk, Heterogeneous agents, Hand-to-mouth, Wealth mobility, Consumption dynamics, Income risk, Marginal propensity to consume, Consumption insurance.

JEL Classification: D12, D14, D15, D31, D52, E21

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1 Introduction

Heterogeneous-agent (HA) macroeconomics can reproduce the cross-sectional and life-cycle distribution of U.S. household wealth, including the large share of households that hold little of it. It does so by organizing the household block around incomplete markets, idiosyncratic income risk, and persistent household heterogeneity (Aguilar, Bils, and Boar, 2025). In this framework, income shocks do more than pin down how much wealth households hold: they are the mechanism that moves households through the wealth distribution, buffer-stock wealth absorbing good income draws and cushioning bad ones.

I document that U.S. household panel data question this mechanism where it matters most: at the bottom of the wealth distribution, where little stands between a household's shocks and its consumption. Within households in the Panel Study of Income Dynamics (PSID), income shocks predict changes in consumption, yet much less whether a household falls into low wealth. The missing driver surfaces in consumption dynamics: household consumption carries large and persistent movements that the permanent and transitory income shocks (Blundell, Pistaferri, and Preston, 2008) cannot account for, a disconnect household microdata increasingly document.¹ A household's consumption and its movements in and out of low wealth, by contrast, are closely tied. What reconciles these joint consumption and wealth-transition dynamics with income-risk-driven heterogeneous-agent macroeconomics?

I identify *idiosyncratic expenditure risk*, a structural household primitive, as the missing driver. Recovered as the persistent, unpredicted component of household non-durable expenditure, it is large, accounting for 15% of the total variance of log consumption, and long-lived, with a half-life near nine years. It concentrates in categories such as schooling and childcare, medical expenses, the upkeep of homes and cars, and trips and recreation. And it moves more than just consumption expenditure: both across households and within them, it shifts the probability of transitions in and out of low wealth. Structurally, it raises every working-age household's exposure to consumption-growth risk by opening a wedge in the household's Euler equation: the covariance between expenditure-risk realizations and consumption growth. Embedded in a calibrated heterogeneous-agent overlapping-generations model, this single primitive generates the wealth transitions and consumption dynamics that income risk and persistent heterogeneity alone miss. Relative to an income-risk economy calibrated to target the same wealth distribution, it closes 78% of the model-data gap on the variance of consumption growth, 85% on escapes from low wealth, and 39% on falls into it. Expenditure risk is what binds a household's consumption dynamics to its movements through the wealth distribution, not merely to its level of wealth.

I make the case for expenditure risk as a household primitive both in the data and in a

¹See Bergholt, Furlanetto, and Mori (2025), Fulford and Low (2024), and Miranda-Pinto, Murphy, Walsh, and Young (2025).

model. In the data, the case rests on two properties of the recovered shocks. They shift the probability of falling into low wealth, and this passthrough strengthens under household fixed effects just as income risk's weakens; and where a shock to income moves a household's consumption and wealth in the same direction, a shock to expenditure raises consumption while drawing down wealth. In the model, I add expenditure risk as persistent marginal-utility shocks to an otherwise standard heterogeneous-agent overlapping-generations economy with income risk and discount-factor heterogeneity; the income-risk economy is that standard benchmark on its own, and both are calibrated to the cross-section and life cycle of U.S. household wealth. With the wealth distribution held in common, the expenditure-risk economy closes much of the gap left open by the income-risk benchmark on consumption-growth volatility and on the transitions households make in and out of low wealth.

Even as median household wealth climbs smoothly over the working life, individual households cross repeatedly in and out of low-wealth status, which I define as non-positive net worth and label hand-to-mouth (H2M).² More households slip in and out of this state than any single wave shows: on average 21% are H2M in a given wave, and 43% at some point over the panel. This churn runs around a persistent core, a sixth of households accounting for nearly three-fifths of all H2M household-years. The income shocks isolated by the standard decomposition account for how a household's consumption and wealth change within its own history, but not for when it falls into low wealth.

To account for those transitions, I build a flexible model of household non-durable expenses, letting income, wealth, and demographics predict it without imposing a functional form, and isolate the part they leave unexplained. The autocovariances of what remains reveal a persistent component buried in i.i.d. noise: purged of that noise, its persistence is about 0.86, a half-life near nine years, and it moves essentially orthogonally to income shocks, marking it as a risk process in its own right rather than mismeasured income risk. A state-space smoother then extracts this persistent component household by household, yielding the persistent expenditure shock. The shock is large: the persistent component alone amounts to 15% of the total variance of log consumption, half of everything the flexible model leaves unexplained.

I take the expenditure shocks back to the wealth transitions, where within a household's own history they shift the probability of crossing the H2M threshold that income shocks leave unexplained. Of the identified income- and expenditure-risk components, the persistent expenditure shock is the only one that passes through to both fall and escape dynamics in the entrapping direction, raising falls and lowering escapes. Its passthrough strengthens rather than washes out under household fixed effects on fall-into-H2M events, just as the

²Zeldes (1989) proxies a binding borrowing constraint by low wealth relative to income, classifying households holding under two months of income in wealth as likely constrained. I draw the line instead at non-positive net worth, a sharper cut that needs no income normalization: a household whose liabilities exceed its assets sits at the bottom of the wealth distribution however income and consumption are measured.

income-shock passthroughs fade, so a force that survives the absorption of everything stable about a household is a within-household shock that comes and goes, not a fixed household trait. And it carries a signature no realized income shock leaves (anticipated income is ruled out separately, in Appendix C.5), raising a household's consumption while drawing its wealth down to fund it, where an income shock raises both, so a residual that splits what income always moves together cannot be income the decomposition missed.

What these shocks capture is concrete: the needs that arrive at a particular household at a particular time. They concentrate in the non-durable categories that tend a household's stocks of physical, human, and memorable capital, the last in the sense of the memorable goods of [Hai, Krueger, and Postlewaite \(2020\)](#): schooling and childcare, medical expenses, the upkeep of homes and cars, and trips and recreation, each falling due as a stock wears, ails, comes of age, or fades. Together they mark a genuine primitive of the household: idiosyncratic needs that move it across the H2M threshold when income shocks do not.

Expenditure risk is modeled as persistent marginal-utility shocks that fall on a risky component of the household consumption basket, leaving a non-risky one untouched, added to an otherwise standard heterogeneous-agent overlapping-generations economy in which persistent heterogeneity enters as fixed, ex-ante differences in discount factors, the margin such models use to match the share of households holding little wealth.³ The shock process inherits the stochastic properties of the household's allocation between risky and non-risky spending in the PSID, matching its variance and autocorrelation in closed form; the share of spending it touches is calibrated so that expenditure risk accounts for the same added fraction of within-household consumption variation as in the data. The economy with these shocks is the expenditure-risk economy; the income-risk economy is the standard benchmark on its own, income risk and discount-factor heterogeneity alone. I calibrate both to the low-wealth shares and age profiles of U.S. household wealth.

I validate the model with the paper's own empirical strategy, run on panels simulated from each economy as if they were the PSID. The steps are the same as in the data: identify the income shocks; let them, together with wealth and demographics, predict consumption expenditure; extract the persistent expenditure shock from what they leave unexplained. The model's single expenditure shock stands in for the data's persistent-and-transitory pair. Only the expenditure-risk economy reproduces the within-household passthroughs to falls, escapes, and wealth changes; in the income-risk economy the estimates collapse or reverse under fixed effects. The income-risk economy turns the exercise into a falsification test. There, no channel disconnects consumption from income realizations, so

³A distinct strand locates idiosyncratic risk in the marginal utility of consumption, modeling health-status shocks to explain retired households' slow wealth drawdown ([Palumbo, 1999](#); [De Nardi, French, and Jones, 2010](#)) and, more recently, over the working life ([Russo, 2025](#)). The expenditure risk here is a marginal-utility shock in that spirit, recovered from a broader measure of consumption expenditure rather than tied to a health process, and disciplined through working-life consumption and wealth-transition dynamics rather than the retirement drawdown.

the extraction can only return income variation the conditioning does not absorb, and it does: the recovered shock moves a household's consumption and wealth together, carrying the sign of the income it is built from, where its counterpart in the data splits them. The passthroughs recovered from the data therefore belong to the expenditure-risk channel, not to the extraction.

Expenditure risk reshapes the model through a single mechanism: in the household's Euler equation, the marginal-utility shock opens a covariance wedge with consumption growth. Through it, the expenditure-risk economy generates the consumption-growth risk income risk cannot: the two economies share the same income process, so the extra consumption-growth volatility belongs to the expenditure-risk channel, and it closes 78% of the income-risk economy's gap to the data. Which households bear that risk is the sharper cut: consumption growth stays volatile across the whole wealth distribution, including among unconstrained households well clear of low wealth, where the income-risk economy, generating volatility largely where the borrowing constraint binds, collapses it toward zero. Among those unconstrained households the variance of consumption growth is 0.02 in the income-risk economy against 0.11 in the data, a gap expenditure risk nearly closes at 0.10.

At the bottom of the wealth distribution, the same wedge closes most of the wealth-transition gap that income risk and discount-factor heterogeneity leave open. The income-risk economy generates a biennial fall into low wealth of 0.045 against 0.085 in the data, and an escape of 0.287 against 0.389; expenditure risk closes 39% of the first gap and 85% of the second. In transitions between wealth quintiles, the closure fades along the wealth distribution, from closing 48% of the gap at the poorest quintile and 64% at the second to 8% at the richest, where the wealthiest households stay put in both economies, a top-tail persistence expenditure risk leaves untouched. The same churn reaches the marginal propensity to consume (MPC): expenditure risk lowers its average, shortens high-MPC spells, and loosens its tie to current wealth, a high MPC becoming a state households pass through rather than a trait they carry. These dynamics belong to the process rather than to the calibration: holding the discount-factor distribution and every other parameter fixed, dialing the shock variance down to zero unwinds the consumption-growth and escape closures in step with it.

This paper fits within a recent literature documenting that household consumption moves beyond income: in the PSID, where consumption growth is nearly as volatile as income growth and only weakly correlated with it ([Miranda-Pinto et al., 2025](#)); in Norwegian transaction records, where shocks orthogonal to income carry roughly ninety percent of consumption-growth variance ([Bergholt et al., 2025](#)); and in surveys linked to credit records, where unexpected expenses are about twice as common as income drops ([Fulford and Low, 2024](#)). Two of the three turn the regularity into a structural model, stochastic thresholds on total consumption ([Miranda-Pinto et al., 2025](#)) and, closest to this paper, a preference shock on

a dedicated expense good (Fulford and Low, 2024); both processes are largely transitory, where the process identified here is a persistent state that tilts the household's allocation every period. I estimate that state alongside income risk, from the joint dynamics of household income and consumption, and follow it, in the data and in general equilibrium, into the transitions households make in and out of low wealth.

A second literature, in heterogeneous-agent macroeconomics, matches the cross-sectional and life-cycle distribution of household wealth by pairing idiosyncratic income risk with persistent household heterogeneity (Castañeda, Díaz-Giménez, and Ríos-Rull, 2003; Hubmer, Krusell, and Smith, 2021; Aguiar et al., 2025).⁴ The model in this paper inherits that machinery, using discount-factor heterogeneity (Carroll, Slacalek, Tokuoka, and White, 2017; Aguiar et al., 2025) to reproduce the wealth distribution as this literature does, but I show that closely matching the distribution does not automatically deliver the dynamics around it. That a stationary cross-section leaves its own dynamics undisciplined is a general point, made sharply by Gabaix, Lasry, Lions, and Moll (2016) for the transition speed of top inequality in random-growth models; here it bites at the opposite end of the distribution and within the stationary economy itself, in the household-level transitions through low wealth. An income-risk economy calibrated to the wealth distribution still misses the churn through it. Expenditure risk closes that dynamics gap without disturbing the distributional fit.

Methodologically, the paper joins the literature that studies the joint dynamics of household income and consumption (Arellano, Blundell, and Bonhomme, 2017; De Nardi, Fella, and Paz-Pardo, 2020; Hall and Mishkin, 1982). I extend the identification strategy of Blundell et al. (2008) (BPP) in a new direction, from a decomposition of income alone to a system that carries an expenditure-risk process alongside the income process. The consumption variation left once income shocks are netted out has been read before as households' advance information about their own future income (Güvenen and Smith, 2014). I split it instead by persistence and test the split: the transitory remainder co-moves positively with household resources, the pattern the advance-information reading fits, while the persistent component does not anticipate higher future income (Appendix C.5), leaving it a structural primitive, identified by stochastic properties of its own and carried into general equilibrium to account for the wealth-transition and consumption dynamics I set out to explain.

The remainder of the paper proceeds as follows. Section 2 establishes the wealth-transition facts; Section 3 identifies and measures expenditure risk; Section 4 carries it to those transitions in the data; Section 5 builds and calibrates the two economies; Section 6 compares their dynamics; Section 7 concludes.

⁴Hubbard, Skinner, and Zeldes (1995) trace low wealth to the saving disincentive of asset-tested social insurance. That channel has weakened as the phenomenon grew: asset-tested transfers have been scaled back (Guner, Rauh, and Ventura, 2024) while the share of households at non-positive net worth rose from about 15% in the 1983 Survey of Consumer Finances (Huggett, 1996) to roughly 21% in recent PSID data.

2 Household wealth dynamics in the data

Substantial within-cohort cycling sits beneath the canonical life-cycle profile of U.S. household wealth: the hand-to-mouth share declines from 39% at age 30 to 8% at age 64, households continually move into and out of low-net-worth status across their working lives, and the wealth-to-income gap between stayers and fallers fans wide with age. These wealth transitions represent the empirical dimension of what the income-risk-driven HA paradigm fails to reconcile. Standard BPP (2008) income-risk identification recovers permanent and transitory shocks whose passthroughs to log-consumption and wealth changes remain significant under household fixed effects, but the analogous passthroughs to transitions into low-net-worth status lose significance or weaken to marginal (10%) significance, leaving the within-household force that moves households across wealth thresholds only loosely identified by income-risk machinery alone.

2.1 Data

PSID's biennial waves from 1999 to 2023 supply household income, wealth, and a non-durable consumption measure as the main variables; the wealth- and income-side analyses use this full window, while the expenditure-risk identification of Section 3 is restricted to waves from 2005 onward, the first wave at which all consumption categories and income-shock components are available.

The empirical analysis uses the PSID, the household survey combining biennial wealth, income, and detailed expenditure modules at the household level. The sample is restricted to households in the Survey Research Center sub-sample (dropping the Survey of Economic Opportunity oversample) headed by a reference person between ages 25 and 64. The income measure is total family money income; net worth is total assets minus total liabilities, aggregated across all household-level financial and physical components of PSID's wealth module. Sample-selection filters follow Aguiar et al. (2025): a consumption floor, an income floor, and within-band thresholds on the food and housing shares of consumption.

The household's flow and stock variables are combined into a single state, log cash-on-hand $x_{i,t} \equiv \ln(W_{i,t-2} + Y_{i,t})$, where $W_{i,t}$ is household net worth and $Y_{i,t}$ is family money income (both in 2001 USD), defined to align with PSID's biennial reporting structure. Respondents report their current assets and liabilities at the time of the wave but their income and expenditure over the preceding year, so the wealth reading $W_{i,t-2}$ is the natural pre-period state for the consumption decision realized over the inter-wave interval $[t-2, t]$. Income and consumption similarly enter as logs throughout, $y_{i,t} \equiv \ln Y_{i,t}$ and $c_{i,t} \equiv \ln C_{i,t}$. After missingness on the underlying variables and the cash-on-hand positivity requirement, the cleaned panel yields 22,497 household-wave transition pairs. The BPP-style first-differencing required for the income-shock identification of Section 2.4 further reduces

the regression sample to 14,762 observations on 3,731 households.

The consumption aggregate is constructed to isolate flows of non-durable consumption from payments tied to durable-asset acquisition. The basket combines food, housing services, transportation, education, childcare, healthcare, clothing, recreation, and trips in 2001 USD, with housing services imputed as rent for renters and 6% of home equity for owners, following [Aguiar et al. \(2025\)](#). The aggregate excludes vehicle capital-building items (vehicle down-payments, lease payments, and loan payments) so that the expenditure-risk identification developed in Section 3 operates on within-period consumption flows rather than on the timing of vehicle finance. Two residual transportation categories ('other vehicle' and 'other public transit') are dropped under the same discipline because the PSID codebook does not specify their composition.

2.2 Wealth transition dynamics

Beneath the level life-cycle profile, U.S. households continually cross wealth thresholds: fall events (transitions into low-net-worth status) and escape events (out of it) remain substantial across the working life, and the hand-to-mouth share declines from 39% at age 30 to 8% at age 64. This churn nonetheless runs around a persistent core of households that recurrently occupy low-net-worth status.

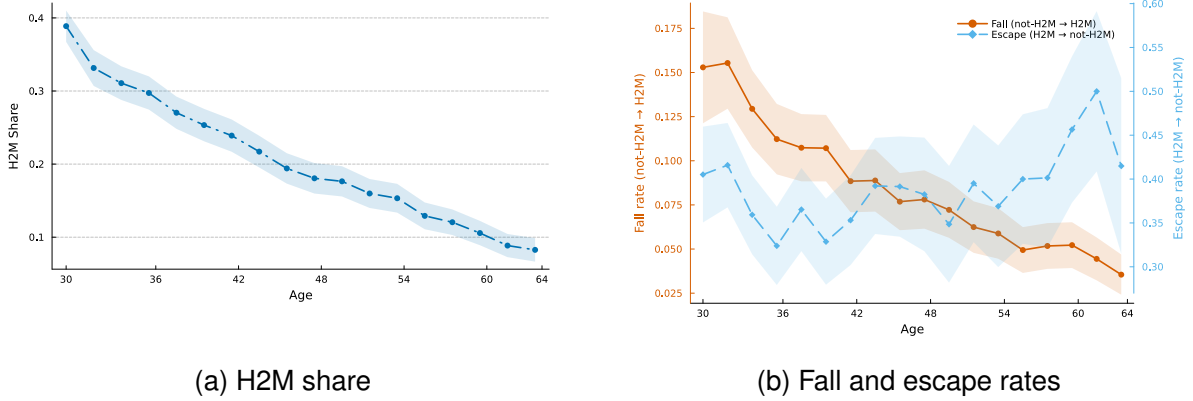
Hand-to-mouth (H2M) status, defined at each wave by a non-positive-net-worth threshold, is the lens through which wealth-side household vulnerability is measured across the working life.⁵ This zero-wealth threshold is robust by construction: a household with more liabilities than assets sits at the bottom of the wealth distribution regardless of how the income process is estimated or how the non-durable consumption aggregate is defined. Under this definition the share of H2M households declines steadily with age, from 38.9% at age 30 to 8.3% at age 64 (panel (a) of Figure 1), adding a life-cycle dimension to the pooled-share statistics commonly reported in the literature (e.g., [Aguiar et al., 2025](#)).

Beneath this declining level profile, households continually cross the H2M threshold in both directions. Biennial fall rates (panel (b) of Figure 1, left axis) decline from 15% at age 30 to 4% at age 64; escape rates (right axis) hover between roughly a third and a half throughout the working life, with no clear age trend. Averaged over ages 26–64, Panel A of Table 1 reports an 8.5% biennial fall rate and a 38.9% biennial escape rate, with 21% of the transition-pair observations starting from H2M, matching the pooled cross-sectional share.⁶ Panel B widens the picture: the cross-sectional H2M share is 20.7%, but the ever-

⁵Alternative H2M definitions in the literature condition on a household-specific income-relative threshold ([Zeldes, 1989](#)) or on liquid wealth: the taxonomy of [Kaplan, Violante, and Weidner \(2014\)](#) additionally captures wealthy hand-to-mouth households, which hold illiquid assets but little cash. The non-positive-net-worth cut used here is closest to their poor hand-to-mouth margin; the object of interest is the passage through the bottom of the net-worth distribution rather than the liquidity-driven consumption response that motivates the liquid-wealth threshold.

⁶All transition rates in the paper follow the age-averaged convention, weighting each age equally over

Figure 1: Hand-to-mouth status by age, PSID.



Notes: PSID 1999–2023 biennial waves, by single-year age. Panel (a): household-level H2M share (net worth ≤ 0); declines steadily from $\sim 39\%$ at age 30 to $\sim 8\%$ at age 64. Panel (b): biennial fall rate (left axis: non-H2M $\rightarrow$ H2M, conditional on starting non-H2M) and escape rate (right axis: H2M $\rightarrow$ non-H2M, conditional on starting H2M).

H2M share, the fraction of households that are H2M in at least one wave, is 42.8%, so the H2M population is rotated through by a wider set of households than the cross-section reveals.

Table 1: H2M transitions and prevalence, PSID.

<i>Panel A: Biennial transition matrix</i>		
State at $t - 2$	State at t	
	Not H2M	H2M
Not H2M	0.915	0.085
H2M	0.389	0.611
<i>Panel B: Prevalence</i>		
Cross-sectional H2M share	20.7%	
Ever H2M share	42.8%	

Notes: PSID 1999–2023, biennial waves. Panel A reports two-state H2M $\leftrightarrow$ non-H2M transition probabilities between consecutive waves, age-averaged over ages 26–64, computed on 22,497 household-wave observations (17,776 non-H2M and 4,721 H2M at $t - 2$); rows give the conditioning state at $t - 2$, columns give the realized state at t . Panel B reports two prevalence measures over the panel: the cross-sectional H2M share is the pooled household-wave H2M-occupancy rate, while the ever-H2M share is the share of distinct households that are H2M in at least one wave during their observation window.

This rotation is not a random reshuffling of the population: beneath it sits a stable core of recurrently H2M households. Among households observed for at least four waves, 44.3% are H2M in at least one wave but only 4.4% in every wave, so H2M status touches a broad

ages 26–64: it is the aggregation under which data and model use identical weights, and the aggregate is the exact average of the age-profile figures. The observation-weighted pooled rates are 0.080 and 0.379; the two conventions differ by half a percentage point on the fall rate and one point on the escape rate.

set of households rather than a fixed group. The H2M household-years, however, are concentrated rather than spread evenly across that broad set: the 15.6% of households that are H2M in more than half their observed waves account for 58.5% of all H2M household-years. The same concentration shows up within the spell distribution itself: most H2M spells last a single biennial wave, but the few running five waves or longer, only 5% of all spells, contain 16% of H2M household-years (Appendix Figure A.1). A between- versus within-household variance decomposition of the binary H2M indicator tells the same story, attributing roughly half of the total variance (an intraclass correlation of 0.47) to permanent between-household differences rather than to within-household movement. The H2M margin is therefore neither a fixed stock of permanently H2M households nor pure churn around a fixed share, but a persistent core of recurrently H2M households surrounded by a churning fringe.

The persistence of this within-cohort churning matters because the household-side response to any shock is conditioned on the wealth position at which the shock lands. A transitory income drop hits a household near the H2M margin very differently from one deep in positive net worth, and the persistent core documented above is exactly the set of households that return to that margin again and again across the working life. Escape is rarely terminal: among observed escapes from H2M, 40% are followed by a re-entry within four years and 54% within a decade (Appendix Figure A.1). The within-household force driving these recurrent transitions is therefore the object an income-risk-driven heterogeneous-agent model must reproduce to deliver realistic consumption-response dynamics.

2.3 Life-cycle profiles and cross-sectional distribution

Median household net-worth accumulation follows a linear age-profile, with the 1st and 3rd quartiles also linear but at different slopes; accordingly, the cross-sectional distribution that fans out with age then exhibits the right-tail concentration that motivates the discount-factor-heterogeneity infrastructure HA-macro models may adopt to match it. Conditional on net-worth status, the life-cycle W/Y profile of stayers diverges sharply from that of fallers.

The cross-section of household wealth fans out at a roughly constant rate across the working life, with the five quantile paths (p_{10} , p_{25} , median, p_{75} , p_{90}) diverging linearly from a near-zero starting point and the two lower quantiles staying near zero throughout (Appendix Figure A.2, panel (a)). The pooled cross-sectional Gini reflects the resulting right-tail concentration: 0.78 for wealth, 0.42 for income, and 0.32 for consumption, a pattern that motivates the discount-factor-heterogeneity infrastructure inherited by the heterogeneous-agent model of Section 5.

Households about to fall into H2M hold systematically lower wealth-to-income ratios two waves earlier than households that will stay non-H2M, and the gap widens sharply across the working life, with the median stayer-to-faller W/Y ratio rising from roughly 3 at age 30 to

roughly 9 at age 64 (Appendix Figure A.2, panel (b)). Part of the divergence is mechanical: W/Y is measured at $t - 2$ conditional on a fall at t , so fallers are drawn from the low-wealth tail of the $t - 2$ distribution by construction.

2.4 Income risk (IR) passthroughs and the fixed-effects attenuation puzzle

Following Blundell et al. (2008), log earnings are decomposed into fixed, permanent, and transitory components. Income shocks significantly predict consumption growth ($\Delta \ln c$) and income-scaled wealth changes ($\Delta W/\bar{y}$, the annualized net-worth change scaled by the household's mean income over its observed waves) under both OLS and household fixed effects, but their passthrough to transitions into low-net-worth status loses significance or weakens to marginal (10%) significance under household fixed effects, leaving largely open what moves a household across wealth thresholds within its own history.

The income-shock series feeding the passthrough regressions are recovered in two steps, detailed in Appendix B. First, log household earnings are residualized against a vector of demographic and labor-market controls to remove deterministic life-cycle variation.⁷ The residualized income process is then decomposed into a household-fixed component, a random-walk permanent component, and an iid transitory component following Blundell et al. (2008) as implemented by Kaplan and Violante (2010); the household-wave-level permanent ($y_{i,t}^P$) and transitory ($y_{i,t}^T$) shocks entering the passthrough regressions are recovered via IV using leads and lags of Δy as instruments.

Passthroughs to consumption growth and income-scaled wealth changes are large, statistically significant under both OLS and household fixed effects (FE), and broadly consistent with the predictions of income-risk-driven heterogeneous-agent models. Panel A of Table 2 reports the standardized coefficients in units of one standard deviation of the shock, so that magnitudes can be compared across risk sources later in the paper.⁸ A one-standard-deviation permanent shock raises $\Delta \ln c$ by 0.051 (OLS) and 0.054 (FE) and raises $\Delta W/\bar{y}$ by 0.046 and 0.041; the analogous transitory-shock passthroughs are 0.042/0.047 on $\Delta \ln c$ and 0.063/0.071 on $\Delta W/\bar{y}$, all significant.

⁷The control set comprises a cubic polynomial in the reference person's age, employment status of the reference person and spouse, education and race of the reference person, state, family size, number of children, and survey-year fixed effects.

⁸In levels units, the regression coefficients on permanent and transitory income shocks for consumption growth are 0.57/0.60 (perm, OLS/FE) and 0.19/0.21 (trans, OLS/FE). The permanent-shock estimates are broadly in line with the partial-insurance ranges reported by Blundell et al. (2008) and Kaplan and Violante (2010); the transitory-shock estimates are higher than the values reported in those papers but are in line with more recent robust treatments of the transitory-income-component estimator that recover non-zero transitory passthroughs of similar magnitude (Commault, 2022; Crawley, 2020). Appendix D.1 (Table D.1) reports a complementary pure-covariance-moment robustness check (BPP-2008 form, identification in Appendix B) that delivers passthroughs of similar magnitude and confirms that the transitory-passthrough estimate is non-zero in this sample.

Table 2: Income-risk passthroughs to consumption, wealth changes, and H2M transitions, PSID.

Panel A: Consumption growth and income-scaled wealth changes

	$\Delta \ln c$		$\Delta W/\bar{y}$	
	OLS	FE	OLS	FE
y^P	0.051*** (0.0031)	0.054*** (0.0046)	0.046*** (0.0090)	0.041*** (0.0122)
y^T	0.042*** (0.0037)	0.047*** (0.0041)	0.063*** (0.0102)	0.071*** (0.0111)
HH-FE	No	Yes	No	Yes
R ²	0.0823	0.0838	0.0230	0.0230
Households	3731			
Observations	14762			

Panel B: H2M fall and escape transitions

	H2M Fall		H2M Escape	
	OLS	FE	OLS	FE
y^P	−0.009*** (0.0029)	−0.004 (0.0031)	0.009 (0.0100)	0.013 (0.0135)
y^T	−0.011*** (0.0027)	−0.004* (0.0024)	0.026*** (0.0099)	0.027** (0.0117)
HH-FE	No	Yes	No	Yes
R ²	0.0654	0.0257	0.0906	0.1732
Households	3154		1188	
Observations	10636		2541	

Notes: 1-SD-units passthrough coefficients of BPP-identified permanent (y^P) and transitory (y^T) income shocks, under OLS and household fixed effects. *Panel A:* passthroughs to consumption growth ($\Delta \ln c$) and income-scaled wealth changes ($\Delta W/\bar{y}$); both shocks remain significant at the 1% level on both outcomes under household fixed effects, and the $\Delta W/\bar{y}$ regression trims the top and bottom 0.5% of its distribution, leaving $\Delta W/\bar{y} \in [-5.24, 6.83]$. *Panel B:* passthroughs to the H2M fall and escape indicators; under household fixed effects the permanent-shock passthrough to FALL loses its significance and the transitory-shock passthrough also weakens substantially, the attenuation pattern motivating the additional household-side primitive introduced in Section 4. Standard errors in parentheses; in Panel B sample sizes differ across columns because FALL conditions on $H2M_{t-2} = 0$ and ESCAPE on $H2M_{t-2} = 1$. PSID 1999–2023, biennial waves. *, **, *** denote significance at the 10%, 5%, 1% levels.

The picture changes for transitions into and out of low-net-worth status. In the pooled-OLS specification, both permanent and transitory income shocks significantly predict falls (1-SD passthroughs of -0.009 and -0.011 respectively, Panel B of Table 2); under household fixed effects, the permanent passthrough loses its significance entirely (-0.004) and the transitory passthrough's significance drops sharply, to -0.004 and significant only at the 10% level. On the escape margin the attenuation is one-sided: the permanent-income passthrough is insignificant under both conditionings, while the transitory-income passthrough to escape survives household fixed effects essentially unchanged. This attenuation does not imply that income shocks play no role in wealth transitions, as they will in any precautionary-savings setting. It does suggest, however, that within-household variation in identified income shocks alone leaves the within-household force driving fall transitions only loosely identified, motivating the search for an additional household-side primitive in the rest of the paper.

3 Measuring expenditure risk

Idiosyncratic expenditure risk (ER) emerges from PSID consumption data in three steps that build on the consumption-residual-as-information idea of [Hall and Mishkin \(1982\)](#) with a flexible empirical model and a state-space identification. First, a random-forest-estimated expenditure model relates log consumption to household cash on hand, realized income-risk components, demographics, and grouped-fixed-effects type, with household-level 5-fold cross-fitting yielding an unpredicted-expenditure residual; this residual is the object on which the rest of the construction operates. Second, the autocovariance structure of this residual reveals a persistent component buried in i.i.d. noise: once the noise is purged, its persistence is $\hat{\rho}_z \approx 0.86$ with a half-life near nine years, and it accounts for half of the unpredicted variation in log consumption, 15% of its total variance; a panel VAR bounds its responses to income shocks an order of magnitude below its own persistence, marking it a structural risk process distinct from income-risk noise (Appendix C.4). Third, a Kalman state-space estimator recovers this persistent component household by household, with an AR(1) extension of the moving-sum instrument of [Blundell et al. \(2008\)](#) measuring the passthrough of its innovations. The persistent expenditure component is sizable, with a cross-sectional standard deviation near 0.24 in log consumption, and Section 4 shows it to be the only risk component that depletes wealth.

3.1 A flexible empirical model for non-durable consumption

The flexible empirical model decomposes log consumption $c_{i,t} = \hat{c}_{i,t} + \xi_{i,t}$ into a non-parametric prediction $\hat{c}_{i,t}$ from cash on hand, permanent and transitory income shocks, demographics, and grouped-fixed-effects type, plus an unpredicted residual $\xi_{i,t}$, a decompo-

sition that rests on mean-independence of latent income and expenditure innovations and on a separability assumption on consumption preferences. A random-forest implementation cross-fit at the household level purges household-specific overfit from the unpredicted-expenditure residual, the raw material on which the expenditure-risk identification operates. Predictor importance is flat across the inputs, empirically warranting the non-parametric flexibility.

Under CRRA-homotheticity of preferences and mean-independence of latent income and expenditure innovations, the household's log-consumption policy admits an additively separable decomposition into a predictable component and an unpredicted residual (derived in Appendix C.1):

$$c_{i,t} = G(\tau_{g(i)}, y_{i,t}^P, y_{i,t}^T, x_{i,t}, X_{i,t}) + h(c_{i,t}^{\text{exp}}) + \epsilon_{i,t}. \quad (1)$$

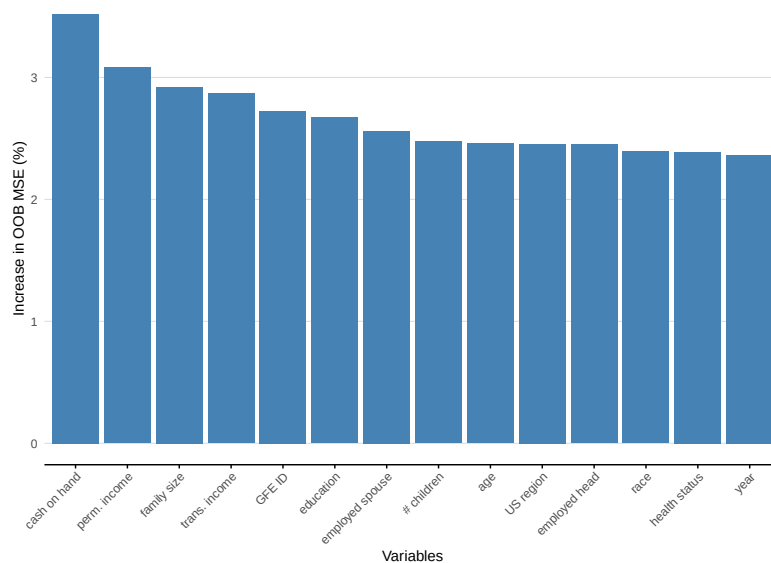
The predictable component $\hat{c}_{i,t} \equiv G(\cdot)$ aggregates a fixed type $\tau_{g(i)}$ recovered via the grouped-fixed-effects method of Bonhomme, Lamadon, and Manresa (2022) (details in Appendix C.2), the BPP-identified permanent and transitory income shocks $y_{i,t}^P$ and $y_{i,t}^T$ from Section 2.4, the log cash-on-hand state $x_{i,t}$ from Section 2.1, and a vector of demographic and wave controls $X_{i,t}$. The unpredicted residual $\xi_{i,t} \equiv h(c_{i,t}^{\text{exp}}) + \epsilon_{i,t}$ consists of a separable function h of the latent expenditure-risk component $c_{i,t}^{\text{exp}}$, the persistent object whose dynamics Section 3.3 models, and an i.i.d. noise term $\epsilon_{i,t}$, and is the object on which the rest of the section operates. One implication of the identifying assumptions, separability under CRRA-homotheticity and mean-independence of the latent expenditure component from (y^P, y^T) , is directly checkable: in a panel VAR of the recovered persistent expenditure shock and the income shocks, the shock's responses to lagged income shocks never exceed 0.010 per standard deviation at any horizon, an order of magnitude below its own serial-correlation response, and, though the permanent-shock responses are statistically nonzero and the transitory one is on impact, they bound any linear dynamic dependence at a quantitatively negligible level (Appendix C.4).

The function G is estimated by a random forest with 5-fold cross-fitting at the household level: households are randomly assigned to exactly one of five folds, and the prediction $\hat{c}_{i,t}$ for household i comes from a 500-tree forest trained on the four folds excluding i . This cross-fit structure guarantees that no household contributes to the model generating its own predictions; any household-specific overfitting therefore contributes noise rather than bias to the residual $\xi_{i,t}$. The five forests produce a 5-fold cross-fit R^2 of 0.70 against an out-of-bag R^2 of 0.72, a ~ 2 pp gap that bounds the household-level overfit at a modest level.

To assess the empirical fit of the flexible specification, I ask whether the consumption response can be summarized by any single input to G . Figure 2 reports a random-forest predictor-importance diagnostic, defined as the percentage increase in prediction error

when a given input is randomly permuted across households. The importance profile is flat. Cash on hand and the permanent income shock are the largest predictors, but only at about 3%; the remaining inputs (the transitory income shock, fixed type, demographic controls, and wave fixed effects) form a tight block at 2.0–2.4%, with no clear dominance. This pattern supports the non-parametric specification: no single argument of G is a sufficient statistic for the household’s consumption response. The random forest therefore lets the data determine the relevant interactions among inputs, a useful feature because buffer-stock saving models imply nonlinear consumption responses.

Figure 2: Predictor importance in the cross-fitted random forest, PSID.



Notes: Each bar reports the percentage increase in out-of-bag mean-squared error when the corresponding predictor’s values are permuted across observations, averaged across the 500 trees of the random forest. Cash on hand and the permanent income shock lead at ~3%; the remaining predictors form a tight 2.0–2.4% block, with no single predictor compressing the household’s consumption response into a sufficient statistic. The fixed-type predictor appears under its axis label GFE ID, the grouped-fixed-effects identifier of Appendix C.2.

3.2 The stochastic properties of unpredicted expenditure

The unpredicted-expenditure residual $\xi_{i,t}$ is near-symmetric and persistent, and its autocovariance function has a telling shape: it falls sharply from lag zero to lag one and then decays geometrically toward zero. That shape is the signature of a persistent component layered with i.i.d. noise, and the decay toward zero rather than toward a positive floor means ξ carries no detectable fixed household component. Fitting the autocovariances to a persistent-plus-noise structure by minimum distance puts the persistence of the underlying component at $\hat{\rho}_z \approx 0.86$ and splits the residual variance into roughly equal persistent and transitory halves.

Pooled across PSID waves 2005–2023, the residual ξ has standard deviation 0.33, skewness 0.04, and excess kurtosis 1.48: near-symmetric, and only mildly heavier-tailed than a Gaussian of the same variance (Appendix Figure C.5). Table 3 collects these moments alongside the persistence summary discussed next.

The persistence is read from the autocovariance function $\gamma_k = \text{Cov}(\xi_{i,t}, \xi_{i,t-k})$. The contemporaneous variance is $\gamma_0 = 0.111$; the first autocovariance is $\gamma_1 = 0.049$, less than half of γ_0 ; and beyond lag one the autocovariances decline geometrically ($\gamma_2 = 0.040$, $\gamma_3 = 0.036$, $\gamma_4 = 0.031$). Two features of this profile fix the structure of ξ . First, the sharp drop from γ_0 to γ_1 is the imprint of an i.i.d. component, which loads on the variance but on no autocovariance: roughly half of γ_0 is this transitory noise. Second, the smooth geometric decay of the surviving autocovariances is the imprint of a persistent, stationary process, and the rate of decay identifies its persistence. The autocovariances tend toward zero rather than flattening at a positive floor, so ξ carries no detectable time-invariant household component: the grouped-fixed-effects prediction of Section 3.1 has already absorbed fixed differences in household spending levels, leaving a mean-reverting residual.

Fitting a persistent-plus-noise structure to the autocovariance function by minimum distance makes these readings precise (estimation in Appendix C.4). The persistent component has autocorrelation $\hat{\rho}_z = 0.86$ (95% CI [0.83, 0.88]),⁹ a half-life of about nine years, and accounts for a share 0.50 of the residual variance, the i.i.d. noise for the other half; in log-consumption units it has a cross-sectional standard deviation of 0.24. Scaled against the broadest denominator, unpredicted expenditure carries 30% of the total variance of log consumption in the estimation sample ($\gamma_0 = 0.111$ against $\text{Var}(\ln c) = 0.37$), so the persistent component alone amounts to 15% of the total variance of log consumption.¹⁰ A test for an additional permanent or fixed component cannot reject its absence, supporting the mean-reverting reading and, with it, that the expenditure-function step left no detectable household-level spending fixed effect in the residual. The next subsection adopts this persistent-plus-noise structure as a state-space model, estimates it by maximum likelihood, and extracts the persistent component household by household.

3.3 Expenditure risk identification

Section 3.2 characterizes ξ as a persistent component buried in i.i.d. noise. I formalize that structure as a dynamic state-space model, estimate it by Kalman maximum likelihood, and extract the persistent component at the household level via the Kalman smoother.

⁹The lag-1 autocorrelation of the observed residual, $\rho_\xi = \gamma_1/\gamma_0 = 0.44$, understates the persistence of the component: the i.i.d. noise inflates γ_0 but not γ_1 , so $\rho_\xi = \rho_z \cdot (\sigma_z^2/\gamma_0) \approx 0.86 \times 0.50$. The component persistence $\rho_z = 0.86$ is the structural object, and the coefficient used to quasi-difference ξ in Section 3.3.

¹⁰The 15% is a ratio of variances, $\sigma_z^2/\text{Var}(\ln c)$. It is not an additive-decomposition share: cross-fitting removes the mechanical orthogonality of an in-sample fit, so the decomposition of log consumption into prediction and residual carries a finite-sample covariance term (7% of the total variance), and allocating any part of that term to the unpredicted side would only raise the share.

Table 3: Stochastic properties of the random-forest residual ξ , PSID.

Property	Value
<i>Distribution</i>	
Standard deviation $\hat{\sigma}_\xi$	0.33
Skewness	0.04
Excess kurtosis	1.48
<i>Persistence</i>	
Persistent-component autocorrelation $\hat{\rho}_z$	0.86
Persistent share of variance	0.50
Half-life (years)	9.1

Notes: Pooled across PSID waves 2005–2023 (25,214 household-wave observations on 5,095 households). The cross-fitted residual $\xi_{i,t}$ is the random-forest residual from equation (1); skewness and excess kurtosis are computed on the pooled distribution. The persistence block reports the autocovariance-minimum-distance estimate of the persistent component of an AR(1)-plus-noise model for ξ ; the persistent share is the fraction of γ_0 attributable to that component. Estimation details, the maximum-likelihood cross-check, and the test against an added permanent or fixed component are in Appendix C.4.

The estimated persistence is $\hat{\rho}_z \approx 0.85$, in line with the autocovariance reading, and the recovered persistent shock is the expenditure-risk object whose passthroughs to wealth and hand-to-mouth status Section 4 measures and the model of Section 5 carries.

Write the residual $\xi_{i,t}$ of equation (1) as a persistent expenditure component plus i.i.d. noise,

$$\xi_{i,t} = z_{i,t} + \varepsilon_{i,t}^{\text{exp}}, \quad z_{i,t} = \rho_z z_{i,t-1} + u_{i,t}^{\text{exp}}, \quad (2)$$

where $z_{i,t}$ is the persistent expenditure component, the state-space counterpart of $h(c_{i,t}^{\text{exp}})$ in equation (1), a stationary AR(1) with i.i.d. innovation $u_{i,t}^{\text{exp}}$ of variance σ_u^2 , and $\varepsilon_{i,t}^{\text{exp}}$ is an i.i.d. transitory term of variance σ_ε^2 absorbing equation (1)’s noise $\epsilon_{i,t}$. The transitory term collects whatever in unpredicted expenditure does not persist: genuinely transitory expenditure shocks, a revision in the household’s information set about its own income process, and measurement error, which the second moments of ξ cannot tell apart. The persistent component, by contrast, is testably free of forward-looking income content: its one-sided filtered estimate does not positively predict the household’s future income growth at any horizon, and the level state, if anything, predicts declines (Appendix C.5). Equation (2) is the AR(1)-plus-noise structure the autocovariance function of Section 3.2 pointed to; here it becomes the data-generating process I estimate.

The system (2) is linear and Gaussian, with a transition equation for the latent state z and a measurement equation $\xi = z + \varepsilon^{\text{exp}}$. For a candidate parameter vector $\theta = (\rho_z, \sigma_u^2, \sigma_\varepsilon^2)$, the Kalman filter passes once through each household’s residual history, returning the one-step-ahead forecast $\hat{\xi}_{i,t|t-1}$, its variance $F_{i,t}$, and the prediction error $v_{i,t} = \xi_{i,t} - \hat{\xi}_{i,t|t-1}$. These assemble into the Gaussian log-likelihood

$$\ell(\theta) = -\frac{1}{2} \sum_{i,t} \left[\ln F_{i,t}(\theta) + \frac{v_{i,t}(\theta)^2}{F_{i,t}(\theta)} \right], \quad (3)$$

which I maximize over θ . The maximum-likelihood estimates put persistence at $\hat{\rho}_z = 0.85$, with the persistent component accounting for half of the residual variance and a half-life near a decade; they closely agree with the autocovariance-minimum-distance estimates of Section 3.2 (the two routes are compared in Appendix C.4). A household block bootstrap that re-estimates the entire measurement chain, from the sample trims through the random-forest and state-space steps, returns the same 95% interval for ρ_z (Appendix D.5).

Given $\hat{\theta}$, the persistent component is recovered household by household by the Kalman smoother, which conditions each estimate on the household's entire residual path,

$$\hat{z}_{i,t} = \mathbb{E} \left[z_{i,t} \mid \{\xi_{i,s}\}_s; \hat{\theta} \right], \quad \hat{u}_{i,t}^{\text{exp}} = \hat{z}_{i,t} - \hat{\rho}_z \hat{z}_{i,t-1}. \quad (4)$$

The smoothed component correlates 0.88 with ξ , and the ratio of recovered to total persistent variance is 0.71, so the extraction captures most of the persistent signal.¹¹ The construction operates on PSID waves 2005–2023,¹² yielding 25,214 household-wave observations on 5,095 households.¹³

The two recovered shocks, the persistent u^{exp} and the transitory ε^{exp} , are the expenditure-risk objects carried into the wealth analysis, and each has its own passthrough to a downstream outcome $Y_{i,t}$. Both are covariance ratios with a common endogenous regressor, the quasi-difference $\Delta^{\rho_z} \xi_{i,t} = \xi_{i,t} - \rho_z \xi_{i,t-1}$, taken at the component persistence ρ_z of equation (2) rather than the attenuated lag-1 autocorrelation ρ_ξ of the observed residual. This quasi-difference carries both shocks at once, $\Delta^{\rho_z} \xi_{i,t} = u_{i,t}^{\text{exp}} + (\varepsilon_{i,t}^{\text{exp}} - \rho_z \varepsilon_{i,t-1}^{\text{exp}})$; an instrument $\zeta_{i,t}^s$ built from the residual path isolates one from the other, so that

$$\beta_{Y|s} = \frac{\text{Cov}(Y_{i,t}, \zeta_{i,t}^s \mid X_{i,t})}{\text{Cov}(\Delta^{\rho_z} \xi_{i,t}, \zeta_{i,t}^s \mid X_{i,t})}, \quad s \in \{u, \varepsilon\}, \quad (5)$$

The persistent shock ($s = u$) is instrumented by the forward, geometrically weighted moving sum of quasi-differences,

$$\zeta_{i,t}^u = \Delta^{\rho_z} \xi_{i,t+1} + \rho_z \Delta^{\rho_z} \xi_{i,t} + \rho_z^2 \Delta^{\rho_z} \xi_{i,t-1},$$

¹¹The smoother shrinks innovations toward zero, so the dispersion of the extracted $\hat{u}^{\text{exp}}$ understates the structural innovation; the calibration of Section 5 takes σ_u^2 from the estimated process, not from the sample moments of $\hat{u}^{\text{exp}}$.

¹²PSID added several non-durable consumption categories starting in 2005, including clothing, recreation, trips, household repairs and furnishings, and internet utilities; the consumption aggregate of Section 2.1 is therefore fully measured only from that wave onward, and the cross-fitted residual ξ inherits the same start year.

¹³The measurement sample exceeds the 22,497 transition pairs of Section 2.1 because the state-space estimation requires no consecutive-wave pairing: every household-wave with a valid residual contributes to the autocovariances at whatever lags its wave pattern supplies.

the AR(1) analog of the three-period moving-sum instrument of [Blundell et al. \(2008\)](#) and [Kaplan and Violante \(2010\)](#), to which it collapses at $\rho_z = 1$; it shares the persistent component with $\Delta^{\rho_z} \xi_{i,t}$ but carries none of the period- t transitory noise, so only $u_{i,t}^{\text{exp}}$ survives in the denominator. The transitory shock ($s = \varepsilon$) is instrumented by the lead quasi-difference $\zeta_{i,t}^\varepsilon = \Delta^{\rho_z} \xi_{i,t+1} = \xi_{i,t+1} - \rho_z \xi_{i,t}$, which picks out the period- t transitory term $\varepsilon_{i,t}^{\text{exp}}$ instead. Section 4 reports both covariance-ratio passthroughs as its baseline and tests them against household fixed effects and a non-linear link function; derivations are in Appendix D.

4 Expenditure risk and household wealth dynamics

This section traces how the two recovered expenditure shocks of Section 3.3 pass through to household wealth, alongside the permanent and transitory income shocks of Section 2.4. The persistent expenditure shock u^{exp} is the empirical heart of the paper’s claim: among the four risk components, it is the only one that depletes wealth, and the only one that moves households across the hand-to-mouth threshold in the entrapping direction on both margins, pushing them in and holding them there. Its imprint is a within-household phenomenon: it survives household fixed effects, and on the transition outcomes amplifies under them, precisely the conditioning under which the income-shock passthroughs to the fall margin attenuate to insignificance or marginal significance. The spending that moves with it concentrates on a basket tied to children, health events, home and vehicle upkeep, and memorable goods. The transitory component ε^{exp} co-moves with the household’s resource flow, with passthroughs of the opposite sign on every wealth outcome; the paper carries only the persistent expenditure shock into the structural model of Section 5.

4.1 Hand-to-mouth transitions: the fixed-effects asymmetry

Income shocks predict transitions into and out of hand-to-mouth status only loosely once household fixed effects absorb cross-household differences; the persistent expenditure shock predicts them robustly, and its imprint strengthens under exactly the conditioning that washes out the income shocks on the fall margin. The transitory expenditure shock’s passthroughs run in the opposite direction on both margins, the signature of a component that co-moves with the household’s resource flow.

I add the two recovered expenditure shocks of Section 3.3 to the income-risk passthrough regressions of Section 2.4, regressing each binary transition indicator jointly on the four risk components $(y_{i,t}^P, y_{i,t}^T, u_{i,t}^{\text{exp}}, \varepsilon_{i,t}^{\text{exp}})$ with the demographic and wave controls of Section 2.4, under both pooled and household-fixed-effects conditioning. Because the persistent and transitory expenditure shocks are recovered as correlated Kalman estimates, I report two estimators: a direct regression on the extracted shocks, and a two-stage least-squares version that instruments each expenditure shock with the noise-orthogonal moving sums

of equation (5), the just-identified covariance-ratio identification written as a regression. Table 4 reports the passthroughs in units of one standard deviation of each shock.

Table 4: Passthroughs to hand-to-mouth transitions, four risk components, PSID.

	Regression		IV (2SLS)	
	OLS	FE	OLS	FE
<i>Panel A: Fall into hand-to-mouth</i>				
y^P	−0.010*** (0.0029)	−0.004 (0.0031)	−0.010*** (0.0032)	−0.002 (0.0036)
y^T	−0.010*** (0.0026)	−0.004* (0.0023)	−0.009*** (0.0028)	−0.002 (0.0026)
u^{exp}	0.024*** (0.0038)	0.029*** (0.0050)	0.062*** (0.0101)	0.073*** (0.0147)
ε^{exp}	−0.025*** (0.0037)	−0.018*** (0.0036)	−0.031*** (0.0060)	−0.025*** (0.0059)
HH-FE	No	Yes	No	Yes
Households	3,096	3,096	2,610	2,610
Observations	10,154	10,154	7,962	7,962
<i>Panel B: Escape from hand-to-mouth</i>				
y^P	0.007 (0.0105)	0.009 (0.0150)	0.014 (0.0120)	0.011 (0.0180)
y^T	0.031*** (0.0103)	0.036*** (0.0121)	0.023** (0.0113)	0.033** (0.0137)
u^{exp}	−0.028** (0.0129)	−0.090*** (0.0215)	−0.117*** (0.0304)	−0.245*** (0.0601)
ε^{exp}	0.067*** (0.0124)	0.084*** (0.0166)	0.099*** (0.0176)	0.121*** (0.0248)
HH-FE	No	Yes	No	Yes
Households	1,109	1,109	874	874
Observations	2,264	2,264	1,707	1,707

Notes: Passthroughs to the probability of falling into (Panel A) and escaping from (Panel B) hand-to-mouth status, in units of one standard deviation of each shock, from a joint regression on the permanent and transitory income shocks (y^P , y^T) and the persistent and transitory expenditure shocks (u^{exp} , ε^{exp}). The *Regression* columns regress on the Kalman-extracted shocks directly; the *IV (2SLS)* columns instrument the two expenditure shocks with the noise-orthogonal moving sums of Section 3.3. *OLS* and *FE* denote pooled and household-fixed-effects conditioning. The fall sample conditions on $H2M_{t-2} = 0$, the escape sample on $H2M_{t-2} = 1$; the IV columns use fewer observations because the instruments require additional adjacent waves. Household-clustered standard errors in parentheses; *, **, *** denote significance at the 10%, 5%, 1% levels.

The fall into hand-to-mouth status shows the asymmetry sharply. Both income shocks predict falls in the pooled regression (each −0.010, significant), but their passthroughs collapse toward insignificance under household fixed effects (−0.004), the attenuation documented in Section 2.4. The persistent expenditure shock moves the opposite way: its

passthrough to falling is positive, significant, and *amplifies* under fixed effects, from 0.024 to 0.029 in the direct regression and from 0.062 to 0.073 once instrumented. Within a household, the waves with a positive persistent expenditure shock are the waves in which it falls into hand-to-mouth status.

Escape from hand-to-mouth status mirrors this. The persistent expenditure shock lowers the probability of escaping, significantly under every specification (-0.028 to -0.090 in the direct regression, -0.117 to -0.245 once instrumented), and the effect again strengthens under fixed effects. Permanent income does not predict escape under any specification, and transitory income retains a significant positive escape passthrough. The persistent expenditure shock is thus the only risk component with adverse passthroughs on both margins of hand-to-mouth status. The transitory expenditure shock's passthroughs run the other way on both margins, negative on entry and positive on escape, consistent with the resource-flow reading established next.¹⁴

4.2 The wealth signature of expenditure risk

On income-scaled wealth changes the four risk components separate by sign: the persistent expenditure shock is the only one that depletes wealth, while both income shocks and the transitory expenditure shock pass through positively. The persistent shock's footprint on consumption is sizable, but a household funds it by running down net worth rather than by saving more.

Table 5 runs the four-component regression of Section 4.1 on consumption growth $\Delta \ln c_{i,t}$ and income-scaled wealth changes $\Delta W_{i,t}/\bar{y}_i$. The wealth panel is unambiguous: the persistent expenditure shock passes through negatively under every specification, from -0.102 (pooled regression) to -0.124 under household fixed effects, and from -0.279 to -0.423 once instrumented. The two income shocks and the transitory expenditure shock all pass through *positively* to wealth, the transitory expenditure shock most strongly ($+0.25$ to $+0.39$).

The persistent shock's consumption footprint is large. A one-standard-deviation shock corresponds to a movement of about 11% in non-durable consumption.¹⁵ Set against a wealth drawdown several times larger in magnitude, the picture is of a household meeting a persistent rise in its expenditure needs by depleting its buffer.

The negative wealth passthrough of the persistent expenditure shock is not the house-

¹⁴Appendix D.2 confirms the transition results under a probit link and a Mundlak (Mundlak, 1978) correlated-random-effects device in place of the linear fixed-effects estimator.

¹⁵This is the noise-orthogonal covariance-ratio estimate; the direct-regression coefficient in the consumption-growth panel (about 0.21 under fixed effects) overstates it because consumption growth mechanically contains the residual from which the expenditure shock is extracted. The figure measures the scale of the shock in consumption units, not an insurance passthrough: the persistent shock is recovered from the consumption residual itself, so its relation to consumption growth is partly definitional. The wealth and hand-to-mouth outcomes, which do not share this construction, carry the structural content.

Table 5: Passthroughs to consumption growth and wealth changes, four risk components, PSID.

	Regression		IV (2SLS)	
	OLS	FE	OLS	FE
<i>Panel A: Consumption growth $\Delta \ln c$</i>				
y^P	0.041*** (0.0024)	0.041*** (0.0031)	0.044*** (0.0026)	0.047*** (0.0039)
y^T	0.040*** (0.0023)	0.046*** (0.0024)	0.038*** (0.0027)	0.044*** (0.0030)
u^{exp}	0.109*** (0.0028)	0.215*** (0.0044)	0.115*** (0.0071)	0.147*** (0.0135)
ε^{exp}	0.152*** (0.0030)	0.111*** (0.0033)	0.192*** (0.0047)	0.199*** (0.0062)
<i>Panel B: Wealth change $\Delta W/\bar{y}$</i>				
y^P	0.050*** (0.0097)	0.043*** (0.0138)	0.050*** (0.0101)	0.035** (0.0137)
y^T	0.057*** (0.0104)	0.064*** (0.0117)	0.054*** (0.0120)	0.055*** (0.0136)
u^{exp}	-0.102*** (0.0095)	-0.124*** (0.0181)	-0.279*** (0.0294)	-0.423*** (0.0571)
ε^{exp}	0.251*** (0.0126)	0.261*** (0.0157)	0.347*** (0.0211)	0.392*** (0.0286)
HH-FE	No	Yes	No	Yes
Households	3,479	3,479	2,943	2,943
Observations	12,418	12,418	9,669	9,669

Notes: Passthroughs to consumption growth $\Delta \ln c$ (Panel A) and income-scaled wealth changes $\Delta W/\bar{y}$ (Panel B), in units of one standard deviation of each shock; columns and estimators as in Table 4. The $\Delta W/\bar{y}$ regression trims the top and bottom 0.5% of its distribution, matching Panel A of Table 2. The consumption-growth panel is reported for completeness; because the expenditure shocks are extracted from the consumption residual, the consumption-growth passthroughs are partly mechanical, and the covariance-ratio estimate discussed in the text (about 0.11 for the persistent shock) is the relevant measure of the shock's scale. Household-clustered standard errors in parentheses; *, **, *** denote significance at the 10%, 5%, 1% levels.

hold budget constraint restated. Conditional on initial resources, any unpredicted rise in spending must lower measured saving within the wave, so if the estimates of Table 5 were accounting, every component of the consumption residual would deplete wealth. The two expenditure shocks are extracted from the same residual by the same machinery, yet they separate in sign: the persistent shock passes through negatively to net worth, the transitory component positively, and the same split runs through the hand-to-mouth margins, where the persistent shock raises entry and lowers escape while the transitory component's passthroughs carry the opposite signs. An accounting identity cannot produce this asymmetry, because it is blind to how long the spending it records will last; the split instead matches the reading of this section, in which the transitory component co-moves with the household's resource flow while only the persistent component is financed by running down the buffer. Two further mechanical channels run in the wrong direction to explain the estimates: the shelter imputation ties measured consumption to home equity and pushes the wealth passthrough toward positive,¹⁶ and the persistent shock carries no positive predictive content for future income growth, so the drawdown is not the front-running of an anticipated income event (Appendix C.5). What remains of the concern is timing alone: the passthroughs measure the within-wave financing of the shock, and the persistent component's distinguishing content is that this financing recurs, which is what the escape margin, households failing to rebuild net worth wave after wave, measures directly. The recurrence is also testable as a prediction: conditional on wealth and income measured at t , the one-sided filtered state still predicts the following wave's fall into hand-to-mouth status and continued wealth drawdown, where a within-wave identity predicts nothing (Appendix D.3).

The persistent expenditure shock's within-household contribution to the variance of wealth changes is on the order of all of income risk, and its passthrough carries the opposite sign. The within-household relationship between the persistent shock and consumption growth also furnishes a calibration moment for the structural model of Section 5, where it pins the marginal-utility weight φ_0 . The standard errors of this section condition on the estimated risk components; a bootstrap of the full estimation chain widens them by 6% at the median, and no passthrough changes sign in any of 500 replications (Appendix D.5).

4.3 The channel: what moves with the persistent shock

Which spending moves with the persistent expenditure shock? Regressing each non-durable category's budget share on the recovered shock $\hat{u}_{i,t}^{\text{exp}}$, within households, sorts the basket cleanly: nine categories load positively and significantly, together about 20%

¹⁶Shelter for owners is imputed at 6% of home equity, following Aguiar et al. (2025), so home-equity movements enter both the consumption aggregate and net worth; that linkage generates a *positive* co-movement between measured consumption and wealth changes, attenuating rather than producing the negative wealth passthrough of the persistent expenditure shock. Consistently, shelter carries the largest negative share loading in Section 4.3.

of non-durable spending, and they service a household's stocks of physical, human, and memorable capital. The categories that contract to make room are committed staples in the spirit of [Chetty and Szeidl \(2007\)](#) consumption commitments, mainly shelter, food at home, and utilities, that households cannot rapidly reschedule at the wave horizon.

For each non-durable category k , I regress its share of total non-durable consumption $s_{i,t}^k$ on the recovered persistent expenditure shock $\hat{u}_{i,t}^{\text{exp}}$, conditional on household and wave fixed effects and the demographic controls of Section 4.1.¹⁷ The decomposition runs on the 24-category finer disaggregation of the PSID expenditure module, which separates, for example, lumpy inpatient and dental outlays from the less volatile health-insurance premia and home repairs from home furnishings; the coarser 10-category aggregation gives a consistent reading and is reported in Appendix D.4. A category is classified *risky* when its loading is positive and significant at the 5% level; beyond organizing this section's reading, the classification defines the risky aggregate whose consumption ratio supplies the allocation moments that discipline the marginal-utility process in Section 5.3.

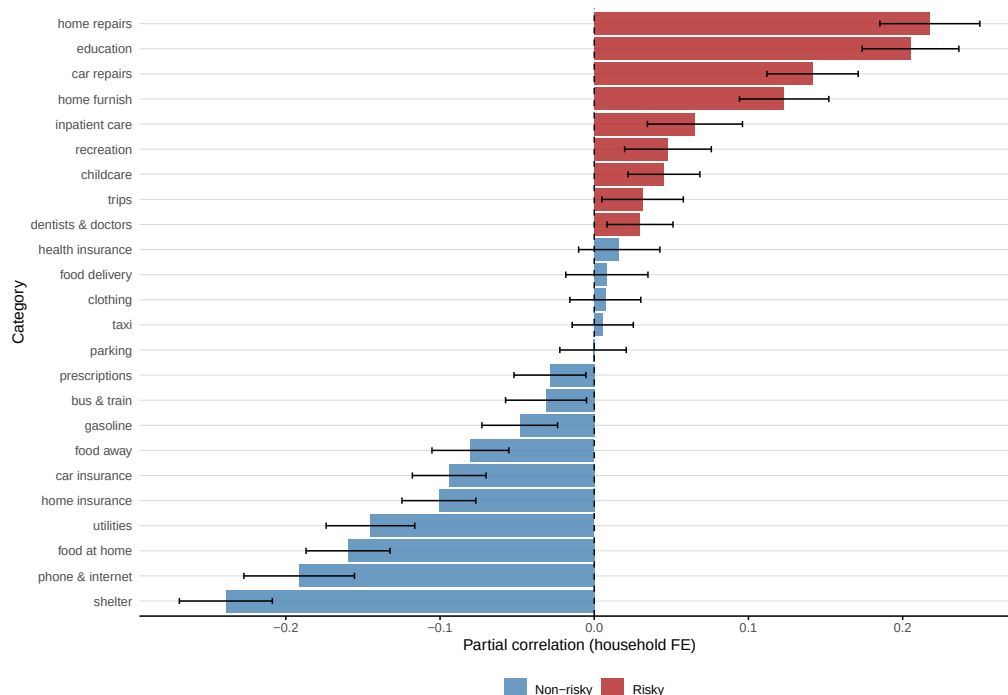
The nine positive loaders fall into four groups: *children* (education, childcare), *lumpy health* (inpatient care, dentists and doctors), *household and vehicle upkeep* (home repairs, home furnishings, car repairs), and *memorable goods* (recreation, trips), the expenditures that build a stock of memories households draw utility from over time ([Hai et al., 2020](#)). Together they make up a 19.8% basket whose aggregate share carries a partial correlation of +0.41 with the persistent shock, larger than any single category's loading. Telling here is what does *not* load: the health-insurance premium is classified non-risky ($t = 1.2$), even as the lumpy inpatient and dental outlays load strongly, so the shock tracks hard-to-reschedule health *events* rather than the contractual premium. The strongest negative loadings are exactly the committed staples whose share must contract when the budget tilts toward the basket: shelter (−0.24), phone and internet, food at home, utilities, and the home- and car-insurance lines. Shelter is the canonical [Chetty and Szeidl \(2007\)](#) consumption commitment; the rest sit in the same spirit as high-adjustment-cost or contractually fixed staples that bind at the two-year wave horizon.

The basket has counterparts in independent data. [Fulford and Low \(2024\)](#), asking households directly about unexpected expenses in a survey linked to credit records, elicit a closely overlapping list, led by medical, vehicle, home, and child-related expenses; [Miranda-Pinto et al. \(2025\)](#) find education and transportation the categories over-represented in the PSID's high-consumption episodes; and the expenses [Bergholt et al. \(2025\)](#) offer to illustrate the income-orthogonal shocks in their Norwegian transaction records, health bills, car and home repairs, a newborn child, read off the same list.

These loadings are not income elasticity in disguise. Appendix Figure D.1 regresses

¹⁷The reported partial correlation is the regression coefficient rescaled by the ratio of standard deviations, $\hat{\beta}^k \cdot \text{sd}(\hat{u}^{\text{exp}}) / \text{sd}(s^k)$, a standardized partial correlation conditional on the household and wave fixed effects and the control set.

Figure 3: Category-level loadings on the persistent expenditure shock, PSID.



Notes: Standardized partial correlations of each non-durable consumption category's budget share with the recovered persistent expenditure shock $\hat{u}_{i,t}^{\text{exp}}$, conditional on household and wave fixed effects and the demographic controls of Section 4.1. Bars are ordered by point estimate; horizontal whiskers are household-clustered 95% confidence intervals. Categories shaded as risky (the expenditure-risk basket) load positively and clear the 5% significance threshold; the negative loaders are the committed staples in the spirit of Chetty and Szeidl (2007) consumption commitments (shelter, phone and internet, food at home, utilities, and the home- and car-insurance lines).

each basket category's share jointly on all three identified shocks, y^P , y^T , and $\hat{u}^{\text{exp}}$; the persistent expenditure shock dominates the loading on every category except childcare, where the transitory income shock co-loads, consistent with the income elasticity of childcare in working-age households.¹⁸

Read together, the positive basket and the contracting commitments are consistent with a household-capital-servicing interpretation of the persistent shock: a positive realization funds the flow expenditures that maintain or expand a household's stocks of physical, human, and memorable capital, with the budget shift absorbed by the contractual categories the household cannot reschedule. The interpretation is a reading of the empirical pattern, not an independently identified mechanism; the structural model of Section 5 parameterizes the same composition through a marginal-utility shock without committing to the categorical reading.

5 A model of income and expenditure risk with fixed preference heterogeneity

An incomplete-markets heterogeneous-agent overlapping-generations model with two calibrated economies, the ER economy (income and expenditure risk) and the IR economy (income risk only, $\rho_\varphi = \sigma_\varphi = 0$), provides the structural framework for quantifying expenditure risk's role in closing the household side of the wealth and consumption dynamics gap. The household problem combines fixed discount-factor heterogeneity (Carroll et al., 2017; Aguiar et al., 2025) over a working-life life-cycle, with expenditure risk entering through persistent marginal-utility shocks on a risky-vs-non-risky consumption split; general equilibrium clears the capital, labor, and social-security markets at biennial frequency, with accidental bequests redistributed across living households. Both economies match the empirical cross-sectional and life-cycle wealth profile within calibration tolerance, sharing the retirement top-tail wealth undershoot characteristic of single-asset β -heterogeneity heterogeneous-agent overlapping-generations (HA-OLG) models. Re-running the empirical passthroughs on simulated panels then validates the calibrated ER economy as the structural counterpart of the data, while the income-risk economy, run through the same estimation pipeline, cannot reproduce the wealth-depleting signature of the persistent expenditure shock.

¹⁸A natural concern is that within-household healthcare-expenditure variation is the joint response of medical spending and labor supply to an illness shock, in which case the healthcare loading would pick up income-driven movement masquerading as expenditure risk. The flexible model of Section 3.1 includes the reference person's contemporaneous health-status variable in its predictor set, so the within-household health-state component enters the fitted prediction $\hat{c}$ rather than the residual ξ on which the identification operates; the multi-shock partial correlations of Appendix Figures D.1 and D.3 confirm that the health-related loadings do not respond to income shocks under this conditioning.

5.1 Households

Each household solves a finite-horizon consumption-saving problem with homothetic preferences over a risky and a non-risky non-durable consumption good, subject to persistent marginal-utility shocks on the risky-vs-non-risky consumption split (in the lineage of [Palumbo, 1999](#); [De Nardi et al., 2010](#); [Russo, 2025](#)). Income is exogenous and decomposes into a fixed-type component plus permanent and transitory shocks; cash-on-hand is the recursive state, and the household's discount factor is drawn from a permanent cross-household distribution ([Carroll et al., 2017](#); [Aguiar et al., 2025](#)). The IR and ER economies share the same income process, β -heterogeneity, and demographic structure; the IR economy is the $\rho_\varphi = \sigma_\varphi = 0$ specialization of ER, in which the marginal-utility shock collapses to a constant and the problem reduces to a one-good consumption-saving model otherwise identical to ER.

The household's flow utility over its two non-durable consumption goods at age j is

$$u(c_{1,i,j}, c_{2,i,j}; \varphi_{i,j}) = \varphi_0 v(c_{1,i,j}) + \exp(\varphi_{i,j}) v(c_{2,i,j}), \quad (6)$$

where $v(x) = (x^{1-\gamma} - 1)/(1 - \gamma)$ is CRRA flow utility with common curvature γ , $c_{1,i,j}$ is the non-risky and $c_{2,i,j}$ the risky good, φ_0 is a calibrated relative-weight parameter, and $\varphi_{i,j}$ is the persistent marginal-utility shock that perturbs the relative weight on the risky good across periods. Homotheticity of the inner aggregator pins the within-period allocation in closed form,

$$\frac{c_{2,i,j}}{c_{1,i,j}} = \left(\frac{\exp(\varphi_{i,j})}{\varphi_0} \right)^{1/\gamma}, \quad (7)$$

so that the two consumption controls collapse to a single effective intertemporal choice of total non-durable consumption $c_{i,j} = c_{1,i,j} + c_{2,i,j}$ at every state. The marginal-utility shock itself follows a Gaussian AR(1),

$$\varphi_{i,j+1} - \mu_\varphi = \rho_\varphi (\varphi_{i,j} - \mu_\varphi) + u_{i,j+1}^\varphi, \quad u^\varphi \sim \mathcal{N}(0, \sigma_\varphi^2), \quad (8)$$

with unconditional mean $\mu_\varphi = -\frac{1}{2}\sigma_\varphi^2/(1 - \rho_\varphi^2)$ corrected so that $\mathbb{E}[\exp(\varphi)] = 1$ in the stationary cross-section.

Labor income at age j is $y_{i,j}(\tau_i, \eta_{i,j}, \varepsilon_{i,j}) = (1 - \theta) w \cdot \exp(\mu_j + \tau_i + \eta_{i,j} + \varepsilon_{i,j})$, with w the equilibrium wage, θ the payroll tax, and the log decomposition following the standard [Storesletten, Telmer, and Yaron \(2004\)](#) structure: a deterministic age profile μ_j , a fixed-type component τ_i drawn at birth, a random-walk persistent component $\eta_{i,j}$, and an i.i.d. transitory shock $\varepsilon_{i,j}$; variance parameters are identified from PSID by the BPP machinery of [Appendix B](#) and held fixed across the two economies. The household's state at age j is the tuple $s_{i,j} = (x_{i,j}, \tau_i, \beta_i, \eta_{i,j}, \varphi_{i,j}, j)$, where cash-on-hand $x_{i,j}$ absorbs the realized transitory income shock at the period boundary and cuts the dimension of the value function

by one relative to carrying assets and the transitory shock separately (Carroll, 2006). The discount factor β_i is drawn once at birth from a truncated Gaussian mixture and held fixed across the life-cycle, with the seven mixture parameters calibrated jointly to the cross-sectional and life-cycle wealth-distribution targets of Section 5.3 below; asset choices are constrained by a state-dependent borrowing limit equal to a fixed pledgeable fraction of the annuitized (per-period-equivalent) value of expected future labor and pension income.

The household chooses total consumption $c_{i,j}$ (and, via equation (7), its within-period allocation across the two goods) to solve the recursive problem

$$V_j(s_{i,j}) = \max_{c_{i,j}} \tilde{u}(c_{i,j}, \varphi_{i,j}) + \beta_i \mathfrak{s}_j \mathbb{E}_j[V_{j+1}(s_{i,j+1})], \quad (9)$$

where $\tilde{u}$ is the indirect flow utility implied by the optimal within-period split in equation (7), $\mathfrak{s}_j$ is the age- j conditional one-period survival probability of the demographic block of Section 5.2, and the next-period state $s_{i,j+1}$ follows the budget constraint $x_{i,j+1} = R(x_{i,j} - c_{i,j}) + y_{i,j+1} + T \cdot w_{j+1}^{\text{beq}}$ together with the transition rules for $\eta_{i,j}$ and $\varphi_{i,j}$; the full optimization problem with explicit borrowing-limit and working / retirement-transition / retired age-phase splits is written out in Appendix E.1. The IR baseline sets $\rho_\varphi = \sigma_\varphi = 0$, collapsing $\varphi_{i,j}$ to its constant mean of zero; under the homothetic CRRA preferences of equation (6), the within-period allocation reduces to a fixed share determined by φ_0 alone, and the household's intertemporal problem becomes the standard one-good consumption-saving model with the same β_i , income process, life-cycle, and borrowing limit as ER.

5.2 General equilibrium and life-cycle

The life-cycle runs at biennial frequency from age 25 to a maximum age of 94, with deterministic retirement at age 64 and age-conditional mortality throughout, and a Huggett-style accidental-bequest pool (Huggett, 1996) tilted geometrically toward younger working-age households, so that entering cohorts do not uniformly start at zero wealth and the youth H2M share can sit below one. Prices clear in a Cobb-Douglas production economy ($\alpha = 0.36$, biennial $\delta \approx 0.15$): the interest rate, wage, and pay-as-you-go (PAYG) social security payroll tax rate are solved jointly to satisfy capital-market, labor-market, and pension-budget conditions; replacement rates are taken from Purcell (2012) and rank-progressive across retirement-age earnings quintiles.

The model runs at biennial frequency on a stationary overlapping-generations demographic structure. A new cohort enters at age 25 each period at constant biennial population growth $n = 0.012$ (approximately 0.6% per year); households face age-conditional one-period survival probabilities $\{\mathfrak{s}_j\}$ from age 25 onward, work for $J^W = 20$ biennial periods through age 64, retire deterministically thereafter, and die with certainty after $J^D = 35$ periods at the maximum age of 94. The survival probabilities are taken from U.S. Social

Security Administration male life tables, and the stationary cross-sectional age distribution $\{\Psi_j\}$ satisfies the standard OLG recursion $\Psi_j \propto s_{j-1} \Psi_{j-1} / (1 + n)$ normalized to sum to one.

Production is Cobb-Douglas, $Y = K^\alpha L^{1-\alpha}$, with capital share $\alpha = 0.36$ and biennial depreciation $\delta \approx 0.1536$ (compounding the standard annual $\delta_{\text{annual}} = 0.08$); the competitive firm sets the rental rate of capital and the wage from the standard FOCs $r + \delta = \alpha (K/L)^{\alpha-1}$ and $w = (1 - \alpha) (K/L)^\alpha$. Accidental bequests from deceased households are aggregated each period into a single pool T and redistributed across the living with a geometric age tilt $w_j^{\text{beq}} \propto (\text{beq_tilt})^j$, generalizing the uniform-redistribution rule of [Huggett \(1996\)](#), which is nested at $\text{beq_tilt} = 1$. The beq_tilt parameter is calibrated jointly with the β -distribution to the life-cycle wealth and H2M targets of Section 5.3.

The government runs a balanced pay-as-you-go social security system: working-age households pay a labor-income tax rate θ on earnings, and retirees receive a pension equal to a rank-progressive replacement-rate schedule $\text{rr}(\eta) \in \{0.778, 0.591, 0.484, 0.422, 0.333\}$ applied to the equilibrium wage and indexed to the household's persistent-income quintile at retirement, with the five replacement rates estimated from U.S. Health and Retirement Study survey data by [Purcell \(2012\)](#). Competitive equilibrium clears three markets simultaneously: capital (K equals the aggregate of household asset holdings over the stationary cross-sectional distribution), labor (L equals the aggregate of working-age effective labor supply), and the pension budget ($\theta \cdot w \cdot L$ equals aggregate pension outlays); the three equilibrium prices (r, w, θ) are pinned by the three clearing conditions, with goods-market clearing following from Walras' law. The formal stationary recursive competitive equilibrium definition is given in Appendix E.2.

5.3 Calibration

Both economies draw on the same set of empirical moments but consume different subsets: the IR calibration targets seven wealth and hand-to-mouth life-cycle moments via global optimization; the ER calibration adds the within-household partial R^2 of the persistent expenditure shock on consumption growth to the optimization (jointly identifying φ_0 alongside the β -distribution parameters) and feeds two additional expenditure-risk moments (the autocorrelation and variance of the risky/non-risky consumption ratio) into the model via the closed-form φ -process mapping. The Gaussian-mixture β distribution is the principal joint margin ([Carroll et al., 2017](#); [Aguiar et al., 2025](#)), calibrated within each economy to its respective moment subset; income-process and demographic parameters are held common across the two economies.

Externally-set parameters and demographic objects are summarized in Appendix Table E.1. The CRRA curvature is fixed at $\gamma = 2$; the Cobb-Douglas technology parameters (α, δ, A) follow the standard [Cooley and Prescott \(1995\)](#) and [Krusell and Smith \(1998\)](#) con-

ventions. The biennial income-process variance parameters ($\sigma_\tau^2, \sigma_\eta^2, \sigma_\varepsilon^2$) are taken directly from the income-components identification of Appendix B on the same PSID sample as the empirical analysis, with $\rho_\eta = 1$ imposing the random-walk limit (Blundell et al., 2008; Kaplan and Violante, 2010). Demographic objects (the life-cycle horizons (J^W, J^D), biennial population growth n , age-conditional survival $\{\mathfrak{s}_j\}$, the deterministic age-income profile $\{\mu_j\}$, and the rank-progressive replacement-ratio schedule of Section 5.2) are pinned by U.S. data sources documented in Appendix Table E.1, and the borrowing-limit pledgeable share is set to 0.20, rounding the 0.185 of Kaplan et al. (2014) and Aguiar et al. (2025).

The internally calibrated parameters and their target moments are reported in Table 6. The principal joint margin is the cross-household discount-factor distribution $\mathcal{F}_\beta$, parameterized as a two-component truncated Gaussian mixture with five parameters plus the lower support ($\beta_{\mu_1}, \beta_{\mu_2}, \beta_{\sigma_1}, \beta_{\sigma_2}, \beta_{p_1}, \beta_{\min}$), in the lineage of Carroll et al. (2017) and Aguiar et al. (2025); the upper support is fixed at $\beta_{\max} = 0.9975$. The IR calibration targets seven life-cycle wealth and hand-to-mouth moments (the youth, midlife, and retirement H2M shares and the midlife and retirement Q_{50} and Q_{75} wealth-to-income ratios) jointly with the `beq_tilt` bequest-tilt parameter, via global optimization of a weighted sum-of-squared-residuals objective; the midlife and retirement Q_{25} ratios are reported alongside the targets but sit outside the objective.

The ER calibration adds three expenditure-risk objects to this structure. The relative-weight parameter φ_0 enters the same global optimization as an eighth free parameter, jointly identified by the within-household partial R^2 of the persistent expenditure shock on consumption growth from Section 4.2. The AR(1)-dynamics marginal-utility parameters ($\rho_\varphi, \sigma_\varphi$) enter through the closed-form mapping of equation (C.8) in Appendix C.1, outside the optimization loop: given the curvature γ and the homotheticity-driven intratemporal allocation of equation (7), the empirical autocorrelation and variance of $\ln(c_2/c_1)$ over the working life pin ρ_φ and σ_φ analytically. Income-process and demographic parameters are held common between the IR and ER calibrations.

The calibrated persistence $\rho_\varphi = 0.347$ and the empirical persistence $\rho_z \approx 0.86$ of Section 3.3 measure different objects, and the model itself supplies the mapping between them. By the intratemporal allocation (7), ρ_φ equals the autocorrelation of $\ln(c_2/c_1)$: it is the persistence of the reallocation between the risky and the non-risky good, which the PSID puts at 0.3473. The persistence ρ_z is instead a property of the residual of *total* log consumption, and a marginal-utility shock reaches total consumption through the household's full policy response, within-period reallocation and intertemporal financing at once, so the persistence of the total-consumption residual is an equilibrium outcome of the model rather than one of its parameters.

Re-running the autocovariance minimum-distance identification of Section 3.2 on the ER economy's simulated random-forest residual of Section 5.5 makes the mapping quantitative: the marginal-utility process with persistence 0.347 generates a total-consumption

residual with estimated persistence 0.68 and a persistent variance share of 0.62, against (0.86, 0.50) in the PSID, at $\gamma_0 = 0.061$ against 0.111, about half the scale. The calibrated economy therefore reproduces the persistent-plus-transitory shape of the residual's autocovariances without targeting it, and roughly doubles the persistence of its marginal-utility input on the way to the measured residual, covering about two thirds of the distance between ρ_φ and ρ_z . Because the simulated panel carries no measurement error, its transitory-looking layer reflects the extraction's smoothing dynamics rather than noise, and the AR(1)-plus-noise split is an approximation there (the overidentification test rejects at the simulated panel's sample size); the naive pooled lag-1 autocorrelation of ξ , about 0.44 in the PSID, comes back at 0.434 on the same panel. The remaining persistence gap should also be read against the sampling uncertainty of its empirical anchor: the fixed-heterogeneity-immune first-difference identification of Appendix C.4 puts ρ_z at 0.77 with a 95% interval of [0.52, 0.90], and the simulated residual's 0.68 lies inside that interval.

Table 6: Internally calibrated parameters and target moments.

	Parameter values			Moment values		
Parameter	IR	ER	Targeted moment(s)	Data	IR	ER
<i>Discount-factor mixture, $\mathcal{F}_\beta$ (truncated 2-component Gaussian)</i>						
β_{μ_1}	0.9791	0.9689	H2M share — midlife	0.2316	0.2307	0.2104
β_{μ_2}	0.8107	0.7419	$Q_{25} W/\bar{y}$ — midlife	0.0041	0.0347	0.0500
β_{σ_1}	0.0600	0.0787	$Q_{50} W/\bar{y}$ — midlife	0.6501	0.7117	0.7106
β_{σ_2}	0.1015	0.1349	$Q_{75} W/\bar{y}$ — midlife	2.1405	2.2638	2.2596
β_{p_1}	0.3288	0.3118	H2M share — retirement	0.0891	0.0821	0.0839
$\beta_{\min}$	0.6071	0.6089	$Q_{25} W/\bar{y}$ — retirement	0.5780	0.4490	0.4348
			$Q_{50} W/\bar{y}$ — retirement	2.1659	1.9672	1.7791
			$Q_{75} W/\bar{y}$ — retirement	6.1472	5.2621	5.1188
<i>Bequest tilt</i>						
beq_tilt	0.7886	0.7799	H2M share — youth	0.3479	0.3585	0.3440
<i>ER marginal-utility process</i>						
φ_0	—	0.5809	Partial R^2 of $\Delta \ln c$ (within-HH)	0.1165	—	0.1269
ρ_φ	—	0.3473	Autocorr. of $\ln(c_2/c_1)$	0.3473	—	0.3473
σ_φ	—	1.1988	Var. of $\ln(c_2/c_1)$	0.4085	—	0.4085

Notes: Parameter columns report the calibrated values; moment columns report the empirical target (Data) and the model-implied moment at the calibrated parameter values (IR, ER). The discount-factor mixture and bequest tilt are calibrated jointly to the youth, midlife, and retirement H2M shares and the midlife and retirement Q_{50} and Q_{75} wealth-quartile ratios (seven life-cycle moments); the Q_{25} rows are displayed for completeness but are not part of the calibration objective. The ER relative-weight parameter φ_0 is additionally calibrated to the within-household partial R^2 of the persistent expenditure shock on consumption growth from Section 4.2; the AR(1) marginal-utility parameters (ρ_φ , σ_φ) are pinned outside the optimization loop by the closed-form mapping to the autocorrelation and variance of $\ln(c_2/c_1)$ derived in Appendix C.1, equation (C.8).

The ER and IR economies land on close β distributions, so the dynamic differences documented in Section 6 do not rest on a wholesale reshaping of discount-factor heterogene-

ity. Appendix Figure E.1 plots the calibrated two-component mixture for both economies, as a density (panel (a)) and as the CDF with the $n_\beta = 5$ discretization the model uses (panel (b)). The lower truncation is essentially identical ($\beta_{\min} = 0.609$ in ER versus 0.607 in IR), the patient-component means sit within a point of each other ($\beta_{\mu_1} = 0.969$ versus 0.979), and the weight on the patient component is comparable ($\beta_{p_1} \approx 0.31$ versus 0.33). The one systematic difference is in the impatient component, which ER calibrates lower and more dispersed than IR ($\beta_{\mu_2} = 0.742$ versus 0.811, $\beta_{\sigma_2} = 0.135$ versus 0.101). Because the marginal-utility-shock channel already generates precautionary saving that income risk alone does not, the ER economy reaches the same hand-to-mouth and wealth targets with a slightly more impatient lower tail rather than a wholesale reshaping of the β distribution. This near-coincidence narrows what the comparison that follows can load onto discount factors: both economies clear the same cross-sectional and life-cycle wealth targets of Section 5.4, yet only the economy with expenditure risk reproduces the passthroughs re-estimated on its simulated panel in Section 5.5. The residual β difference is handled directly in Section 6.5, which dials σ_φ at a fixed β distribution, with the consumption-growth and escape-rate results scaling with σ_φ alone and the fall rate reading the same way once the corner's larger hand-to-mouth stock is accounted for.

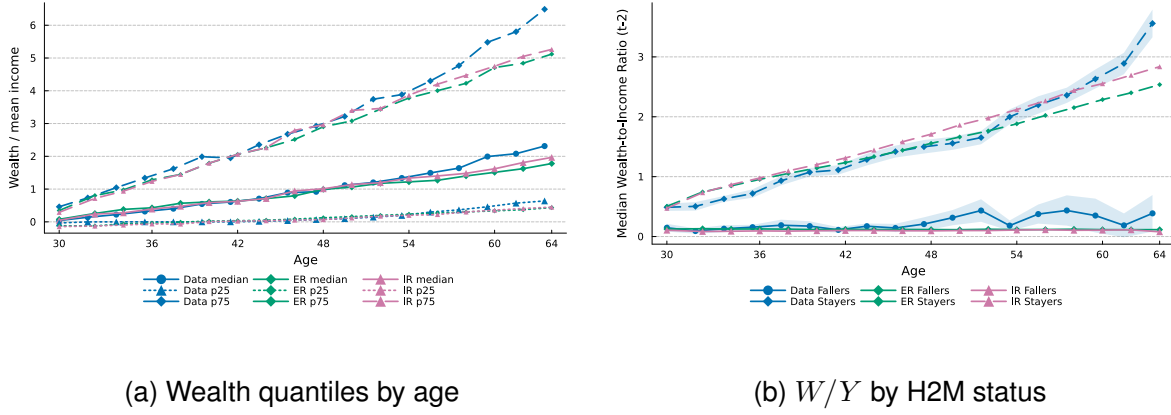
5.4 Time-preference heterogeneity fits cross-sectional and life-cycle dimensions of the wealth distribution

Both economies hit the targeted life-cycle moments within calibration tolerance (the youth, midlife, and retirement H2M shares and the wealth quantiles by age) and reproduce the untargeted cross-sections closely: the wealth, income, and consumption Gini coefficients come within a few hundredths of the data, and the conditional W/Y profile by H2M status matches qualitatively. Both economies undershoot the top tail of the wealth distribution upon retirement, a limitation characteristic of incomplete-market economies in the absence of extra incentives for wealth accumulation beyond patience and preparing for retirement. The level fit, shared symmetrically across the two economies, sharpens the dynamic comparison of Section 6: differences in dynamics cannot be attributed to differential level-moment fit.

The hand-to-mouth life-cycle profile is hit at all three targeted stages in both economies: youth (ER 0.344 / IR 0.359 vs data 0.348), midlife (ER 0.210 / IR 0.231 vs data 0.232), and retirement (ER 0.084 / IR 0.082 vs data 0.089) all land within 2.2 percentage points of their empirical targets (Appendix Figure E.2). The youth margin is the demanding one, because new cohorts enter at age 25 with zero assets and only the geometrically-tilted slice of the accidental-bequest pool as starting wealth; the calibrated bequest tilt is the margin that lets both economies reach the data's 0.348 without an explicit intergenerational wealth channel. Because the fit is shared symmetrically across the two regimes, the H2M side of the life-

cycle profile reveals no differential level fit between ER and IR.

Figure 4: Cross-sectional wealth profile by age: data vs calibrated economies.



Notes: Model counterpart of Figure A.2. Panel (a): cross-sectional wealth quantiles (p_{25} , median, p_{75}) in mean-household-income units, with data (PSID 1999–2023) and the two calibrated economies overlaid by single-year age. Panel (b): median wealth-to-income ratio at $t - 2$ conditional on H2M status at t , with data, IR, and ER overlaid; the data series is shown with its 95% bootstrap confidence interval.

Panel (a) of Figure 4 extends the life-cycle reading to the cross-section of wealth itself. The median and p_{25} wealth-to-mean-income paths overlay the data through the early 50s in both economies, with the data median rising above both in the final pre-retirement waves, a milder version of the top-tail undershoot below; the p_{75} path tracks the data closely through midlife (Q_{75}^{midlife} ER 2.26 / IR 2.26 vs data 2.14) and starts to undershoot at the upper end of the working life ($Q_{75}^{\text{retirement}}$ ER 5.12 / IR 5.26 vs data 6.15, a ~16% retirement top-tail undershoot). The pooled cross-sectional Gini coefficients reflect the same fit pattern (wealth Gini ER 0.73 / IR 0.73 vs data 0.78, income Gini ER 0.39 / IR 0.39 vs data 0.42), with both economies undershooting by a few hundredths through the retirement top-tail margin and no IR-vs-ER gap.¹⁹

The level fit extends beyond the internal-calibration target set. Panel (b) of Figure 4 reports the median wealth-to-income ratio at $t - 2$ conditional on H2M status at t (a cross-section that is not in the calibration objective), and both economies reproduce its qualitative structure: stayer households fan out roughly linearly from $W/Y \approx 0.5$ at age 30 to $W/Y \approx 3$ upon retirement, while faller households (those who fall into H2M two waves later) sit near zero throughout, with the stayer–faller wedge widening with age in both data and model. The cross-sectional consumption Gini, also outside the calibration objective, lands close to its empirical counterpart in both economies (IR 0.32 and ER 0.34 vs data 0.32), the consumption-side concentration reproduced without explicit targeting. Both economies match the data not only on the labeled targets but on the broader level

¹⁹The aggregate accidental-bequest pool is not a separate calibration target; it nonetheless emerges symmetrically across the two economies, leaving the dynamic comparison of Section 6 free of cross-regime bequest-flow asymmetries.

structure those targets imply, both in the cross-section and over the life-cycle dimensions, and any difference in dynamics across regimes is therefore attributable to the proposed expenditure-risk mechanism rather than to differential level fit.

5.5 Empirical validation: the calibrated ER economy reproduces the observed expenditure-risk passthroughs

Re-estimating the Section 4.2 passthroughs on the ER economy’s simulated panel recovers an expenditure shock whose within-household imprint matches the data’s persistent expenditure risk: it raises consumption growth and hand-to-mouth entry depleting wealth, reproducing the sign on all four outcomes and the wealth-change magnitude almost exactly. Running the identical identification on the income-risk economy, which carries no genuine expenditure risk, does not reproduce that pattern: its extracted shock reverses sign on wealth changes and on hand-to-mouth escape once household fixed effects absorb latent fixed-type variation. The contrast shows the recovered passthrough is a property of the marginal-utility-shock channel, not an artifact of the identification strategy.

I simulate each calibrated economy drawing $B = 50,000$ households at age 25 from the stationary distribution of fixed types (τ_i, β_i) and propagating them through $J^W = 20$ biennial periods of working life. At each age, the persistent income shock η , transitory shock ε , and marginal-utility shock φ are drawn from their respective calibrated conditional distributions, and the household policy functions of the GE solution determine consumption and asset choices. The resulting household-period panel mirrors the PSID panel in age range and biennial spacing, and I feed it into the same empirical pipeline run on PSID in Sections 2.4–4.2: the BPP income-shock identification of Section 2.4 recovers the simulated permanent and transitory income shocks (y^P, y^T) , and the random-forest expenditure function of Section 3.1, followed by the quasi-difference projection of its residual on a three-wave moving-sum instrument, recovers the simulated persistent expenditure shock $\hat{u}^{\text{exp}}$, with the residual persistence re-estimated on each simulated panel. The extraction runs in its single-shock form deliberately: the state-space split of Section 3.3 exists to strip from the persistent component a transitory term that is present in the data but absent by construction in the model, whose only expenditure-risk primitive is a *persistent* marginal-utility shock. The joint passthrough regressions of Sections 4.1–4.2 are then re-estimated on this simulated regressor block under both OLS and household fixed effects.

Panel A of Table 7 reports the H2M-flow passthroughs, in units of one standard deviation of each shock. On the fall margin, the ER expenditure-shock passthrough survives household FE with a sign-consistent, statistically significant estimate ($\hat{u}^{\text{exp}}$: OLS 0.027, FE 0.015), consistent with the positive within-household empirical estimate of Section 4.1 (+0.029 per SD under FE); the IR analogue collapses by an order of magnitude, from 0.018 under OLS to 0.001 under FE, so the fall margin is consistent with the ER economy but

Table 7: Simulated-panel passthroughs: H2M flows (Panel A) and consumption and wealth (Panel B).

	IR		ER	
	OLS	FE	OLS	FE
Panel A: H2M flow transitions				
	H2M Fall			
y^P	−0.026*** (0.0003)	−0.012*** (0.0003)	−0.016*** (0.0003)	−0.009*** (0.0003)
y^T	−0.045*** (0.0004)	−0.036*** (0.0004)	−0.039*** (0.0004)	−0.033*** (0.0004)
u^{exp}	0.018*** (0.0003)	0.001*** (0.0003)	0.027*** (0.0003)	0.015*** (0.0003)
R^2	0.0504	0.0469	0.0420	0.0355
HHs	49,592		49,865	
Obs.	564,615		584,999	
	H2M Escape			
y^P	0.183*** (0.0013)	0.110*** (0.0015)	0.196*** (0.0014)	0.170*** (0.0016)
y^T	0.278*** (0.0011)	0.242*** (0.0011)	0.273*** (0.0012)	0.257*** (0.0013)
u^{exp}	−0.006*** (0.0015)	0.059*** (0.0021)	−0.163*** (0.0022)	−0.164*** (0.0029)
R^2	0.3109	0.3424	0.2643	0.2864
HHs	22,554		23,835	
Obs.	135,385		115,001	
Panel B: Consumption growth and wealth changes				
	$\Delta \ln c$			
y^P	0.113*** (0.0003)	0.104*** (0.0003)	0.115*** (0.0004)	0.108*** (0.0005)
y^T	0.109*** (0.0004)	0.106*** (0.0004)	0.111*** (0.0005)	0.109*** (0.0005)
u^{exp}	0.012*** (0.0002)	0.025*** (0.0002)	0.026*** (0.0003)	0.042*** (0.0003)
R^2	0.3179	0.3141	0.1318	0.1299
	$\Delta W/\bar{y}$			
y^P	0.161*** (0.0004)	0.138*** (0.0003)	0.173*** (0.0005)	0.165*** (0.0004)
y^T	0.258*** (0.0004)	0.246*** (0.0004)	0.265*** (0.0005)	0.261*** (0.0005)
u^{exp}	−0.013*** (0.0006)	0.054*** (0.0006)	−0.141*** (0.0007)	−0.118*** (0.0006)
R^2	0.3730	0.5249	0.3492	0.3875
HHs	50,000			
Obs.	700,000			

Notes: Passthrough coefficients in units of one standard deviation of each shock, from joint regressions on the BPP-identified permanent (y^P) and transitory (y^T) income shocks and the recovered persistent expenditure shock (u^{exp}), re-estimated on the calibrated IR and ER simulated panels under OLS and household fixed effects. Panel A reports passthroughs to the H2M-flow indicators (Fall and Escape); Panel B reports passthroughs to consumption growth ($\Delta \ln c$) and biennial wealth changes ($\Delta W/\bar{y}$). Standard errors in parentheses, clustered at the household level; *, **, *** denote significance at the 10%, 5%, 1% levels.

separates the two economies less sharply than the outcomes that follow. On the escape margin, ER's passthrough is significantly negative under both OLS and FE (-0.163 and -0.164), matching the within-household sign of the empirical estimate (-0.090 per SD) while overshooting its magnitude; the IR analogue is near zero under OLS (-0.006) and turns significantly positive under FE ($+0.059$), the wrong-signed within-household estimate the placebo contrast is designed to expose.

Panel B reports the analogous passthroughs to consumption growth ($\Delta \ln c$) and bien-nial wealth changes ($\Delta W/\bar{y}$). On $\Delta \ln c$ both economies deliver a positive passthrough that strengthens under FE (IR: $0.012 \rightarrow 0.025$; ER: $0.026 \rightarrow 0.042$), so consumption growth does not discriminate between them; this is expected, because $\hat{u}^{\text{exp}}$ is extracted from consumption-side variation in the first place and the outcome partly contains the regressor, in the model as in the data. On $\Delta W/\bar{y}$ the IR-vs-ER contrast resurfaces, and it is the sharpest of the four outcomes: ER's passthrough is significantly negative under both OLS and FE (-0.141 and -0.118), with the FE estimate landing almost exactly on the empirical -0.124 per SD of Section 4.2, while IR's passthrough flips from -0.013 under OLS to $+0.054$ under FE, a second OLS-to-FE sign reversal mirroring the H2M-escape failure. Income-shock passthroughs survive the OLS-to-FE transition with stable signs in both economies on all four outcomes, leaving the expenditure-shock channel as the only diagnostic that discriminates IR from ER on the wealth-side outcomes.

The IR-vs-ER divergence on the wealth-side outcomes reflects what $\hat{u}^{\text{exp}}$ identifies in each simulated panel. In ER, the extraction recovers an object that tracks the economy's true marginal-utility process φ , which has direct within-household effects on saving, so the recovered passthrough is coherent with the model's own primitives and its sign is preserved across OLS and FE on every outcome. In IR, no source of within-household consumption variation exists beyond income risk itself, but the pipeline still produces a non-trivial $\hat{u}^{\text{exp}}$ object: the random-forest residuals capture spending variation tied to latent household types and to income-shock realizations that the conditioning set does not absorb. Under household fixed effects the fixed-type part is removed, and the within-household variation that remains in $\hat{u}^{\text{exp}}$ comoves with the income-shock channel, so its coefficient on the wealth-side outcomes inherits the income shocks' sign rather than its own OLS pattern. The placebo analysis run on the IR-only economy validates the ER economy: absent any specific channel breaking the strict income-consumption link of the incomplete-markets framework, any recovered residual-consumption measure is a spurious object capturing either cross-sectional fixed-type latent variation or within-household income-side variation.

6 The ER footprint on wealth and consumption dynamics

With both economies pinned to the same level benchmark and the calibrated ER economy validated on the empirical passthroughs of Section 5.5, expenditure risk reshapes the calibrated HA-OLG economy through a single force: a covariance wedge in the household's Euler equation between the marginal-utility shock and consumption growth, its magnitude pinned by the expenditure risk identified in Section 3 and absent in the income-risk-only economy by construction. Through this wedge the ER economy reproduces the consumption-growth risk dynamics income risk alone cannot generate, matching the variance, dispersion, and wealth-profile spread of $\Delta \ln c$, including the high volatility of wealth-ier, unconstrained households the income-risk economy leaves far too smooth.

The same wedge moves the wealth-transition matrices toward the data that an economy with income risk and discount-factor heterogeneity alone misses, closing most of the hand-to-mouth escape-rate gap and a substantial part of the fall-rate gap, with the mobility lift concentrated in the bottom half of the wealth distribution. Holding the calibrated β distribution fixed and dialing σ_φ to zero retraces the consumption-growth results and the escape-rate closure dose-responsively, with the $\sigma_\varphi \rightarrow 0$ corner landing below the income-risk economy on the escape rate, so those dynamics are σ_φ -driven and therefore wedge-driven; the raw fall rate moves less across the dial, but only because the corner economy carries a larger hand-to-mouth stock, and per unit of that stock the wedge carries the fall closure as well.

6.1 The covariance wedge

Expenditure risk alters the income-risk economy through the covariance wedge in the household's Euler equation between the marginal-utility shock and consumption growth, derived in closed form from the risky-good intertemporal optimality condition. Because a positive marginal-utility realization tilts next period's basket toward the risky good lowering its marginal-utility growth, the wedge is signed positive: households face higher expected marginal-utility growth than the deterministic benchmark and save more against it. The wedge is trivially absent in the income-risk-only economy ($\sigma_\varphi = \rho_\varphi = 0$) by construction, and it alone drives the consumption-growth risk and wealth-transition dynamics the next two subsections document.

Starting from the household's Euler equation in the risky-good consumption c_2 at interior optima (Appendix F.2, equation (F.1)), the covariance identity isolates a structural wedge:

$$\mathbb{E}_j \left[\left(\frac{c_{2,j+1}}{c_{2,j}} \right)^{-\gamma} \right] - \frac{\exp(\varphi_j)}{R\beta \mathfrak{s}_j \cdot \mathbb{E}_j[\exp(\varphi_{j+1})]} = - \frac{\text{Cov}_j(\exp(\varphi_{j+1}), (c_{2,j+1}/c_{2,j})^{-\gamma})}{\mathbb{E}_j[\exp(\varphi_{j+1})]} \geq 0, \quad (10)$$

with $R \equiv 1 + r$ the gross interest rate and $\mathfrak{s}_j$ the age- j conditional survival probability of

the demographic block of Section 5.2; the derivation details are in Appendix F.2.

The right-hand side of equation (10) is the covariance wedge, and its sign is unambiguous. The intratemporal first-order condition of equation (7) pins the within-period allocation at $c_2/c_1 = (\exp(\varphi)/\varphi_0)^{1/\gamma}$, so a positive realization of φ_{j+1} tilts the household's basket toward the risky good at $t + 1$, raising $c_{2,j+1}$ for any given total expenditure and lowering the marginal-utility (MU) growth ratio $(c_{2,j+1}/c_{2,j})^{-\gamma}$. Conditional on φ_j , the random variables $\exp(\varphi_{j+1})$ and $(c_{2,j+1}/c_{2,j})^{-\gamma}$ therefore comove negatively, so $\text{Cov}_j(\cdot, \cdot) < 0$ and the wedge is positive: expected MU growth in c_2 exceeds its deterministic- φ benchmark, and households save more to internalize the ex-ante exposure to φ -realizations that will reshape their intratemporal allocation. In the IR economy $\sigma_\varphi = \rho_\varphi = 0$, the conditional distribution of φ_{j+1} collapses to a point mass at zero; the covariance vanishes (the wedge in equation (10) attains the lower bound with equality) and the equation reduces to the canonical CRRA Euler $\mathbb{E}_j[(c_{2,j+1}/c_{2,j})^{-\gamma}] = 1/(R\beta \mathfrak{s}_j)$.

6.2 Expenditure risk spreads consumption-growth volatility across the wealth distribution

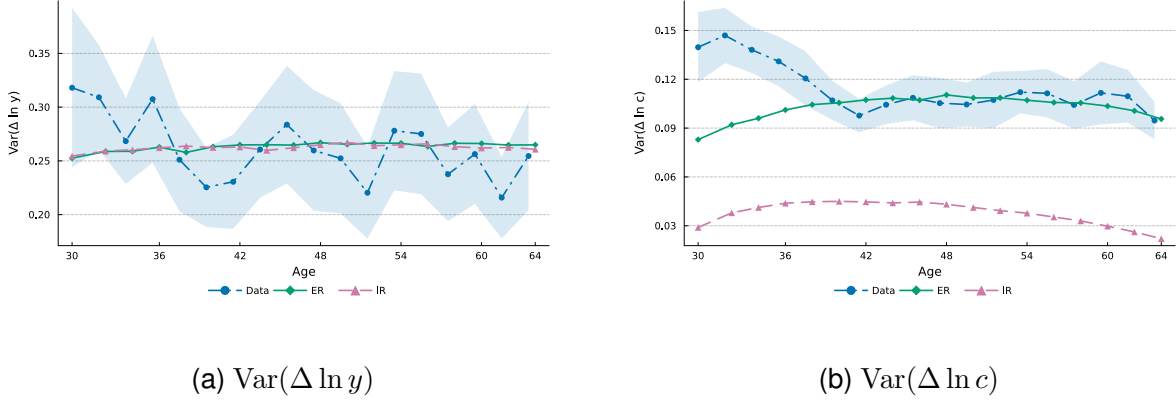
Through the covariance wedge, the ER economy reproduces the consumption-growth volatility dynamics that income risk alone cannot generate, matching the variance of $\Delta \ln c$, its lifecycle profile, and its full cross-sectional dispersion (the $P_{90} - P_{10}$ sits within the PSID band). The decisive cut is who bears that risk: consumption-growth volatility stays high across the whole wealth distribution and among unconstrained, non-hand-to-mouth households, which ER matches while the income-risk economy, generating volatility primarily where the borrowing constraint binds, collapses toward zero off the constraint.

The covariance wedge generates consumption-growth volatility that income risk alone cannot, and the gap is wide. Pooled over the working life, ER closes 78% of the IR-Data gap²⁰ on the aggregate $\text{Var}(\Delta \ln c)$ moment (IR 0.038, ER 0.102, PSID 0.121) and 85% on the pooled $\text{autocorr}(\ln c)$ (IR 0.945, ER 0.866, PSID 0.852), and reproduces the cross-sectional $\text{Var}(\ln c)$ (IR 0.345, ER 0.385, PSID 0.369). The lifecycle profiles in Figure 5 show two economies sharing the same idiosyncratic income-risk, both sitting by construction inside or at the edge of the PSID confidence band at every age (panel a), yet exhibiting severely different rates of consumption-growth volatility (panel b): the ER-economy profile, *unmatched*, traces the data from middle age onward, while the IR profile runs at roughly a third of the empirical level. The two economies share the income process, demographics, and technology, and differ in the household block only through the marginal-utility shock and the recalibrated discount-factor mixture that accompanies it; Section 6.5 separates the

²⁰Gap-closure percentages here and in Section 6.3 are computed from the unrounded underlying moments; recomputing them from the rounded values displayed in the text can differ by one to two percentage points.

two, showing the consumption-growth gap scales with σ_φ alone at a fixed β distribution.

Figure 5: Age profile of income-growth and consumption-growth variances: PSID, IR, and ER.



Notes: Biennial cross-sectional variances by age, with the PSID 95% confidence band shaded. Panel (a): income-growth variance. The IR and ER profiles overlap by construction (both economies share the income process of Section 5.1) and both sit inside or at the edge of the PSID band across the working life, the same-input control. Panel (b): consumption-growth variance. The ER profile sits inside the PSID band from age 38 onward, while the IR profile runs at roughly a third of the empirical level at every age (pooled ratio ≈ 0.31).

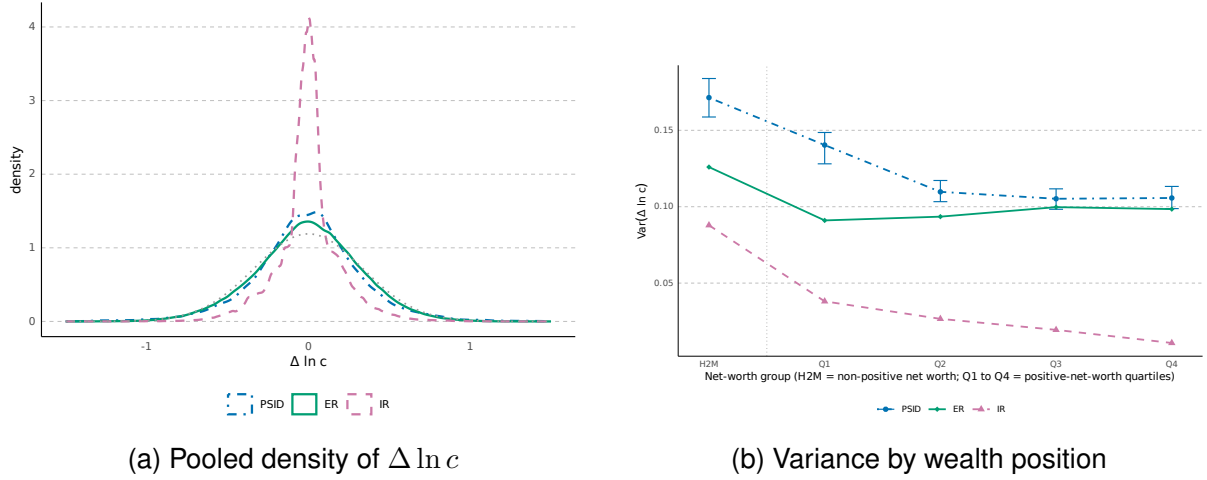
The *uncalibrated* match runs deeper than the variance: ER reproduces the full cross-sectional dispersion of consumption growth. The robust $P_{90} - P_{10}$ spread of $\Delta \ln c$ is 0.823 in PSID (95% bootstrap interval $[0.807, 0.837]$), 0.807 in ER, at the interval's lower edge, and only 0.443 in IR, so the ER density overlays PSID across the entire central region while the IR density is a tall, narrow spike (Figure 6, panel a).

The decisive evidence stands in who bears the consumption-growth risk. In the data, $\text{Var}(\Delta \ln c)$ stays high across the whole wealth distribution: hand-to-mouth households carry the most (0.171), but even the top positive-net-worth quartile retains roughly 0.10. The ER economy reproduces this flat-but-elevated profile, while IR collapses it (Figure 6, panel b): across the positive-net-worth quartiles ER stays flat near 0.10, tracking the PSID level over the upper quartiles, while IR falls from 0.038 to 0.011, and among non-hand-to-mouth households ER carries 0.096 against PSID's 0.115 but IR only 0.024. The reading is that income risk delivers consumption-growth volatility primarily where the borrowing constraint still bites, so an income-risk-only economy leaves its wealthy and unconstrained households with far too smooth consumption dynamics, whereas the expenditure shock reaches every household regardless of its balance sheet.

6.3 Expenditure risk closes the wealth-transition gap

The same wedge that spreads consumption-growth risk also closes the wealth-transition gap an economy with income risk and discount-factor heterogeneity alone misses, with the

Figure 6: Consumption-growth dispersion and exposure across the wealth distribution: PSID, IR, and ER.



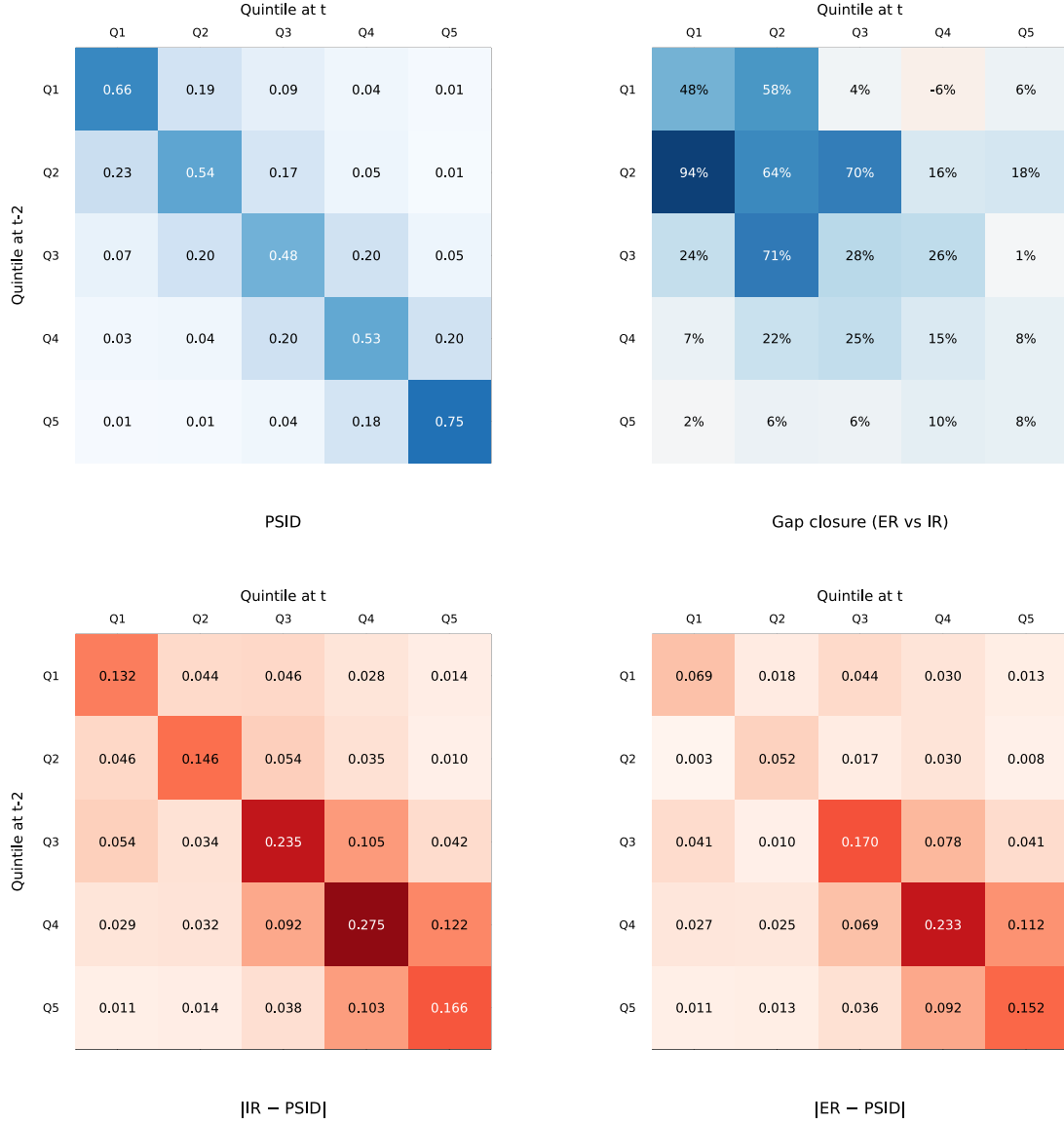
Notes: Panel (a): pooled density of (residualized) consumption growth $\Delta \ln c$ over the working life; ER overlays the PSID density across the central region while IR is a narrow spike. Panel (b): $\text{Var}(\Delta \ln c)$ by wealth position, with hand-to-mouth households (non-positive net worth, $a \leq 0$) at the far left and positive-net-worth households conditionally split into quartiles. The raw and log-scale density variants and the per-quintile cuts are in Appendix F.6.

H2M margin offering the cleanest cut: the ER economy closes 85% of the IR-Data gap on the biennial escape rate out of low-net-worth status and 39% on the fall rate. The improvement is not confined to the H2M margin: the full 5×5 wealth-quintile transition matrix and the age-conditional rank-rank wealth autocorrelation move the same way, ER sitting closer to PSID than IR at every row and every age band. The cell-by-cell closure is sharpest at the bottom of the distribution (48% at the lowest quintile, 64% at the second-lowest), tapers in the middle, and is smallest at the top (8% closure), where the top-tail persistence puzzle of Section 5.4 carries through unchanged. The mobility carries a consumption-side corollary explored in Section 6.4: the same churn makes a high marginal propensity to consume transient and loosens its tie to wealth.

Table 8 reports the biennial 2×2 hand-to-mouth fall and escape rates for PSID and the two calibrated economies. Both economies are calibrated to the same stationary H2M-share targets in Section 5.4, so this comparison is a stress test of *flows* across economies with closely matched H2M stocks: the escape-rate closure is mechanism-driven, not calibration-driven, and the fall-rate closure largely so once stocks are held comparable.²¹ The PSID escape rate is 0.389, the IR economy generates 0.287, and ER lifts

²¹The $\sigma_\varphi \rightarrow 0$ corner of the Section 6.5 grid, recomputed in this table's transition convention, makes the attribution exact (Appendix Table F.3). With the ER β distribution held fixed and the ER wedge turned off, the escape rate drops to 0.256, below IR's 0.287: the wedge step (+11.8 pp) exceeds the full IR-to-ER escape difference (+8.7 pp), and the β -composition step works against it (−3.1 pp). On the raw fall rate the wedge step is +0.4 pp of the +1.6 pp difference, but the corner is not stock-comparable, its H2M share rising to 0.259 against 0.181 in ER and 0.210 in IR as the wedge's precautionary saving is removed; per unit of the H2M stock, the wedge carries 96% of the IR-to-ER fall difference.

Figure 7: Biennial 5×5 wealth-quintile transitions: PSID anchor, ER-vs-IR gap closure, and residual model errors.



Notes: All four panels are 5×5 heatmaps over wealth quintiles, with rows indexing the quintile at $t - 2$ and columns the quintile at t ; each row of every transition-probability panel sums to one. *Top-left:* PSID biennial transition probabilities, the empirical anchor. *Top-right:* cell-by-cell IR-Data gap closure under ER, $(P_{ij}^{IR} - P_{ij}^{ER}) / (P_{ij}^{IR} - P_{ij}^{PSID})$, expressed in percent; blue indicates ER moves the cell toward PSID, red away. *Bottom-left* and *bottom-right:* absolute residual errors $|P_{ij}^{IR} - P_{ij}^{PSID}|$ and $|P_{ij}^{ER} - P_{ij}^{PSID}|$. The IR and ER raw transition matrices are reported alongside PSID in Appendix Figure F.1.

Table 8: Biennial hand-to-mouth (H2M) fall and escape rates: PSID, IR, and ER.

	Fall rate (Not H2M $\rightarrow$ H2M)	Escape rate (H2M $\rightarrow$ Not H2M)
IR (no φ -process)	0.045	0.287
ER	0.060	0.374
PSID	0.085	0.389

Notes: Biennial transition probabilities across the H2M margin. *Fall rate:* probability of transitioning from non-H2M at $t - 2$ to H2M at t . *Escape rate:* probability of transitioning from H2M at $t - 2$ to non-H2M at t . H2M defined as non-positive net worth, as in Section 2.2; rates are age-averaged over the working life (ages 26–64), with each transition dated to the wave at which the destination status is observed, matching the PSID-rates construction. IR refers to the calibrated income-risk economy with discount-factor heterogeneity but no φ -process; ER adds the persistent marginal-utility shock. Implied stay probabilities (1 minus each off-diagonal) and the full 2×2 matrices are in Appendix F.1.

it to 0.374, closing 85% of the IR-Data gap. The fall rate is 0.085 in PSID, an understated 0.045 in IR, and 0.060 in ER, closing 39% of the gap. The flow closure carries over to durations: the mean H2M spell is 5.73 years in PSID (corrected for the censoring of spells at the panel edges), 7.85 years in IR, and 5.64 years in ER, so the ER economy reproduces the data’s spell length within about 2% without targeting it. Escape is, moreover, frequently not permanent: the share of escaped households that have re-entered H2M rises steadily with the horizon since escape, and the ER economy tracks the PSID recurrence curve more closely than the income-risk baseline, though both economies generate less re-entry than the data (Appendix Figure F.2).

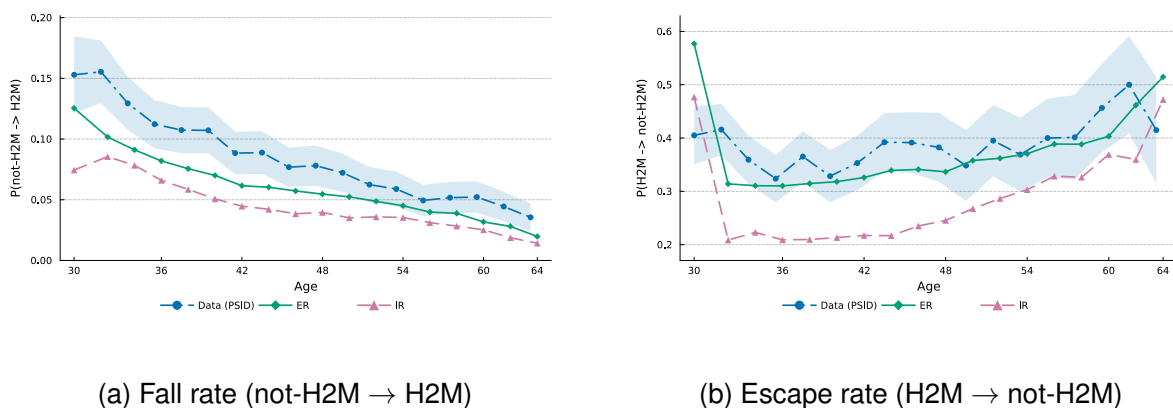
This same direction extends across the full wealth distribution. The PSID 5×5 wealth-quintile transition matrix (Figure 7, top-left) shows substantial cross-quintile churn: diagonal persistence ranges from 0.48 at Q_3 to 0.75 at Q_5 , with adjacent-quintile transitions in the 17–23% range. Both calibrated economies undershoot this mobility, with IR diagonals uniformly above PSID and ER diagonals in between. The cell-by-cell IR-Data gap closure under ER (top-right) is sharply tilted across the wealth distribution: 48% closure at $Q_1 \rightarrow Q_1$ and 64% at $Q_2 \rightarrow Q_2$ in the lower half, 28% at $Q_3 \rightarrow Q_3$ and 15% at $Q_4 \rightarrow Q_4$ in the middle, and little closure at $Q_5 \rightarrow Q_5$ (8%). The same pattern repeats in the off-diagonals: the bottom adjacent cells close most of the way ($Q_2 \rightarrow Q_1$ at 94%, $Q_1 \rightarrow Q_2$ at 58%) and $Q_2 \rightarrow Q_3$ closes 70%, in the rows where households are most likely to move in either direction across a two-year window, while the top-row off-diagonals reach at most 10% closure.

The bottom-row panels of Figure 7 report absolute model errors relative to PSID: ER’s errors are lighter than IR’s in all but two cells (one tie, one small exception noted above), but the heaviest remaining errors sit at $Q_3 \rightarrow Q_3$ (0.170 under ER vs 0.235 under IR) and $Q_4 \rightarrow Q_4$ (0.233 vs 0.275). The age-conditional rank-rank wealth autocorrelation tells a parallel story (Appendix Table F.2): ER closes 20% of the IR-Data gap in the young age tercile, 15% in the middle, and 10% in the old. Across both cross-sections, ER’s closures

concentrate where the empirical mobility is highest, at the bottom of the wealth distribution and the youngest ages, and fade where it is lowest. At the top of the distribution this fade is decisive: only 8% of the Q_5 persistence gap closes under ER, so the top-tail puzzle of Section 5.4's wealth distribution carries through unchanged into the dynamic moments. Structural routes for addressing it sit outside the scope of the present paper; I return to the residual in Section 7.

The H2M result of Table 8 also extends to the lifecycle. Figure 8 reports the biennial fall and escape rates by age across PSID, IR, and ER: ER tracks the PSID age profile closely, inside or near the 95% confidence band over most of the working life on the escape rate and closing most of the age-by-age fall-rate gap, while IR sits below ER at nearly every age on both rates, the same lifecycle shape as PSID but shifted further down. The gap closure of Table 8 is not a pooling artifact, then, but a tracking of the empirical age profile that IR alone misses. This wealth-side closure and the consumption-growth footprint of Section 6.2 are two faces of the same covariance wedge, as Section 6.1 derives analytically; Section 6.4 reads the same wedge through the household's policy function, and Section 6.5 isolates it, dialing σ_φ at a fixed β distribution.

Figure 8: Biennial H2M fall and escape rates by age: PSID, IR, and ER.



Notes: Biennial transition probabilities by age, with the H2M state defined as non-positive net worth (Section 2.2). Shaded bands are the PSID 95% confidence interval. ER (green) tracks the PSID profile closely on both rates across the working life, sitting between IR and the data at nearly every age; IR (pink) runs below both over most of the age range, the same lifecycle shape shifted down. The pooled rates from the same data are reported in Table 8.

6.4 Expenditure risk loosens the tie between wealth and the marginal propensity to consume

The wealth mobility of Section 6.3 has an important consumption-side corollary: expenditure risk turns a household's marginal propensity to consume from a fixed trait into a transient state. Holding the calibrated β distribution and every other parameter at its baseline

ER value and scaling only the shock variance through $\sigma_{\varphi}^{\text{cf}} = (1 - \alpha) \sigma_{\varphi}^{\text{base}}$, the re-calibration-free dial that Section 6.5 develops fully, the mean MPC falls from 0.301 at the $\sigma_{\varphi} \rightarrow 0$ corner to 0.269 at the calibrated value, the prudence direction of the covariance wedge. At this level the model errs toward too little marginal propensity rather than too much: a biennial windfall response of 0.27–0.30 sits below what the quarterly non-durable rebate responses of roughly 0.1–0.4 (Johnson, Parker, and Souleles, 2006; Parker, Souleles, Johnson, and McClelland, 2013) would cumulate to over two years, the familiar undershoot of one-asset economies without wealthy hand-to-mouth households (Kaplan and Violante, 2014). The novel move is in persistence rather than level: the within-household autocorrelation of the MPC falls from 0.825 to 0.755, the mean high-MPC spell, a run of consecutive biennial waves with $\text{MPC} \geq 0.5$, shortens from 6.3 to 4.3 years, and the share of those spells lasting a single wave rises from 0.43 to 0.50, monotonically across the grid and more sharply at the lower $\text{MPC} \geq 0.3$ threshold (Appendix F.4). In a single-asset economy this is the hand-to-mouth churn of Table 8 read through the policy function, boundary crossings and movements through the MPC distribution being the same object, so the empirical anchor is the passthrough evidence of Section 4.1.

The transience has a cross-sectional counterpart: expenditure risk loosens the tie between current wealth and the marginal propensity to consume. A wealth specification, within-age quintiles plus a linear net-worth term, explains 0.594 of the variance in simulated MPCs with the wedge off but 0.487 at the calibrated value, and the loosening does not ride on the two economies’ separately calibrated discount-factor mixtures: recomputed under the IR mixture with the φ process switched on, the reverse path left open by the flow decomposition of Appendix F.3, the wedge lowers wealth’s explanatory power by the same 11 pp from either direction (Appendix F.5). The state does the untying at every wealth level: conditional on age, a household in the top tercile of the φ distribution carries the mean MPC of a household one wealth quintile poorer, so high propensities persist into the upper quintiles without illiquid balance sheets doing the work (Kaplan and Violante, 2014).

The dispersion also shares the corollary’s state-like character: the between-household share of age-adjusted MPC variance falls from 0.69 under income risk alone to 0.58 under expenditure risk, with movements in φ accounting for a third of the within-household remainder. Read against the evidence that observables price household MPCs poorly, at most 8% of their variation in Lewis, Melcangi, and Pilossoph (2026) (LMP) and little of it through measured wealth (Fuster, Kaplan, and Zafar, 2021; Boehm, Fize, and Jaravel, 2025), the wedge works in the same direction, cutting the explanatory power of an LMP-style covariate set (income terciles, the average propensity to consume, age) from about 7% to 4%, and it names part of the latent heterogeneity those authors infer: a persistent expenditure state rather than a fixed type. What would price the state is a change rather than a level: conditional on age, income, and current wealth, next-wave wealth recovers most of the φ signal while lagged wealth recovers almost none, the footprint in wealth data

of a shock that finances spending by drawing the household's buffer down; end-of-period wealth embeds the same consumption choice, making this a statement about where the transient state is visible, not a forecasting design.

6.5 Robustness: wedge-driven dynamics scale with σ_φ

The σ_φ dial already used for the MPC corollary of Section 6.4, holding the calibrated β distribution and every other parameter fixed and scaling only $\sigma_\varphi^{\text{cf}} = (1 - \alpha) \sigma_\varphi^{\text{base}}$ across $\alpha \in \{0, 0.25, 0.5, 0.75, 1\}$ ($\alpha = 1$ the $\sigma_\varphi \rightarrow 0$ corner that turns the wedge of equation (10) off), isolates that wedge as the force behind the consumption-growth results and the escape-side wealth transitions of Section 6.2–Section 6.3. Applied to the headline moments it produces a clean dose-response: $\text{Var}(\Delta \ln c)$ falls 56% and $\text{Var}(\Delta W/\bar{y})$ falls 32% across the grid. Because σ_φ is the single moving primitive, every response is attributable to the marginal-utility-shock channel alone, a re-calibration-free reading of how much each observable depends on σ_φ . The economy is solved in general equilibrium and simulated at each grid point; Appendix Figure F.3 reports the dose-response curves as percent changes from baseline.

Panel (a) of Appendix Figure F.3 traces the two variance declines across the grid, both monotonic. The percent-change asymmetry is a baseline effect: in absolute variance units the two outcomes decline in lockstep across the grid, both losing ≈ 0.06 variance units between $\alpha = 0$ and $\alpha = 1$ (Appendix Figure F.4, panel a), but the same absolute reduction is a larger share of the smaller baseline (consumption-growth variance 0.102 vs wealth-change variance 0.184). The wedge contributes the same absolute quantity of variance to both outcomes; what differs across the two is the income-risk-driven baseline that σ_φ does not move. Panel (b) reports the wedge's footprint on the H2M margin in the simulated panel's pooled event-share convention, the share of working-life household-waves at which a fall or an escape occurs: the fall-event share declines 17% across the grid and the escape-event share 9%, in absolute terms 0.7 and 0.8 percentage points (Appendix Figure F.4, panel b). The pooled event share blends the at-risk stock with the conditional transition rate, and the two move in opposite directions as the wedge turns off (the H2M share rises 8.0 pp while conditional escape falls), so it understates the wedge's effect on the transition rates themselves: recomputed in the destination-indexed, age-averaged convention of Table 8, the conditional escape rate falls from 0.374 at baseline to 0.256 at the $\sigma_\varphi \rightarrow 0$ corner, below the IR economy's 0.287, while the conditional fall rate barely moves (Appendix Table F.3). Turning off the wedge dampens consumption-growth volatility, lowers precautionary saving, and slows household churn across the low-net-worth boundary.

Four indirect outcomes not plotted move in the directions the wedge interpretation predicts: across the grid the panel-level H2M share rises 8.0 pp, the wealth Gini rises 4.0 pp, the Shorrocks mobility index falls by 0.051, and the general-equilibrium interest rate rises

1.8 pp as the wedge’s precautionary-saving contribution unwinds.²² Together with Appendix Figure F.3, the exercise supports the wedge interpretation of Section 6.2–Section 6.3: every ER-vs-IR divergence is retraced in direction inside the ER calibration by varying σ_φ alone, with no other parameter free to move, and the consumption-growth-variance response is close to the full ER-vs-IR gap in magnitude. On the transition rates the corner is sharp once stocks are held comparable: the conditional escape-rate response exceeds the full ER-vs-IR gap (a +11.8 pp wedge step against a +8.7 pp gap, the corner landing below IR), and while the raw conditional fall-rate response is only +0.4 pp of a +1.6 pp gap, the raw comparison is confounded by the corner’s larger hand-to-mouth stock (an age-averaged share of 0.259 against 0.181 in ER and 0.210 in IR), which mechanically raises falls; per unit of the H2M stock the wedge carries 96% of the fall difference alongside the full escape difference, so both flow closures of Section 6.3 are wedge-driven on stock-comparable terms (Appendix Table F.3).

7 Conclusion

This paper identifies idiosyncratic expenditure risk as a structural household primitive, large and persistent in U.S. household data and the only risk component that predicts both the fall into and the failure to escape low-net-worth status under household fixed effects. Embedded in the household’s Euler equation, ER enters through a covariance wedge between the marginal-utility shock and consumption growth; in a calibrated heterogeneous-agent overlapping-generations (HA-OLG) model this single wedge generates the household wealth-transition and consumption dynamics that income-risk-only baselines miss.

Two residuals locate where this account remains incomplete. The top-tail gap in wealth persistence of Section 5.4 carries largely unchanged into the dynamic moments, with only 8% of the Q_5 IR-Data gap closing under ER, and the cross-sectional consumption-growth variance of Section 6.2 undershoots PSID at the youngest working ages by 0.03–0.06 in variance units.

Two extensions are concrete. The top-tail residual calls for preferences richer than a single marginal-utility shock, such as the extreme-value taste shocks of [Briglia, Chatterjee, Corbae, Dempsey, and Ríos-Rull \(2026\)](#); and as a structural primitive rather than a reduced-form correction, expenditure risk carries into heterogeneous-agent settings with aggregate risk, where the covariance wedge would interact with business-cycle and monetary shocks. The transience of high marginal propensities to consume (Section 6.4) gives those settings a concrete stake: transfers targeted on current low-wealth status would reach a population that recycles faster under expenditure risk than an income-risk economy would imply.

²²The [Shorrocks \(1978\)](#) mobility index reduces a wealth-transition matrix to a single number: it is zero when every household stays in its starting wealth state, and rises as more households move between states.

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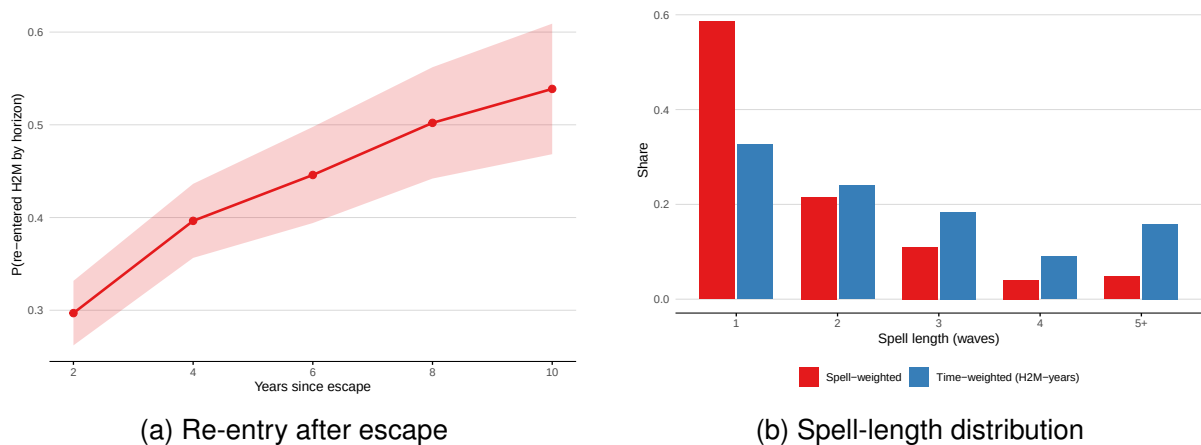
A Supplementary empirical evidence

Figure A.1 presents two cuts of the hand-to-mouth (H2M) margin behind the churn-around-a-persistent-core reading of Section 2.2. The recurrence panel follows households after they escape H2M. Re-entry is common and accumulates with time: 30% of observed escapes return to H2M within two years and 54% within a decade, with the eligible sample narrowing at longer horizons as it conditions on a full post-escape observation window. For a large share of households, escape from low net worth is less an exit than an interval between spells.

The spell-length panel distributes H2M spells under two weightings. Counting each spell once, the spell-weighted distribution describes a typical episode: a single biennial

wave is the modal and majority outcome. Counting each H2M household-year once, so that a spell of k waves enters with weight k , the time-weighted distribution describes how the time households spend at low net worth is spread across spell lengths, and it shifts the mass sharply to the right: spells of five waves or longer, only 5% of all episodes, hold 16% of all H2M household-years. The two weightings would coincide if every spell had the same length; their divergence is the quantitative content. The steep spell-weighted profile is the churn, the brief passages that dominate the count of episodes; the flat time-weighted profile is the persistent core, the long spells that absorb most of the cumulative time at low net worth. Because a spell still in progress at a household's last observed wave is measured at its truncated length, this concentration in long spells is a lower bound.

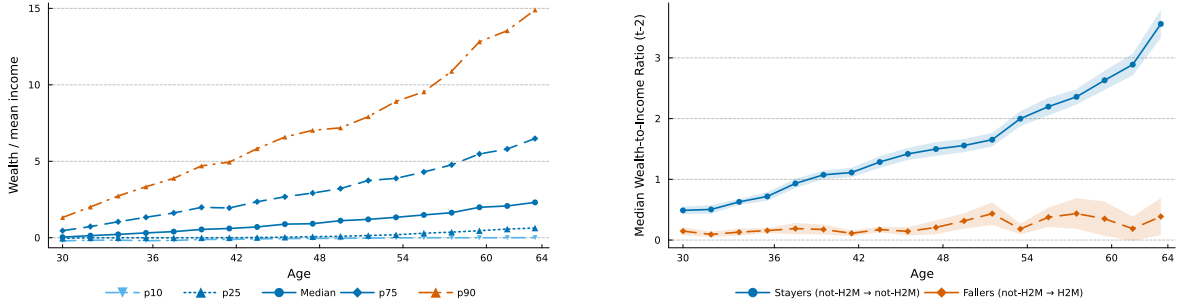
Figure A.1: Hand-to-mouth recurrence and spell length, PSID.



Notes: PSID 1999–2023, biennial waves; H2M status is non-positive net worth. Panel (a): among observed escapes from H2M, the share that have re-entered H2M by each horizon; the eligible sample conditions on observing the full post-escape horizon, so it shrinks as the horizon lengthens; the shaded band is a 95% confidence interval from a household block-bootstrap. Panel (b): the distribution of H2M spell lengths, a spell being a maximal run of contiguous H2M waves (a wave gap breaks a run); the spell-weighted series counts each spell once and the time-weighted series counts each H2M household-year once, so a spell of length k receives weight k . Spells are measured at their observed in-panel length, censored spells included. The time-weighted bar at 5+ exceeds the bar at 4 because 5+ is an open category averaging roughly six waves.

Figure A.2 reports the cross-sectional wealth profile by age behind the life-cycle reading of Section 2.3. Panel (a) plots the five wealth quantiles in mean-household-income units: the two lower quantiles stay near zero across the working life (p_{10} moves from -0.20 at age 30 to roughly zero at age 64 and p_{25} from -0.03 to 0.64), while the upper three climb steeply, the median from 0.04 to 2.31 , p_{75} from 0.46 to 6.49 , and p_{90} from 1.33 to nearly 15 . Panel (b) plots the median wealth-to-income ratio at $t - 2$ conditional on H2M status at t ; the stayer-to-faller gap widens with age and is partly mechanical (see the figure note).

Figure A.2: Cross-sectional wealth profile by age, PSID.



(a) Wealth quantiles by age

(b) W/Y by H2M status

Notes: PSID 1999–2023, by single-year age. Panel (a): cross-sectional household-wealth quantiles (p_{10} , p_{25} , median, p_{75} , p_{90}) in mean-household-income units; the five paths diverge linearly at roughly constant rate, with p_{10} and p_{25} staying near zero throughout the working life. Panel (b): median wealth-to-income ratio conditional on H2M status at each wave with bootstrap confidence intervals; the stayer-vs.-faller divergence widens with age and is partly mechanical, since W/Y is measured at $t - 2$ conditional on whether a fall occurs at t , drawing fallers by construction from the low-wealth tail.

B Income risk identification

This appendix details the identification of the income-risk components used throughout the paper. The body sections call out only the pieces they need: Section 2.4 uses the recovered permanent and transitory shock series in passthrough regressions, while Section 5 uses the variance-component estimates as calibration inputs.

Residualization. Log household earnings $\ln \tilde{y}_{i,t}$ are decomposed as $\ln \tilde{y}_{i,t} = X_{i,t}^\top \beta + y_{i,t}$, where the control vector $X_{i,t}$ comprises a cubic polynomial in the reference person's age, employment status of the reference person and spouse, education and race of the reference person, state, family size, number of children, and survey-year fixed effects. The residualized income process $y_{i,t}$ is the object on which the rest of the construction operates.

Income process. The residualized income process is

$$\begin{aligned} y_{i,t} &= \alpha_i + z_{i,t}^P + \varepsilon_{i,t}^T, \\ z_{i,t}^P &= z_{i,t-2}^P + \eta_{i,t}^P, \end{aligned} \tag{B.1}$$

with α_i a household-fixed effect, $z_{i,t}^P$ a random-walk persistent component driven by the biennial innovation $\eta_{i,t}^P$, and $\varepsilon_{i,t}^T$ an iid transitory shock. The three random components are mutually orthogonal both across households and over time.

Variance components. The variance of the fixed component, σ_α^2 , is identified from the cross-sectional variance of $y_{i,t}$ at the youngest ages observed in the panel, before the permanent component has had time to accumulate dispersion. The biennial-innovation variances $\sigma_{\eta^P}^2$ and $\sigma_{\varepsilon^T}^2$ are identified from the autocovariance structure of $\Delta y_{i,t}$ via standard [Blundell et al. \(2008\)](#) moment conditions:

$$\begin{aligned}\sigma_{\varepsilon^T}^2 &= -\text{Cov}(\Delta y_{i,t}, \Delta y_{i,t-2}), \\ \sigma_{\eta^P}^2 &= \text{Var}(\Delta y_{i,t}) - 2\sigma_{\varepsilon^T}^2.\end{aligned}\tag{B.2}$$

The point estimates are $\hat{\sigma}_\alpha^2 = 0.186$, $\hat{\sigma}_{\eta^P}^2 = 0.045$, and $\hat{\sigma}_{\varepsilon^T}^2 = 0.107$, broadly in line with the two-year-equivalent estimates of [Storesletten et al. \(2004\)](#); they feed the calibration of the heterogeneous-agent model in Section 5 (Table E.1).

Permanent and transitory shock series. Following [Kaplan and Violante \(2010\)](#), household-wave-level estimates of the permanent and transitory shocks ($\hat{\eta}_{i,t}^P$ and $\hat{\varepsilon}_{i,t}^T$, written as $y_{i,t}^P$ and $y_{i,t}^T$ in the body) are recovered as linear projections of $\Delta y_{i,t}$ onto two BPP instruments:

$$\begin{aligned}\hat{y}_{i,t}^P &= \hat{\mathbb{E}}[\Delta y_{i,t} \mid f_{i,t}^P], & f_{i,t}^P &= \Delta y_{i,t-2} + \Delta y_{i,t} + \Delta y_{i,t+2}, \\ \hat{y}_{i,t}^T &= \hat{\mathbb{E}}[\Delta y_{i,t} \mid f_{i,t}^T], & f_{i,t}^T &= \Delta y_{i,t+2}.\end{aligned}\tag{B.3}$$

Recovering a persistent-shock observation requires four consecutive waves of data; recovering a transitory-shock observation requires three, the contemporaneous growth $\Delta y_{i,t}$ plus its lead difference. The recovered shock series enter the passthrough regressions of Section 2.4 and the expenditure-risk identification of Section 3.3.

BPP-2008 covariance-ratio passthroughs. The IV-based shock recovery of equation (B.3) admits a direct robustness check via the original [Blundell et al. \(2008\)](#) covariance-moment identification of consumption passthroughs:

$$\begin{aligned}\phi_{\text{perm}} &\equiv \frac{\text{Cov}(\Delta c_{i,t}, f_{i,t}^P)}{\text{Cov}(\Delta y_{i,t}, f_{i,t}^P)}, \\ \psi_{\text{trans}} &\equiv \frac{\text{Cov}(\Delta c_{i,t}, \Delta y_{i,t+2})}{\text{Cov}(\Delta y_{i,t}, \Delta y_{i,t+2})},\end{aligned}\tag{B.4}$$

where $f_{i,t}^P$ is the three-difference instrument of equation (B.3). The control-conditioned moments in equation (B.4) are essential for the transitory passthrough: the raw covariance ratio computed on level-form Δy and Δc flips sign relative to the controlled estimate, reproducing the small-or-negative transitory estimate of older BPP treatments, while the residualized version brings the transitory passthrough into alignment with the modern partial-insurance ranges of [Commault \(2022\)](#) and [Crawley \(2020\)](#). The estimated covariance-ratio passthroughs and their bootstrap confidence intervals are reported in Appendix D.1

(Table D.1), where the corresponding expenditure-risk passthrough of Section 3.3 is also collected for a unified comparison.

C Expenditure-risk identification

This appendix collects supporting material for Section 3. Subsection C.1 derives the log-additive decomposition of equation (1) from the model's preferences and marginal-utility-shock process; Subsection C.2 details the grouped-fixed-effects construction of the type $\tau_{g(i)}$; Subsection C.3 reports auxiliary diagnostics from the random-forest expenditure-function fit; Subsection C.4 reports the AR(1)-plus-noise estimation of the persistent expenditure component of Section 3.2, its robustness, and a panel-VAR orthogonality test; and Subsection C.5 tests the advance-information alternative, showing the persistent component does not predict future income.

C.1 Derivation of the expenditure-function decomposition

This subsection derives the log-additive decomposition of equation (1) from the two-good CRRA preferences and AR(1) marginal-utility-shock process of the structural model in Section 5. The derivation is two exact steps followed by one first-order approximation; the only inexact step is the log-linearization of the consumption Euler equation in Step 3.

Step 1: Within-period homotheticity (exact). Flow utility on total non-durable expenditure $c_{i,t}$ with marginal-utility shock $\phi_{i,t}$ is

$$u(c, \phi) = \phi_0 v(c_1) + e^\phi v(c_2), \quad v(x) = \frac{x^{1-\gamma} - 1}{1 - \gamma}, \quad c_1 + c_2 = c, \quad (\text{C.1})$$

where (c_1, c_2) is the household's intratemporal allocation across the two consumption goods (non-risky and risky, respectively, in Section 5) and ϕ_0 is the relative-weight parameter on c_1 . The intratemporal first-order condition $\phi_0 c_1^{-\gamma} = e^\phi c_2^{-\gamma}$ pins down a share $\Omega(\phi)$ such that $c_1 = \Omega(\phi) c$ and $c_2 = (1 - \Omega(\phi)) c$, with Ω depending on ϕ alone, not on c , x , or any other state. Substituting the optimal allocation back into u gives the indirect-utility form

$$\tilde{u}(c, \phi) = c^{1-\gamma} H(\phi) + K(\phi), \quad (\text{C.2})$$

where $H(\phi)$ collects the ϕ -dependent multiplicative weights and $K(\phi)$ is a c -independent additive term. The c -dependence of $\tilde{u}$ is concentrated in the $c^{1-\gamma} H(\phi)$ term, separating the level of total spending from the marginal-utility shock through the homotheticity of the inner aggregator.

Step 2: Euler equation in total expenditure (exact). With state $s_{i,t} = (\tau_i, \beta_i, z_{i,t}^P, x_{i,t}, \phi_{i,t}, j)$, where z^P is the persistent income level whose innovation Appendix B recovers as y^P , the Bellman equation is

$$V_j(s) = \max_c \tilde{u}(c, \phi) + \beta \mathbb{E}[V_{j+1}(s') | s] \quad \text{s.t.} \quad x' = R(x - c) + y(\tau, z^P, y^T), \quad (\text{C.3})$$

and applying the envelope theorem on the inner allocation yields the interior Euler equation in total expenditure:

$$M(\phi_t) c_t^{-\gamma} = \beta R \cdot \mathbb{E}_t[M(\phi_{t+1}) c_{t+1}^{-\gamma}], \quad M(\phi) \equiv \phi_0 \Omega(\phi)^{-\gamma}. \quad (\text{C.4})$$

Step 3: Log-linearization (first-order). Under joint log-normality of $(M(\phi_{t+1}), c_{t+1}/c_t)$ conditional on date- t information, the realized growth differs from its conditional expectation by a martingale-difference innovation that decomposes into the canonical BPP-style channels plus a marginal-utility-shock channel:

$$\Delta \ln c_{i,t+1} - \mathbb{E}_t[\Delta \ln c_{i,t+1}] = \psi_P y_{i,t+1}^P + \psi_T y_{i,t+1}^T + \psi_\phi u_{i,t+1}^\phi + \Xi_{i,t+1}^E, \quad (\text{C.5})$$

where $u_{i,t+1}^\phi$ is the AR(1) innovation of the marginal-utility-shock process, Ξ^E is the higher-order remainder, and $(\psi_P, \psi_T, \psi_\phi)$ are the partial-derivative loadings at the linearization point. Log-normality (equivalently a second-order Taylor expansion around a deterministic path) is required for the log-Euler form; higher-order remainders flow into Ξ^E and ultimately into the residual ϵ of equation (1).

Step 4: Iterating to the level of log consumption. Iterating Step 3 forward from period $t - 1$ and projecting on the date- t observables of the household state gives

$$\ln c_{i,t} = G(\tau_i, y_{i,t}^P, y_{i,t}^T, x_{i,t}) + h(u_{i,t}^\phi) + \epsilon_{i,t}, \quad (\text{C.6})$$

where $G(\cdot)$ is the predictable component (the conditional expectation of $\ln c_{i,t}$ given the state observable to the econometrician), $h(\cdot)$ is the contribution of the unpredicted marginal-utility-shock innovation alone (a function of $u_{i,t}^\phi$ only, evaluating to $\frac{1}{\gamma}(\ln M(\phi_{i,t}) - \mathbb{E}_{t-1} \ln M(\phi_{i,t}))$ at the linearization point), and $\epsilon_{i,t}$ is the residual martingale-difference component absorbing the higher-order remainders. The empirical specification of equation (1) corresponds to this structural decomposition with three substitutions: the marginal-utility-shock innovation $u_{i,t}^\phi$ is replaced by its empirical analog $u_{i,t}^{\text{exp}}$, the innovation of the persistent expenditure component identified in Section 3.3 as the latent state of the state-space model (2), whose accumulation the state carries and whose smoothed estimate the Kalman recursion recov-

ers, the structural fixed type τ_i is identified by the GFE label $\tau_{g(i)}$ of Subsection C.2, and a vector of demographic and wave controls $X_{i,t}$ is added to absorb deterministic life-cycle variation that the structural model does not carry.

Step 5: Closed-form calibration mapping $(\rho_\varphi, \sigma_\varphi^2) \leftrightarrow (\text{Autocorr}, \text{Var})$ of $\ln(c_2/c_1)$. The intratemporal first-order condition of Step 1, written in logs, makes the observed log-ratio of risky to non-risky consumption a linear rescaling of the latent marginal-utility shock by $1/\gamma$:

$$\ln\left(\frac{c_{2,i,t}}{c_{1,i,t}}\right) = \frac{1}{\gamma} (\varphi_{i,t} - \ln \varphi_0). \quad (\text{C.7})$$

The AR(1) law for φ of equation (8) is therefore inherited by $\ln(c_2/c_1)$ one-for-one, with the curvature γ rescaling the innovation variance but not the autocorrelation (both numerator and denominator of the autocovariance scale by $1/\gamma$). Equating the AR(1)-implied second moments of $\ln(c_2/c_1)$ to their empirical counterparts then pins $(\rho_\varphi, \sigma_\varphi^2)$ analytically:

$$\rho_\varphi = \text{Autocorr}\left(\ln \frac{c_2}{c_1}\right), \quad \sigma_\varphi^2 = (1 - \rho_\varphi^2) \cdot \gamma^2 \cdot \text{Var}\left(\ln \frac{c_2}{c_1}\right), \quad (\text{C.8})$$

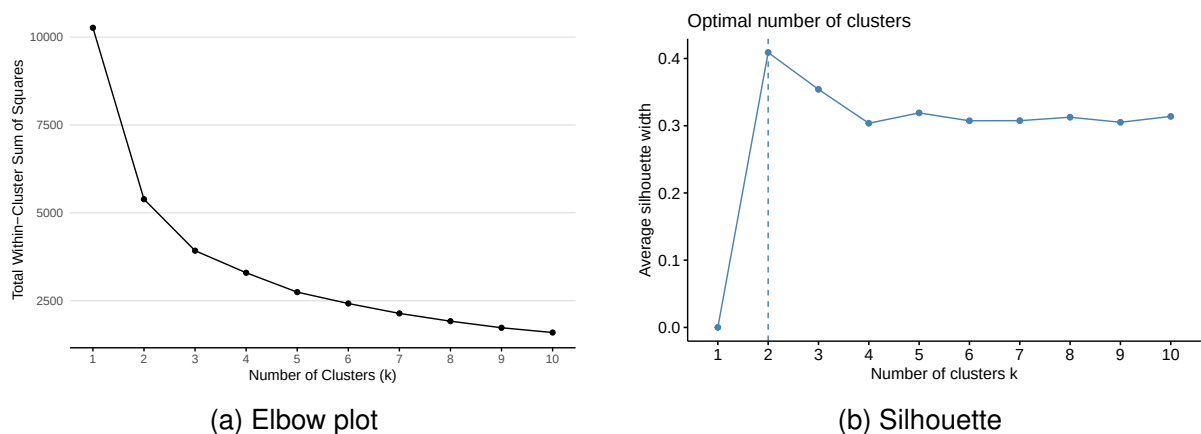
so that the AR(1) marginal-utility parameters $(\rho_\varphi, \sigma_\varphi^2)$ are calibrated outside the global optimization loop of Section 5.3, given γ and the two empirical moments computed on PSID. The same equation (C.7) is also the bridge between the latent structural $\varphi_{i,t}$ and its empirical counterpart recovered in Section 3.3: the innovation in $\ln(c_2/c_1)$ tracks, up to the constant $1/\gamma$ scaling, the same economic margin as the persistent expenditure innovation $u_{i,t}^{\text{exp}}$ that the Kalman smoother extracts; both are empirical counterparts of the AR(1) innovation $u_{i,t}^\varphi$ that drives $h(\cdot)$ in equation (C.6), the former measured on the allocation between the two goods and the latter on total expenditure, and because the two are measured on different objects with different noise exposures, their fitted processes need not coincide (see the footnote in Section 5.3).

C.2 Grouped fixed-effect type

The categorical type $\tau_{g(i)}$ entering the random-forest expenditure function G of equation (1) is recovered via two-step grouped-fixed-effects clustering in the spirit of Bonhomme et al. (2022). For each household, time-averaged residuals from first-stage regressions of log income and log consumption on the income-risk control vector are further residualized against household-level wealth and income-volatility features and standardized; K -means clustering is then applied to the resulting two-dimensional space of standardized income- and consumption-residual means, and the cluster label enters G as a categorical feature. The two-dimensional clustering captures persistent household heterogeneity in both income and spending levels.

The number of clusters is set to $K = 2$ as the silhouette-method optimum and the elbow-plot kink on this two-dimensional feature space (Figure C.1). A finer $K = 4$ specification, reported alongside the $K = 2$ baseline as a robustness check, identifies sub-clusters within the two main types but does not change the qualitative life-cycle or time profiles of log income and log consumption that the baseline picks up (Figure C.2).

Figure C.1: Choosing K : elbow and silhouette diagnostics on the two-dimensional residual-mean space.



Notes: K -means clustering on the two-dimensional space of standardized household-level income- and consumption-residual means. Panel (a): total within-cluster sum of squares as a function of K ; the elbow at $K = 2$ is the standard elbow-method signal that additional clusters explain rapidly diminishing within-cluster variation. Panel (b): average silhouette width as a function of K ; $K = 2$ maximizes silhouette in this two-dimensional space.

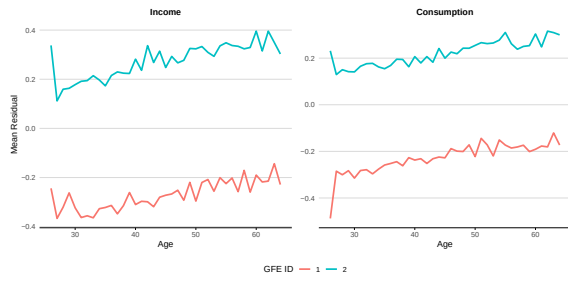
C.3 Auxiliary expenditure-function estimation results

Two diagnostics from the random-forest expenditure-function fit support the modelling choices of Section 3.1.

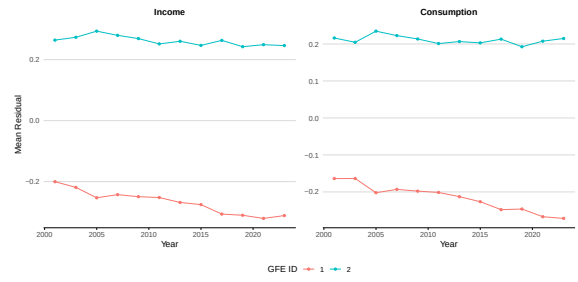
Figure C.3 plots the mean predicted log consumption against log family income with Q_5/Q_{95} bands from a quantile random-forest variant of the same specification. The profile is monotonically increasing and very close to linear over the entire log-income support, with stable quantile bands at both tails. The near-linearity indicates that the fitted prediction surface is smooth and monotone in income over the entire support, with no erratic behavior where the forest could overfit; because the profile plots fitted values rather than a partial-dependence object, it summarizes the prediction along the income dimension rather than a conditional marginal effect. The band stability indicates that the predictive distribution does not widen where data are thinner.

Figure C.4 reports the pointwise difference between the mean and the median prediction from the same fit, across the log-income distribution. The 95% confidence bands cover zero at every percentile; the point estimate drifts mildly from positive (low income)

Figure C.2: Cluster profiles, $K = 2$ vs $K = 4$, PSID.



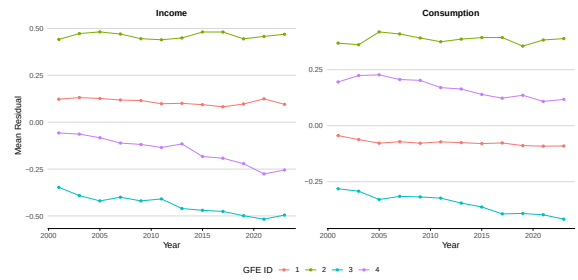
(a) $K = 2$, life-cycle profiles



(b) $K = 2$, time profiles



(c) $K = 4$, life-cycle profiles

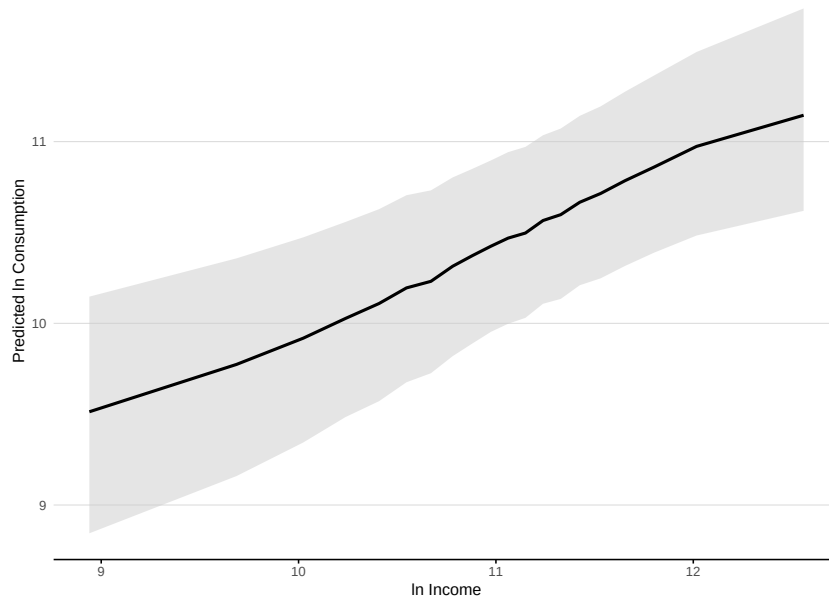


(d) $K = 4$, time profiles

Notes: Mean log-income and log-consumption profiles by cluster, with the $K = 2$ baseline (top row) and the $K = 4$ robustness check (bottom row). Left column: profiles by reference-person age (life-cycle). Right column: profiles by PSID wave (time). The $K = 4$ specification subdivides each of the two main $K = 2$ clusters but preserves the qualitative life-cycle and time patterns.

to negative (high income) but is not statistically distinguishable from zero. The absence of significant skewness in the conditional predicted distribution sanity-checks the use of the cross-fitted mean prediction $\hat{c}_{i,t}$ for downstream residual construction.

Figure C.3: Fitted consumption against log income, with quantile bands.



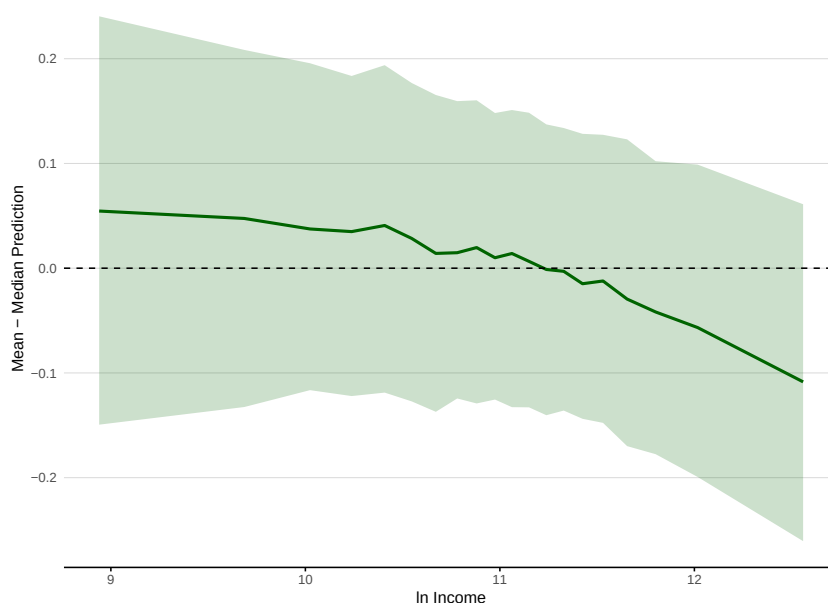
Notes: Mean predicted log non-durable consumption (solid line) against log family income, with Q_5/Q_{95} bands from the quantile random-forest variant of the expenditure-function specification. The profile is monotonically increasing and approximately linear over the entire log-income support; the quantile bands are stable across the income distribution.

C.4 Estimating and validating the persistent expenditure component

This subsection reports the estimation behind the persistence summary of Section 3.2 together with three validation exercises. Figure C.5 documents the near-Gaussian shape of the cross-fitted residual ξ . Table C.1 reports the AR(1)-plus-noise decomposition of ξ by two independent routes, autocovariance minimum distance and Kalman maximum likelihood, alongside the overidentification and permanent-component tests. Figure C.6 plots the autocovariance profile that identifies the persistence, and Table C.2 checks its robustness to the estimation choices. Table C.3 and Figure C.7 report a panel-VAR test of the orthogonality between the persistent expenditure shock and lagged income shocks, the operational form of the mean-independence assumption behind equation (1).

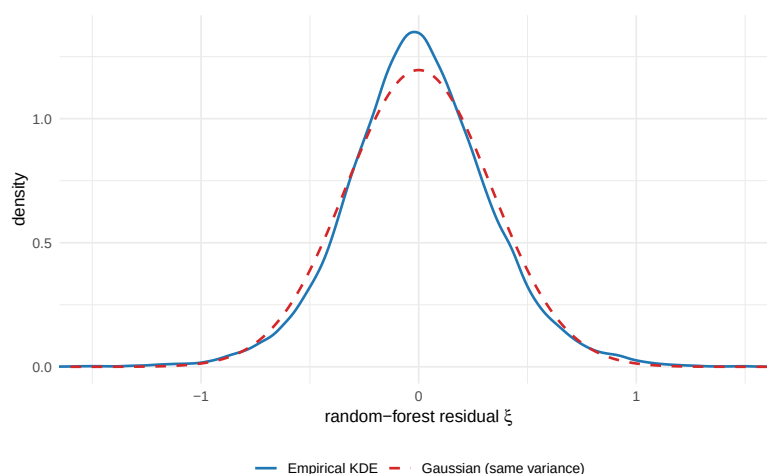
Distribution. Figure C.5 plots the pooled density of ξ against a Gaussian of the same variance. The two nearly coincide, with only mild excess kurtosis, supporting the linear-Gaussian state-space model adopted in Section 3.3.

Figure C.4: Mean vs. median predicted consumption.



Notes: Pointwise difference between the mean prediction and the median prediction of the random-forest expenditure-function fit, plotted across the log-income distribution with 95% confidence bands. The bands cover zero at every percentile; the point estimate drifts mildly from positive at low income to negative at high income but is not statistically distinguishable from zero.

Figure C.5: Empirical density of ξ versus a Gaussian of the same variance, PSID.



Notes: Pooled empirical density of the cross-fitted residual $\xi_{i,t}$ (solid) overlaid with the Gaussian density $\mathcal{N}(0, \gamma_0)$ of the same total variance, $\gamma_0 = \hat{\sigma}_z^2 + \hat{\sigma}_\varepsilon^2 = 0.111$ (dashed). The two nearly coincide: skewness is 0.04 and excess kurtosis is 1.48, so ξ is near-Gaussian with only mildly heavier tails. The near-normality supports the linear-Gaussian state-space model that Section 3.3 estimates by Kalman maximum likelihood.

AR(1)-plus-noise estimates: two routes. The persistence summary of Section 3.2 comes from the state-space model of equation (2), estimated two ways. The autocovariance route fits the model-implied moments $\gamma_0 = \sigma_z^2 + \sigma_\varepsilon^2$ and $\gamma_k = \rho_z^k \sigma_z^2$ for $k \geq 1$ to their empirical counterparts by diagonally weighted minimum distance, with household block-bootstrap standard errors. The Kalman route maximizes the Gaussian log-likelihood (3) over the same parameters. The two agree closely (Table C.1): persistence $\hat{\rho}_z = 0.86$ and 0.85, a persistent share of the residual variance near one half, and a half-life of roughly nine years. The AR(1)-plus-noise structure is not rejected by an overidentification test. A remaining time-invariant household component, one the expenditure-function step of Section 3.1 might have left in ξ , is addressed twice: freeing its variance in the levels fit returns an estimate whose interval includes zero but leaves the model weakly identified, so the sharper check re-estimates the process from first-differenced autocovariances, in which any fixed component drops out mechanically rather than being estimated. That route returns $\hat{\rho}_z = 0.77$ with a 95% interval of [0.52, 0.90] and a persistent variance share of 0.62, wide by construction but consistent with the levels estimates (third column of Table C.1).

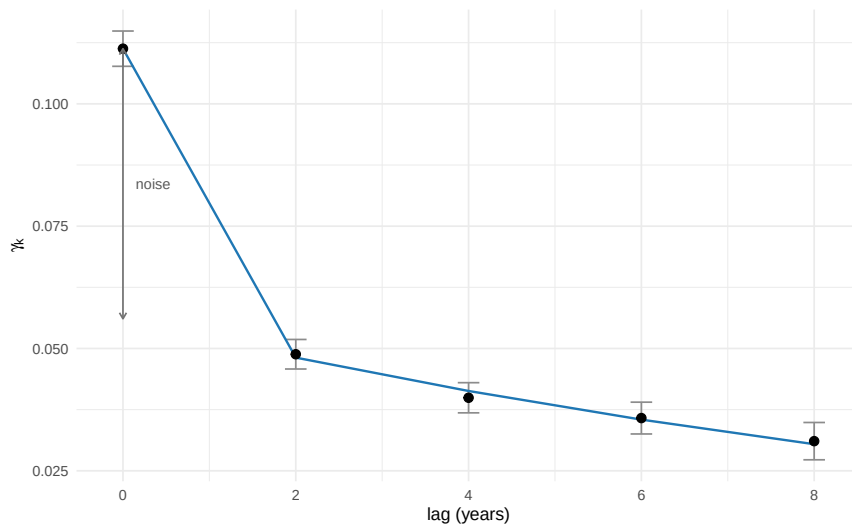
Table C.1: AR(1)-plus-noise estimates of the cross-fitted residual ξ : levels and first-difference routes.

Quantity	Autocovariance MD	Kalman ML	First-difference MD
Persistence $\hat{\rho}_z$	0.858 (0.013)	0.846 (0.011)	0.771 (0.109)
Persistent variance $\hat{\sigma}_z^2$	0.056 (0.002)	0.062	0.068
Transitory (noise) variance $\hat{\sigma}_\varepsilon^2$	0.055 (0.002)	0.051 (0.001)	0.042
Innovation variance $\hat{\sigma}_u^2$	0.015 (0.002)	0.017 (0.001)	0.027
Persistent share $\hat{\sigma}_z^2 / (\hat{\sigma}_z^2 + \hat{\sigma}_\varepsilon^2)$	0.50 (0.01)	0.55	0.62
Half-life (years)	9.1	8.3	5.3

Notes: Levels and first-difference routes to the AR(1)-plus-noise model $\xi_{i,t} = z_{i,t} + \varepsilon_{i,t}^{\text{exp}}$, $z_{i,t} = \rho_z z_{i,t-1} + u_{i,t}^{\text{exp}}$ of equation (2). The autocovariance route fits $\gamma_0 = \sigma_z^2 + \sigma_\varepsilon^2$ and $\gamma_k = \rho_z^k \sigma_z^2$ ($k \geq 1$) by diagonally weighted minimum distance; standard errors (in parentheses) come from 500 household block-bootstrap resamples. The Kalman route maximizes the Gaussian log-likelihood (3). The Kalman-route persistence standard error is the full-pipeline bootstrap value of Appendix D.5; the ML Hessian value (0.009) understates it. The 95% confidence interval for $\hat{\rho}_z$ (MD) is [0.830, 0.883] and for the half-life [7.4, 11.1] years. The naive lag-1 autocorrelation $\gamma_1/\gamma_0 = 0.44$ (0.01) is the noise-attenuated benchmark: the i.i.d. term inflates γ_0 but no γ_k for $k \geq 1$, so $\gamma_1/\gamma_0 = \rho_z \cdot (\sigma_z^2/\gamma_0)$. The AR(1)-plus-noise fit is not rejected (overidentification $J = 1.16$, 2 degrees of freedom, $p = 0.56$). Allowing an additional permanent (household-fixed) component returns $\hat{\sigma}_\alpha^2 = 0.025$, 95% CI [-0.020, 0.032]; the interval includes zero but also values nontrivial relative to γ_0 , and the freed specification is weakly identified ($\hat{\rho}_z = 0.65$, 95% CI [0.44, 0.90]). The first-difference route is therefore the informative fixed-component check: it fits the same model to the autocovariances of $\Delta\xi$ over adjacent waves, $g_0 = 2\sigma_z^2(1 - \rho_z) + 2\sigma_\varepsilon^2$, $g_1 = -\sigma_z^2(1 - \rho_z)^2 - \sigma_\varepsilon^2$, and $g_k = -\sigma_z^2(1 - \rho_z)^2 \rho_z^{k-1}$ for $k \geq 2$, in which any time-invariant component drops out mechanically rather than being estimated (18,700 adjacent-wave differences; overidentification $p = 0.42$). Because its $k \geq 2$ moments are scaled by $(1 - \rho_z)^2$, the route is far less precise by construction: the 95% interval for $\hat{\rho}_z$ is [0.52, 0.90], containing the levels estimate, and for the persistent share [0.54, 0.74]. Pooled across PSID waves 2005–2023, 25,214 household-wave observations on 5,095 households.

Autocovariance profile. Figure C.6 shows the empirical autocovariance function and its fitted AR(1)-plus-noise counterpart. The drop from γ_0 to γ_1 is mostly the i.i.d. transitory component, which loads on the contemporaneous variance alone (the AR(1) decay contributes the residual 0.008); the geometric decay of the surviving autocovariances identifies the persistence, and their convergence toward zero rather than a positive floor is consistent with the absence of a fixed component.

Figure C.6: Autocovariance function of the cross-fitted residual ξ , PSID.



Notes: Empirical autocovariances $\gamma_k = \text{Cov}(\xi_{i,t}, \xi_{i,t-k})$ (points, ± 2 SE) and the fitted AR(1)-plus-noise function (line), against the lag in years. The drop from $\gamma_0 = 0.111$ to $\gamma_1 = 0.049$ (marked “noise”) is the i.i.d. transitory component, which loads on the contemporaneous variance but on no autocovariance; the geometric decay of γ_k for $k \geq 1$ (0.049, 0.040, 0.036, 0.031) identifies the persistence $\hat{\rho}_z = 0.86$ and a half-life near nine years. The autocovariances decay toward zero rather than a positive floor, so ξ carries no detectable time-invariant household component.

Robustness of the persistence. The persistence is stable across the estimation choices (Table C.2): adding lags, switching to identity weighting, and restricting to a composition-fixed balanced subpanel all leave $\hat{\rho}_z$ near 0.86, ruling out a long-lag composition artifact.

Orthogonality to income shocks. The expenditure-function decomposition of equation (1) assumes the latent expenditure innovation is mean-independent of the permanent and transitory income shocks. A panel vector autoregression operationalizes this as a test of whether the persistent expenditure shock responds to lagged income shocks. Figure C.7 and Table C.3 report the generalized impulse responses of the persistent shock u^{exp} from a one-lag panel VAR in the persistent shock and the two income shocks. The own-shock response dominates and decays toward zero by the ten-year horizon, while the responses to both income shocks are an order of magnitude smaller on impact and remain small in absolute terms (at most 0.010) at every horizon, though statistically nonzero at every horizon

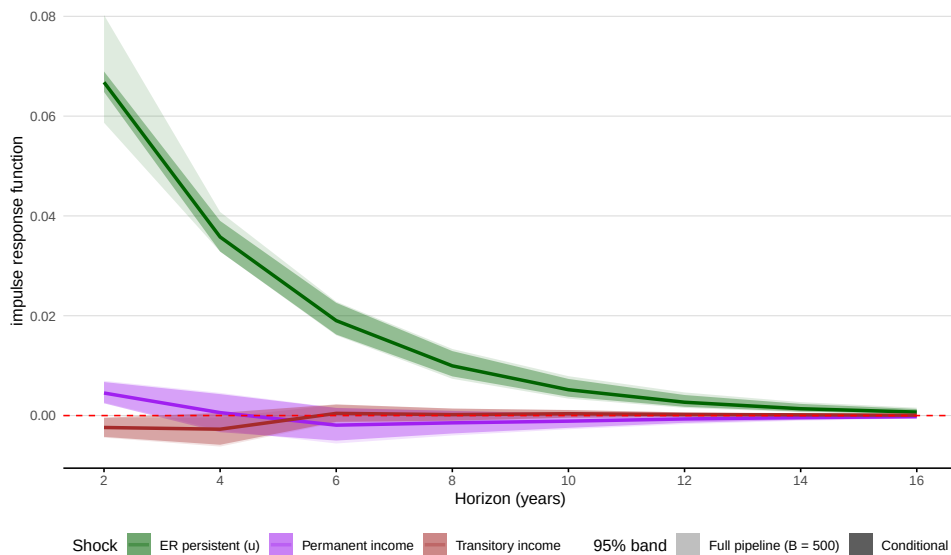
Table C.2: Robustness of the persistence $\hat{\rho}_z$.

Specification	$\hat{\rho}_z$	Households
Baseline ($K = 4$ lags, diagonal weighting)	0.858 (0.013)	5,095
More lags ($K = 6$)	0.880 (0.013)	5,095
Identity weighting	0.860 (0.013)	5,095
Balanced block (8 waves, fixed composition)	0.862 (0.018)	575

Notes: Autocovariance-minimum-distance estimate of the persistent-component autocorrelation under alternative lag counts, weighting matrices, and a composition-fixed balanced subpanel (households observed in all of the last eight PSID waves), which rules out a long-lag composition artifact. Household block-bootstrap standard errors in parentheses. The persistence lies between 0.86 and 0.88 across the four specifications.

for the permanent shock and on impact for the transitory one, bounding any linear dynamic dependence on income shocks at a quantitatively negligible level.

Figure C.7: Panel-VAR impulse responses of the persistent expenditure shock, PSID.



Notes: Generalized impulse-response functions of the persistent expenditure shock u^{exp} to a one-standard-deviation innovation in itself, the permanent income shock, and the transitory income shock, from a one-lag panel vector autoregression in $(u^{\text{exp}}, y^{\text{p}}, y^{\text{T}})$ estimated by first-difference GMM. Horizons in years (biennial PSID cadence). Lines are the full-sample point GIRFs: the own-shock response decays from 0.067 at the two-year horizon toward zero, while the responses to the two income shocks are an order of magnitude smaller on impact and never exceed 0.010 at any horizon. For each response, the darker band is the conditional 95% household block-bootstrap interval (500 replications) and the lighter band the 95% percentile interval across the $B = 500$ full-pipeline bootstrap replications of Appendix D.5, in which the entire estimation chain is re-estimated; the two nearly coincide. The dashed line marks zero.

Table C.3: Panel-VAR impulse responses of the persistent expenditure shock u^{exp} , PSID.

Horizon	Years	$u^{\text{exp}} \rightarrow u^{\text{exp}}$	$y^{\text{P}} \rightarrow u^{\text{exp}}$	$y^{\text{T}} \rightarrow u^{\text{exp}}$
1	2	0.067 [0.065, 0.069]	0.006 [0.003, 0.009]	−0.008 [−0.014, −0.001]
2	4	0.036 [0.033, 0.039]	0.010 [0.005, 0.014]	−0.010 [−0.018, 0.000]
3	6	0.019 [0.016, 0.023]	0.008 [0.004, 0.012]	−0.001 [−0.003, 0.001]
4	8	0.010 [0.008, 0.013]	0.005 [0.003, 0.008]	−0.001 [−0.003, 0.001]
5	10	0.005 [0.004, 0.007]	0.003 [0.002, 0.005]	0.000 [−0.001, 0.001]

Notes: Generalized impulse-response functions of the persistent expenditure shock u^{exp} to a one-standard-deviation innovation in each variable of the one-lag panel VAR $(u^{\text{exp}}, y^{\text{P}}, y^{\text{T}})$, estimated by first-difference GMM. Horizons indexed in biennial PSID periods; brackets are 95% household block-bootstrap confidence intervals (500 replications). The own-shock response dominates and decays to near zero by horizon five. The responses to the permanent and transitory income shocks never exceed 0.010 per standard deviation at any horizon, an order of magnitude below the own-response on impact, though the permanent-shock responses are estimated precisely enough to be statistically nonzero at every horizon (positive throughout) and the transitory one on impact (negative): the mean-independence assumed in equation (1) is violated only within a tight quantitative bound rather than holding as an exact zero, and the test bounds linear dynamic dependence only. The reverse direction is equally negligible: the response of permanent income to a u^{exp} impulse is +0.005 [0.002, 0.007] at the two-year horizon and statistically indistinguishable from zero at every later one, and the transitory-income response is −0.002 [−0.004, 0.000] on impact and zero thereafter; Appendix C.5 tests the anticipation alternative directly with valid one-sided regressors. Adding the transitory expenditure component ε^{exp} as a fourth shock leaves the persistent shock's orthogonality to income intact, with income-shock responses below 0.005 per standard deviation at every horizon.

C.5 Risk versus advance information: the persistent component does not predict future income

The persistent component z could in principle reflect advance information rather than risk: a household that privately knows its income will grow consumes ahead of that income (Guvenen and Smith, 2014), generating a persistent consumption residual without any expenditure risk. The alternative has a testable signature, that the household's persistent expenditure state should positively predict its future income growth, and this subsection runs that projection. The smoothed estimates of equation (4) cannot be used for it: $\hat{z}_{i,t}$ conditions on the household's entire residual path, including future waves, so a projection of future income growth on a smoothed state embeds future information by construction. The test therefore uses the one-sided filtered analogues, $\hat{z}_{i,t}^{\text{filt}} = \mathbb{E}[z_{i,t} \mid \xi_{i,1}, \dots, \xi_{i,t}; \hat{\theta}]$ and $\hat{u}_{i,t}^{\text{filt}} = \hat{z}_{i,t}^{\text{filt}} - \hat{\rho}_z \hat{z}_{i,t-1}^{\text{filt}}$, whose information set stops at t ; the smoothed regressor is reported alongside them only to quantify the look-ahead artifact. Table C.4 reports local projections of income growth between t and $t+k$, purged of the observables of the income regression, on each period- t regressor at $k = 2, 4, 6$ years.

Table C.4: Local projections of future income growth on the persistent expenditure state, PSID.

	Residualized income growth over horizon k		
	$k = 2$ years	$k = 4$ years	$k = 6$ years
Filtered state $\hat{z}_{i,t}^{\text{filt}}$	−0.004 (0.003)	−0.018 (0.006)***	−0.025 (0.007)***
Filtered innovation $\hat{u}_{i,t}^{\text{filt}}$	0.011 (0.005)**	0.005 (0.007)	−0.003 (0.008)
Smoothed innovation $\hat{u}_{i,t}$ (look-ahead)	0.020 (0.004)***	0.025 (0.006)***	0.022 (0.008)***

Notes: Coefficients per one standard deviation of the regressor, from local projections of income growth between t and $t+k$, purged of the observables of the income regression, on the household's period- t expenditure state, with the first-difference demographic and wave controls of Section 4.1. Household-clustered standard errors in parentheses; *, **, *** denote significance at the 10%, 5%, 1% levels. The filtered rows use the one-sided Kalman estimates, whose information set stops at t ; their samples are 16,412, 11,761, and 8,315 household-waves for the state and 13,671, 9,668, and 6,687 for the innovation, which additionally requires the preceding wave. The smoothed row conditions on the household's full residual path, including future waves, and is reported only to quantify the look-ahead artifact a smoothed-state implementation would produce.

The advance-information signature does not appear. The filtered level state, the object a household carrying private information would hold, never predicts income growth positively: its coefficient is −0.004 at two years and turns significantly negative from four years on, the opposite sign of the alternative; beyond the table's window the slope deepens further (−0.031, SE 0.009, at eight years and −0.035, SE 0.012, at ten), still the opposite sign, on samples that tilt progressively toward long-tenure households. The filtered innovation shows a one-wave +0.011 co-movement that dies by four years (+0.016 at two years when

the raw income shocks at t and $t - 1$ enter as controls, gone thereafter), a transient arrival-wave correlation rather than the growing income profile advance information implies. The smoothed row is the cautionary contrast: the same projection run on the smoothed innovation returns $+0.020$ to $+0.025$, significant at every horizon, a pure look-ahead artifact, so any implementation of this test on smoothed states, including the reverse-direction column of a panel VAR estimated on them, would manufacture the anticipation it looks for.

An assumption-light companion requires no generated regressor at all: under the AR(1)-plus-noise model of equation (2), $\text{cov}(\xi_{i,t}, \Delta y_{i,t \rightarrow t+k})$ identifies the persistent component's predictive content directly, and divided by σ_z^2 it reads $+0.040$ (95% household block-bootstrap interval $[-0.004, 0.081]$) at two years, -0.030 $[-0.086, 0.028]$ at four, and -0.086 $[-0.157, -0.013]$ at six. Against *raw* income growth the same object starts positive ($+0.117$) and decays to zero, so what predictive content exists runs through the predictable income component the observables absorb, not through private information. The direction of the feasible biases reinforces the reading: measurement error in income depresses $\xi_{i,t}$ and measured subsequent growth together, pushing the projection toward a positive coefficient, so the null is conservative for the risk interpretation. The advance-information reading therefore fits at most the transitory term ε^{exp} , as Section 3.3 assigns it; the persistent component carries no forward-looking income content.

D Passthrough estimate robustness

This appendix collects robustness checks for the passthrough estimates of the main body. Subsection D.1 reports the [Blundell et al. \(2008\)](#) covariance-ratio passthroughs for all four risk components, the pure-moment counterpart to the regression-based identification of Section 4 and cross-referenced also from Sections 2.4 and 3.3. Subsection D.2 checks the fall and escape passthroughs of Section 4.1 under a probit link and a Mundlak correlated-random-effects device in place of the linear probability model and the fixed-effects transformation. Subsection D.3 tests the timing of the wealth passthroughs with a predetermined-predictor design, the filtered persistent state at t predicting the following wave's transition and wealth change conditional on the household's position at t . Subsection D.4 reproduces the channel decomposition of Section 4.3 at the coarser ten-category aggregation and reports the multi-shock partial-correlation diagnostic. Subsection D.5 prices the estimation of every generated input into the paper's inference, re-running the full estimation chain under a household block bootstrap.

D.1 Covariance-ratio passthroughs

The passthroughs identified for the income shocks in Section 2.4 and for the expenditure shocks in Section 3.3 are, in pure-moment form, the covariance ratios of [Blundell et al.](#)

(2008): $\beta = \text{Cov}(Y, \zeta) / \text{Cov}(X, \zeta)$ for an outcome Y , an endogenous shock proxy X , and an instrument ζ . For a single instrument this is exactly the just-identified two-stage least-squares estimator applied component by component; because the covariance ratio does not condition on the other, correlated components, its estimand coincides with the joint IV columns of Section 4 only under mutual orthogonality of the shocks, and it is reported here as the assumption-light single-component companion. The covariance ratio estimates each component separately, instrumenting the quasi-difference $\Delta^{\rho_z} \zeta$ with the noise-orthogonal moving sum for the persistent component and the lead quasi-difference for the transitory one, rather than entering the two jointly. Table D.1 reports the four-component covariance ratios, residualized against the control set, in units of one standard deviation of each shock.

Table D.1: Covariance-ratio passthroughs, four risk components, PSID.

	$\Delta \ln c$	$\Delta W / \bar{y}$	H2M fall	H2M escape
y^P	0.067 [0.052, 0.089]	0.054 [0.005, 0.100]	−0.015 [−0.030, 0.000]	0.023 [−0.029, 0.086]
y^T	0.036 [0.027, 0.046]	0.078 [0.053, 0.104]	−0.012 [−0.019, −0.005]	0.039 [0.016, 0.065]
u^{exp}	0.106 [0.099, 0.114]	−0.207 [−0.269, −0.153]	0.048 [0.030, 0.066]	−0.098 [−0.196, −0.047]
ε^{exp}	0.161 [0.156, 0.166]	0.311 [0.274, 0.348]	−0.031 [−0.042, −0.024]	0.090 [0.063, 0.121]

Notes: Covariance-ratio passthroughs $\beta = \text{Cov}(Y, \zeta) / \text{Cov}(X, \zeta)$ of each outcome on the four risk components, in units of one standard deviation of each shock, residualized against the demographic and wave controls of Section 4.1; the endogenous regressor X is income growth Δy for the income rows and the quasi-difference $\Delta^{\rho_z} \zeta$ for the expenditure rows. The income shocks use their three-difference and lead instruments; the persistent expenditure shock uses the forward moving sum ζ^u and the transitory shock the lead quasi-difference ζ^ε of Section 3.3. Brackets are 95% household block-bootstrap confidence intervals (1,000 replications); estimation samples vary by component and outcome (N between 1,707 and 17,545).

The covariance ratios reproduce the signs of the body throughout: the persistent expenditure shock is the only wealth-depleting component (−0.21 on $\Delta W / \bar{y}$, interval excluding zero) and the only one that raises hand-to-mouth entry and lowers escape, while both income shocks and the transitory expenditure shock pass through positively. Estimated separately, without conditioning on the correlated transitory component, the persistent shock's magnitudes are more conservative than the joint two-stage least-squares estimates of the body, but the signs and significance agree. The two income passthroughs to consumption growth, 0.067 and 0.036 per standard deviation, are positive and of the same order as the regression-based estimates of Section 2.4; the transitory passthrough is positive and bounded away from zero, in the partial-insurance range of Commault (2022) and Crawley (2020).²³

²³The covariance ratio has no within-household variant: instrumenting the persistent shock alone under

D.2 Link function and within-household conditioning

The transition results of Section 4.1 are checked against two robustness exercises on the direct-regression specification. First, the linear probability model is replaced by a probit reporting average marginal effects, addressing the concern that a linear conditional probability can stray outside the unit interval.²⁴ Second, the household fixed-effects transformation is replaced by a Mundlak (Mundlak, 1978) correlated-random-effects device, which augments the pooled regressor set with the household-level means of the four shocks; the coefficient on the contemporaneous shock then recovers the within-household passthrough, providing a tractable within-household analog for the probit that sidesteps the incidental-parameters problem on short panels.²⁵ Table D.2 reports the four-component fall and escape passthroughs across the link-function and conditioning combinations, in units of one standard deviation of each shock.

The pattern of Section 4.1 survives both exercises. The probit average marginal effects sit within 0.002 of the linear-probability coefficients throughout. The Mundlak device reproduces the within-household reading: on fall, the persistent expenditure passthrough amplifies relative to the pooled estimate ($0.024 \rightarrow 0.029$) while the income passthroughs attenuate to roughly half their pooled magnitude (near -0.006); on escape, the persistent expenditure passthrough sharpens from -0.028 to -0.094 , mirroring the fixed-effects estimate of Table 4, while permanent income stays insignificant and transitory income holds. The within-household amplification of the persistent expenditure shock and the attenuation of the income shocks are thus neither a linear-probability artifact nor an artifact of the fixed-effects estimator.

D.3 Predetermined-predictor design: the persistent state predicts the following wave

The passthroughs of Section 4 are estimated over the same biennial interval in which the shock realizes, so their wealth results could in principle be read as the household budget constraint restated (Section 4.2). This subsection tests the timing concern directly with a predetermined-predictor design: the one-sided filtered persistent state $\hat{z}_{i,t}^{\text{filt}}$, whose infor-

household fixed effects is weakly identified, because the fixed-effects transformation removes the persistent variation the instrument exploits. The single-component fixed-effects two-stage estimate of the wealth passthrough is an uninformative -1.5 with a standard error near 0.9 . This is why the body reports the two expenditure components jointly, where the transitory component supplies the within-household identifying variation; in the pooled cross-section the joint and separate estimates agree in sign.

²⁴Logit average marginal effects are within two basis points of probit throughout and are omitted to keep the table compact.

²⁵The basic Mundlak device (household means only) coincides with the within transformation in the linear case; Chamberlain (1982) generalizes the projection to include leads and lags of the time-varying regressors, the form usually called Mundlak-Chamberlain. The household-means variant is sufficient for the within-versus-pooled comparison here.

Table D.2: Link-function and within-household robustness of the H2M-flow passthroughs, PSID.

	Pooled		Within-HH (Mundlak CRE)	
	LPM	probit	LPM	probit
<i>Panel A: H2M fall</i>				
y^P	−0.010*** (0.0029)	−0.010*** (0.0030)	−0.007** (0.0031)	−0.008** (0.0032)
y^T	−0.010*** (0.0026)	−0.010*** (0.0025)	−0.006** (0.0023)	−0.006*** (0.0023)
u^{exp}	0.024*** (0.0038)	0.022*** (0.0034)	0.029*** (0.0051)	0.029*** (0.0048)
ε^{exp}	−0.025*** (0.0037)	−0.023*** (0.0033)	−0.016*** (0.0037)	−0.016*** (0.0034)
<i>Panel B: H2M escape</i>				
y^P	0.007 (0.0105)	0.007 (0.0105)	−0.003 (0.0145)	−0.003 (0.0148)
y^T	0.031*** (0.0103)	0.032*** (0.0102)	0.034*** (0.0122)	0.036*** (0.0123)
u^{exp}	−0.028** (0.0129)	−0.028** (0.0131)	−0.094*** (0.0217)	−0.094*** (0.0228)
ε^{exp}	0.067*** (0.0124)	0.068*** (0.0125)	0.086*** (0.0165)	0.088*** (0.0171)
HH-mean controls	No	No	Yes	Yes
Observations (fall)			10,173	
Observations (escape)			2,278	

Notes: Passthroughs to the H2M fall and escape indicators from joint regressions on the four risk components (y^P , y^T , u^{exp} , ε^{exp}), in units of one standard deviation of each shock, under two link functions (linear probability model; probit average marginal effects) and two conditioning schemes (pooled; Mundlak correlated-random-effects). The Mundlak columns augment the regressor set with the household-level means of the four shocks. Household-clustered standard errors in parentheses; *, **, *** denote significance at the 10%, 5%, 1% levels. The pooled block reproduces the OLS reading of Section 4.1 and the Mundlak block the within-household reading; logit average marginal effects are within two basis points of probit and are not displayed.

mation set runs through wave t (consumption and income through t , wealth through $t - 2$), predicting outcomes dated $t + 2$, conditional on the household's position at t , measured by its within-wave net-worth percentile rank (and its square) and log family income, alongside the demographic and wave controls of Section 4.1. Under within-wave accounting the state carries no information about the following wave conditional on that position, because wealth at t is measured after the spending interval $\xi_{i,t}$ covers, so the shock's own contemporaneous financing is already embedded in the conditioning variable; under the AR(1) process of Section 3.3 the still-elevated state keeps draining resources and predicts the following wave. Measurement error in wealth at t pushes the following wave toward mean reversion, against the design finding continued deterioration, so the estimates are conservative for the risk reading.

Table D.3 reports the estimates. The filtered state predicts the following wave's fall into hand-to-mouth status, $+0.007$ to $+0.008$ per standard deviation at the 1% level against a biennial fall rate of 0.079 (pooled within that estimation sample), and the following wave's wealth change, -0.031 to -0.047 at the 5% and 1% levels. The raw residual ξ_t delivers the same reading on both outcomes, so the result does not ride on the Kalman extraction. Escape is the underpowered margin of this design: the origin sample is a fifth the size of the fall sample and the standard errors four times larger, the state's coefficient is indistinguishable from zero, and only the fresh innovation moves the margin (-0.017 to -0.026 , at the 10% and 5% levels); the escape evidence of the body accordingly rests on the contemporaneous within-household estimates of Section 4.1, not on this design.

The table reports pooled estimates only. The household fixed-effects transformation is not the right conditioning for a predetermined-predictor question: with roughly five waves per household, demeaning a persistent lagged state induces Nickell-type bias, and the within transformation removes precisely the persistent variation whose predictive content the design tests. The two-sided smoothed state is likewise excluded as a predictor, because it embeds future residuals by construction (Appendix C.5).

D.4 Channel decomposition: finer multi-shock and main-category robustness

Figure D.1 reports the within-household multi-shock partial correlations of each finer expenditure-risk-basket category's share on the three identified shocks ($y^P, y^T, \hat{u}^{\text{exp}}$) entered jointly, the companion to the cross-correlation decomposition (Figure 3) discussed in Section 4.3. Expenditure risk dominates the loading on every basket category except childcare, where the transitory-income-shock loading is comparable in magnitude, and trips, where the income-shock loadings are comparable or larger at the point estimates, consistent with the fragility of that category's classification flagged in Subsection D.5.

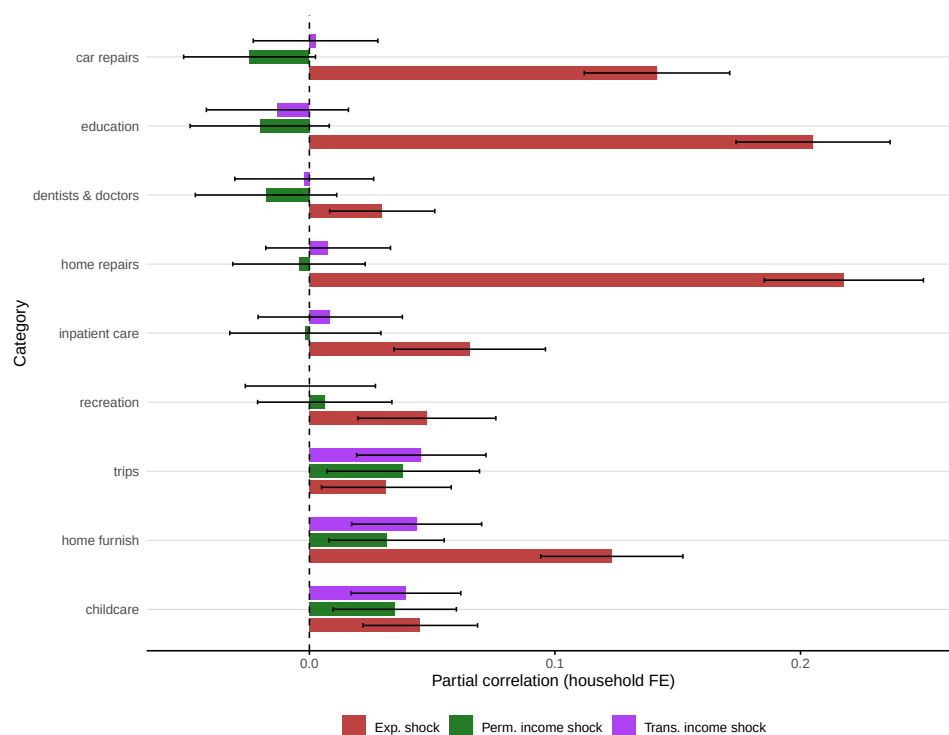
The channel decomposition of Section 4.3 is reported on the 24-category finer dis-

Table D.3: Forward $t \rightarrow t + 2$ outcomes on the lagged filtered state, PSID.

	(1) Baseline	(2) + income shocks
<i>Panel A: Fall into H2M at $t + 2$ (origin $H2M_t = 0$)</i>		
$\hat{z}_t^{\text{filt}}$	0.0072*** (0.0023)	0.0083*** (0.0024)
$\hat{u}_t^{\text{filt}}$	0.0081*** (0.0026)	0.0078*** (0.0027)
ξ_t	0.0082*** (0.0023)	0.0088*** (0.0024)
Observations	13,458	11,649
<i>Panel B: Escape from H2M at $t + 2$ (origin $H2M_t = 1$)</i>		
$\hat{z}_t^{\text{filt}}$	0.0012 (0.0098)	−0.0035 (0.0117)
$\hat{u}_t^{\text{filt}}$	−0.0165* (0.0095)	−0.0258** (0.0108)
ξ_t	−0.0052 (0.0093)	−0.0117 (0.0109)
Observations	2,954	2,230
<i>Panel C: Wealth change $(W_{t+2} - W_t)/\bar{y}$</i>		
$\hat{z}_t^{\text{filt}}$	−0.0310** (0.0132)	−0.0467*** (0.0153)
$\hat{u}_t^{\text{filt}}$	−0.0174 (0.0160)	−0.0194 (0.0181)
ξ_t	−0.0310** (0.0139)	−0.0427*** (0.0160)
Observations	16,240	13,733

Notes: Pooled regressions of outcomes dated $t + 2$ on predictors dated t , in units of one standard deviation of each predictor, conditional on the within-wave net-worth percentile rank at t (and its square), log family income at t , and the demographic and wave controls of Section 4.1; column (2) adds the raw income shock at t and $t - 1$. Predictors: the one-sided filtered persistent state $\hat{z}_t^{\text{filt}}$, its filtered innovation $\hat{u}_t^{\text{filt}}$, and the raw residual ξ_t (assumption-light companion). The wealth-change outcome is trimmed at its pooled 0.5/99.5 percentiles, matching Table 5. The reported observation counts refer to the $\hat{z}_t^{\text{filt}}$ and ξ rows; the $\hat{u}_t^{\text{filt}}$ rows lose one additional adjacent wave (N between 1,883 and 13,540). Household-clustered standard errors in parentheses; *, **, *** denote significance at the 10%, 5%, 1% levels.

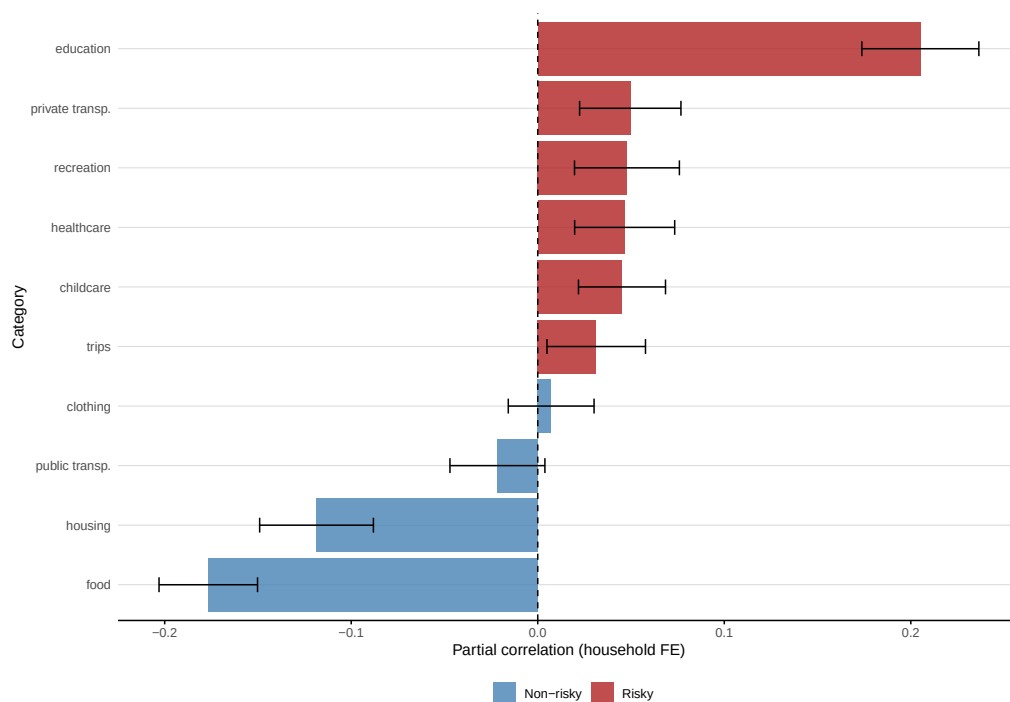
Figure D.1: Multi-shock partial correlations on the expenditure-risk basket, PSID.



Notes: Within-household partial correlations of each risky category's share with the three identified shocks ($y^P, y^T, \hat{u}^{exp}$) entered jointly. Bars grouped by category; whiskers are household-clustered 95% confidence bands. Expenditure risk dominates the loading on every category except childcare, where the transitory-income-shock loading is comparable in magnitude, and trips, where both income-shock loadings are comparable or larger. Same sample and controls as Figure 3.

aggregation of the PSID expenditure module, the underlying variables that the module's standard 10-category main aggregation collapses. This subsection also reproduces the channel figures at the coarser main-aggregation level, in which health insurance and inpatient care collapse into a single *healthcare* category, home repairs and home furnishings collapse into *housing maintenance and furnishings*, and the remaining categories collapse analogously. Figure D.2 reports the decomposition of the share of each main category on the recovered persistent expenditure shock $\hat{u}_{i,t}^{\text{exp}}$, the main-category counterpart to Figure 3; Figure D.3 reports the multi-shock partial correlations of each positively-loading main category on $(y^p, y^t, \hat{u}^{\text{exp}})$ jointly, the main-category counterpart to Figure D.1.

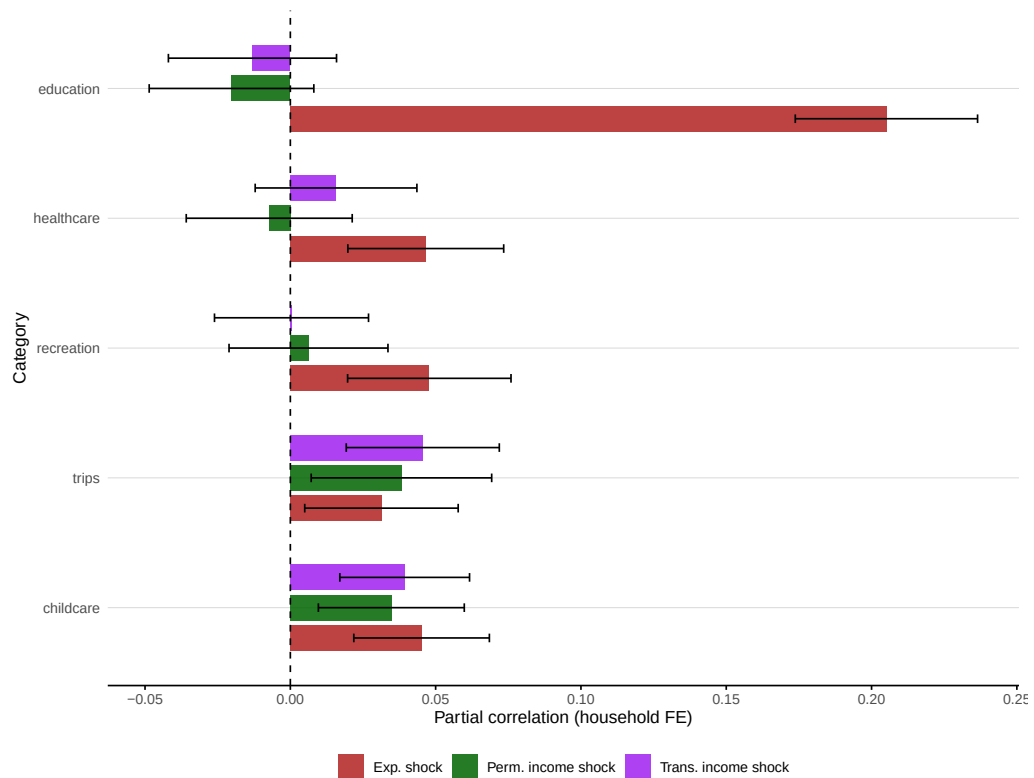
Figure D.2: Main-category loadings on the persistent expenditure shock, PSID.



Notes: Main-category counterpart to Figure 3. Standardized partial correlations of each non-durable consumption main-category share with $\hat{u}_{i,t}^{\text{exp}}$, conditional on household and wave fixed effects and the demographic controls of Section 4.1. Six main categories load positively (private transport, education, childcare, healthcare, recreation, trips), summing to a 15.9% basket whose aggregate partial correlation is +0.22, in line with the 19.8% basket and +0.41 aggregate partial correlation of the finer-level baseline.

The main-category and finer-category readings agree on the qualitative composition of the channel: the positively-loading basket is concentrated in expenditures servicing kids, sickness, and household capital, with childcare the one category where an income-shock co-loading is comparable to the expenditure-risk loading. The finer breakdown is preferred as the main-body baseline because it lets the empirical reading distinguish health-insurance premiums from inpatient-care outlays and home repairs from home furnishings, both informative dimensions for the structural interpretation of Section 5; the main-category aggregation is reported here as a coarser robustness for readers who prefer the standard

Figure D.3: Multi-shock partial correlations on the main-category basket, PSID.



Notes: Main-category counterpart to Figure D.1. Within-household partial correlations of each main-category share with the three identified shocks ($y^P, y^T, \hat{u}^{\text{exp}}$) entered jointly. The qualitative reading of Section 4.3 survives the coarser aggregation: expenditure risk dominates the loading on education and healthcare, with income-shock loadings near zero or of the opposite sign; childcare and trips are the categories where income-shock loadings are comparable to the expenditure-risk loading.

PSID expenditure-module aggregation.

D.5 Inference under generated regressors: a bootstrap of the full estimation chain

The expenditure-risk estimates of Sections 3 and 4 are built on generated regressors: the consumption residual $\xi_{i,t}$, the fitted income shocks, the household types, and the smoothed states $\hat{z}_{i,t}$ and $\hat{u}_{i,t}^{\text{exp}}$ are all estimated objects, and the standard errors reported so far condition on them. This appendix asks whether the paper’s inference survives when the estimation uncertainty of every one of these inputs is priced in. The answer is a household block bootstrap of the entire estimation chain: each replication draws 5,235 households with replacement from the post-selection PSID sample²⁶ and re-runs every step of the pipeline on the redrawn panel, from the sample trims, through the income-shock projections, the household-type clustering, and the cross-fitted random-forest consumption function, to the autocovariance and Kalman estimates of the expenditure-risk process and all downstream passthrough regressions. Percentile confidence intervals across $B = 500$ replications therefore reflect sampling and first-stage uncertainty jointly, the object that conditional standard errors miss (Pagan, 1984; Murphy and Topel, 1985).

Two implementation choices matter. First, because a resampled panel contains repeated copies of the same household, the cross-fitting folds of the random-forest step are assigned at the level of the original household, so that all copies of a household share a fold; without this restriction the forest trains on exact copies of its test households and the out-of-fold residual is no longer honest. Second, all algorithmic seeds (fold assignment, forest growing, clustering) are held at their baseline values across replications, so the intervals isolate sampling uncertainty. The Kalman likelihood failed to converge in 8 of 500 replications; these are excluded from the statistics that depend on the smoothed states.

Holding the algorithmic seeds fixed could in principle understate total uncertainty, because the random-forest fit, the fold assignment, and the clustering step carry Monte Carlo randomness of their own. A companion run that redraws all of these seeds in every replication ($B = 500$) shows this component is negligible: across the 1,381 statistics collected by the exercise, the ratio of the redrawn-seed to the fixed-seed bootstrap standard error has median 1.001 (10th–90th percentile range $[0.990, 1.017]$), the implied algorithmic share of total variance is 0.1 percent at the median and 3.3 percent at the 90th percentile, and every conclusion of this appendix is unchanged. The intervals reported here can therefore be read as total-uncertainty intervals.

²⁶The resampling frame is the panel remaining after the deterministic, row-wise selection filters alone (the consumption and income floors and the food- and housing-share bands): 5,235 households, 36,702 household-wave observations. It exceeds the 5,095-household, 25,214-observation measurement sample of Section 3 because the household draw is taken at the top of the pipeline, before the sample-dependent quantile trims and the post-2005 estimation window that each replication re-applies.

Every headline estimate survives. The persistence of the expenditure-risk process is $\hat{\rho}_z = 0.86$ with a 95 percent pipeline interval of $[0.83, 0.88]$ (autocovariance route) and 0.85 with $[0.82, 0.87]$ (Kalman route); the persistent component's share of unpredicted variation is 0.50 with $[0.49, 0.54]$, and its share of the total variance of log consumption is 0.151 with $[0.148, 0.169]$. All passthroughs of the persistent shock u^{exp} and the transitory shock ε^{exp} to consumption growth, wealth changes, and H2M transitions retain their sign in 100 percent of replications, and each interval excludes zero, in the direct regressions, the instrumented versions, the covariance-ratio estimates, and the Mundlak specifications alike. The anticipation tests are also unaffected: the one-sided filtered state never positively predicts future income growth. Figure D.4 summarizes the thirty headline quantities tracked in the figure.

The panel-VAR evidence receives the same treatment. Refitting the panel VAR once on each replication's panel and reading confidence bands off the cross-replication distribution of point GIRFs prices the estimation of the smoothed states into the bands without any inner bootstrap. The resulting bands nearly coincide with the conditional block-bootstrap bands reported in Appendix C.4 for every response of u^{exp} to the income shocks, and the orthogonality reading is unchanged: the pipeline point responses and bands stay within the 0.010–0.018 range of the conditional estimates, an order of magnitude below the own-persistence response. The one band that widens is the own-response at impact ($[0.059, 0.080]$ against the conditional $[0.065, 0.069]$), which scales with the estimated innovation variance and plays no role in the argument (Figure C.7, which overlays both band families).

The exercise also shows where the paper's reported uncertainty was already adequate and where it was not. Across the regression coefficients, the pipeline standard errors exceed the household-clustered analytic ones by 6 percent at the median (17 percent at the 90th percentile), and the autocovariance-based process parameters reproduce their conditional block-bootstrap standard errors to within 8 percent, so the generated regressors contribute little additional uncertainty to these objects. The one exception is the Kalman maximum-likelihood step, whose Hessian-based standard errors understate the pipeline ones by 28 to 48 percent; the persistence standard error quoted in Appendix C.4 for that route is therefore the bootstrap one. Finally, the risky-basket classification of Section 4.3 is stable: eight of the nine risky categories are re-selected in more than 90 percent of replications, while trips is re-selected in 69 percent.

E Quantitative model details

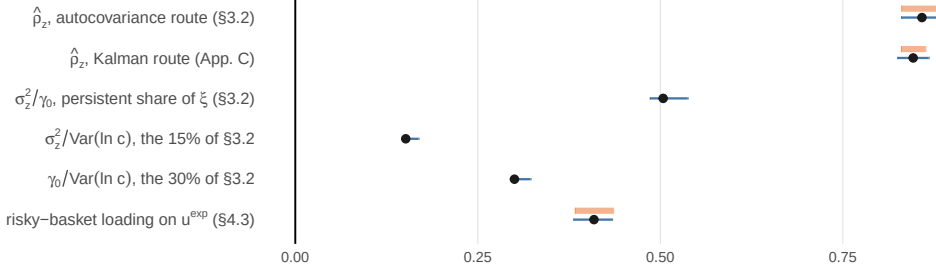
This appendix collects specification details of the model of Section 5 that the body summarizes in compressed form. Subsection E.1 writes out the full household optimization problem of Section 5.1 with explicit state transitions, budget constraints, borrowing limits, and the working / retirement-transition / retired phase splits. Subsection E.2 states the sta-

Figure D.4: Full-pipeline household bootstrap: survival of the paper's estimates.

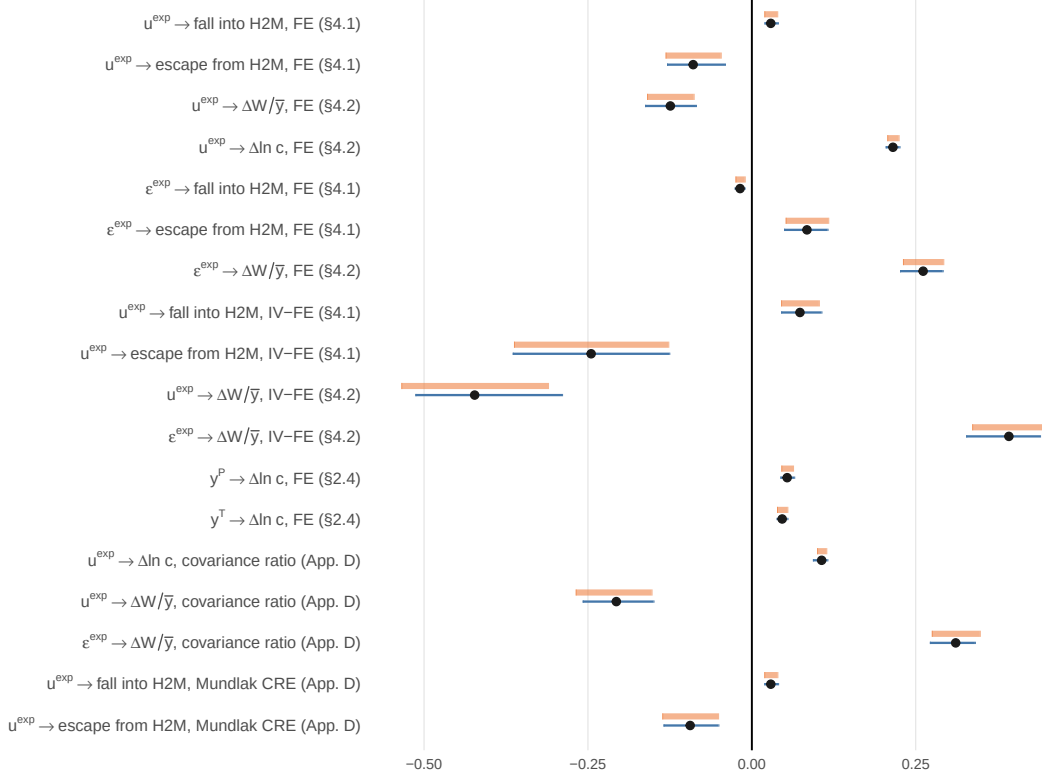
Full-pipeline household bootstrap: headline estimates

B = 500 replications; whole estimation chain re-run per replication

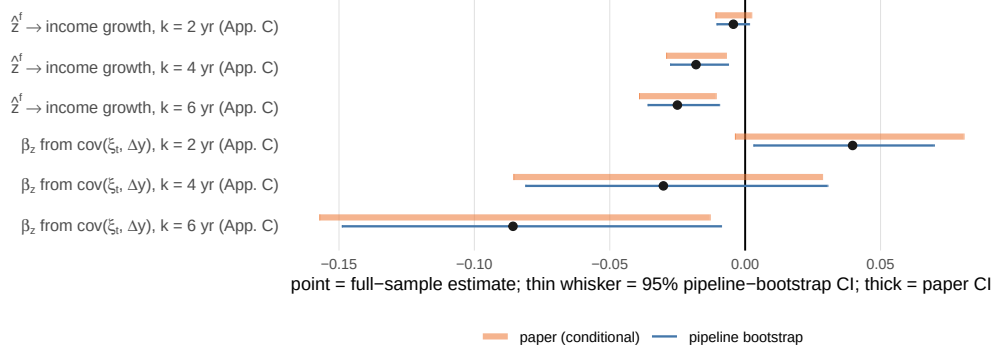
ER process, variance decomposition & channel (§3, §4.3)



Passthroughs, per 1 SD of the shock (§2.4, §4, App. D)



Anticipation falsifications (App. C)



Notes: Full-pipeline household block bootstrap ($B = 500$). Points are full-sample estimates; thin whiskers are 95 percent percentile intervals across replications in which the entire estimation chain, from sample trims to the smoothed expenditure-risk states, is re-estimated; thick whiskers are the intervals reported in the paper (analytic household-clustered or conditional-bootstrap), where one exists.

tionary recursive competitive equilibrium of Section 5.2 formally. Subsection E.3 collects the externally-set-parameter table, the calibrated discount-factor-distribution figure, and the hand-to-mouth-by-age target-fit figure summarized in Section 5.3.

E.1 Full household optimization problem (discretized)

The continuous household problem of Section 5.1 is discretized for numerical solution. The random-walk persistent income shock η is Tauchen-discretized on an age-conditional grid $\mathcal{E}_\eta$ with transition matrix Π_η ; the i.i.d. transitory income shock ε is Gauss-Hermite-quadrature-discretized on $\mathcal{E}_\varepsilon$ with probabilities π_ε ; the AR(1) marginal-utility shock φ is Tauchen-discretized on $\mathcal{E}_\varphi$ with transition matrix Π_φ . The three shock processes are mutually independent, so the joint one-step transition factors as $\Pi(\eta', \varepsilon', \varphi' \mid \eta, \varphi) = \Pi_\eta(\eta' \mid \eta) \pi_\varepsilon(\varepsilon') \Pi_\varphi(\varphi' \mid \varphi)$. The fixed type τ and discount factor β are drawn at birth from discrete fixed grids $\mathcal{T}$ and $\mathcal{B}$ (the latter the truncated Gaussian-mixture support of Section 5.3) and held constant. Labor income at age j is written compactly as $y_j(\tau, \eta, \varepsilon) = (1 - \theta) \exp(\mu_j + \tau + \eta + \varepsilon) w$, with deterministic age profile $\{\mu_j\}_{j \leq J^w}$, equilibrium wage w , and payroll tax θ (Section 5.1); all income objects below, including the borrowing-limit expectation $\tilde{y}$, are after-tax. Writing $\mathcal{S}_j = \mathcal{X} \times \mathcal{T} \times \mathcal{B} \times \mathcal{E}_\eta \times \mathcal{E}_\varphi \times \{j\}$ for the age- j state space, with cash-on-hand x on an endogenous grid $\mathcal{X}$, a household at age j in state $s_{i,j} = (x_{i,j}, \tau_i, \beta_i, \eta_{i,j}, \varphi_{i,j}, j) \in \mathcal{S}_j$ solves

$$\begin{aligned}
 V_j(s_{i,j}) = & \max_{c_{1,i,j}, c_{2,i,j}, a_{i,j+1}} u(c_{1,i,j}, c_{2,i,j}; \varphi_{i,j}) \\
 & + \beta_i \mathfrak{s}_j \sum_{(\eta', \varepsilon', \varphi')} \Pi(\eta', \varepsilon', \varphi' \mid \eta_{i,j}, \varphi_{i,j}) V_{j+1}(s_{i,j+1}) \\
 \text{s.t. } & a_{i,j+1} = x_{i,j} - c_{1,i,j} - c_{2,i,j} \\
 & a_{i,j+1} \geq -BL(j, \tau_i, \eta_{i,j}) \\
 & x_{i,j+1} = (1 + r) a_{i,j+1} + y_{j+1}(\tau_i, \eta', \varepsilon') + T \cdot w_{j+1}^{\text{beq}} \\
 & \eta' \in \mathcal{E}_\eta, \quad \varepsilon' \in \mathcal{E}_\varepsilon, \quad \varphi' \in \mathcal{E}_\varphi \\
 & s_{i,j+1} = (x_{i,j+1}, \tau_i, \beta_i, \eta', \varphi', j + 1),
 \end{aligned} \tag{E.1}$$

where $u(\cdot)$ is the two-good CRRA flow utility of equation (6) (with the intratemporal-allocation FOC of equation (7) pinning the within-period split between c_1 and c_2), $\mathfrak{s}_j$ is the age- j conditional one-period survival probability from the demographic block of Section 5.2, r is the equilibrium interest rate, and T is the aggregate accidental-bequest pool with age-tilt weights $w_{j+1}^{\text{beq}} \propto (\text{beq_tilt})^{j+1}$.

The state-dependent borrowing limit $BL(j, \tau, \eta)$ pledges a repay-share of the annuitized (per-period-equivalent) value of expected labor and pension income over the household's remaining life, the discounted-survival-weighted average of expected income from age j onward:

$$BL(j, \tau, \eta) = \text{repay} \cdot \frac{\sum_{s=0}^{J^D-j} (1+r)^{-s} S(j, s) \tilde{y}_{j+s}(\tau, \eta)}{\sum_{s=0}^{J^D-j} (1+r)^{-s} S(j, s)}, \quad (\text{E.2})$$

with $S(j, s) = \prod_{k=0}^{s-1} \mathfrak{s}_{j+k}$ the cumulative probability of surviving from age j to age $j+s$ (so $S(j, 0) = 1$), and $\tilde{y}_{j+s}(\tau, \eta)$ the expected labor or pension income at age $j+s$ conditional on the fixed type τ and the persistent income state η at age j . The working-age component uses $\tilde{y}_{j+s}(\tau, \eta) = \mathbb{E}[y_{j+s}(\tau, \eta_{j+s}, \varepsilon) \mid \eta_j = \eta]$ integrating over future paths of the random-walk persistent shock and the i.i.d. transitory shock; the retirement-window component uses the deterministic Social Security payment $\tilde{y}_{j+s}(\tau, \eta) = \text{rr}(\eta) \cdot w$ for $j+s > J^W$. The repay-share parameter is set to $\text{repay} = 0.20$, following [Kaplan et al. \(2014\)](#) and [Aguiar et al. \(2025\)](#) (who use 0.185); BL is recomputed at every general-equilibrium iteration via the equilibrium (r, w) .

The problem partitions across three age phases. For working-age households $j < J^W$ the formulation in equation (E.1) applies as written. In the age- J^W problem, the labor-income block inside $x_{i,j+1}$ is replaced by the deterministic Social Security payment $y_{J^W+1} = \text{rr}(\eta_{i,J^W}) \cdot w$ implied by the household's final persistent earnings state and the rank-progressive replacement-rate schedule of Section 5.2, with ε and the transition probabilities over it dropped from the expectation. For retired households $j \geq J^W + 1$, the persistent earnings state η is frozen at its retirement-boundary value (its transition becomes degenerate), and the marginal-utility shock φ contracts deterministically toward zero according to its AR(1) decay rate, capturing the gradual resolution of expenditure-risk uncertainty over the retirement window of the workingLifeER specification. Age-conditional mortality applies throughout the life-cycle: at every age j the household survives to the next period with probability $\mathfrak{s}_j$ and dies with probability $1 - \mathfrak{s}_j$, entering the Bellman through the survival-adjusted continuation factor $\beta_i \mathfrak{s}_j$ of equation (E.1), with its remaining assets passing into the accidental-bequest pool T in the death event. At the terminal age $j = J^D$, $\mathfrak{s}_{J^D} = 0$: death is certain and the continuation value collapses to zero.

The IR specialization of Section 5.1 obtains by setting $\rho_\varphi = \sigma_\varphi = 0$ in the AR(1) law for φ , eliminating the φ state dimension and collapsing the $\mathcal{E}_\varphi$ -summation to its degenerate point mass at $\varphi = 0$; under homothetic CRRA preferences, the intratemporal allocation rebalances to a fixed share determined by φ_0 alone (equation (7)), and the resulting problem is the standard incomplete-markets life-cycle consumption-saving model with discount-factor heterogeneity, random-walk plus transitory income shocks, accidental bequests, and the same state-dependent borrowing limit as in the ER baseline.

E.2 Stationary recursive competitive equilibrium

The competitive equilibrium of Section 5.2 is stated formally as a stationary recursive equilibrium of the discretized economy of Subsection E.1.

A *stationary recursive competitive equilibrium* for the two-good ER economy consists of (i) age-indexed value and policy functions $\{V_j, g_j^{c1}, g_j^{c2}, g_j^{a'}\}_{j=1}^{J^D}$, (ii) a stationary cross-sectional probability distribution λ on the joint state (s, j) with $s \in \mathcal{S}_j$, (iii) prices (r, w, θ) , and (iv) aggregates (K, L, T) such that:

1. *Household optimization.* Given prices (r, w, θ) and the bequest pool T , the value functions V_j solve the Bellman equation (E.1) and the policies $(g_j^{c1}, g_j^{c2}, g_j^{a'})$ achieve its maximum.
2. *Firm optimization.* Prices satisfy the competitive firm's first-order conditions, $r + \delta = \alpha (K/L)^{\alpha-1}$ and $w = (1 - \alpha) (K/L)^\alpha$.
3. *Aggregation.* K and L aggregate household asset holdings and working-age effective labor supply over the stationary distribution λ ; the accidental-bequest pool aggregates the end-of-period assets of the non-surviving households, $T = \sum_j (1 - s_j) \int g_j^{a'} d\lambda_j$, and is redistributed to the living via the age-tilt weights w_j^{beq} .
4. *Government budget.* The pay-as-you-go payroll tax balances the pension budget, $\theta w L$ equal to aggregate pension outlays integrated against λ over the retired population.
5. *Stationarity of the distribution.* λ is invariant under the joint operator induced by the household savings policy $g_j^{a'}$, the discretized shock transitions $(\Pi_\eta, \pi_\varepsilon, \Pi_\varphi)$, and survival-weighted age progression, mass s_j carried from age j to $j + 1$ and the complement $1 - s_j$ exiting into the bequest pool, with new-cohort entry at $j = 1$ replacing the exiting mass so that the age distribution reproduces the demographic block of Section 5.2.
6. *Goods-market clearing.* By Walras' law, the goods market clears at $Y = C + \delta K$ in stationary equilibrium and follows automatically from conditions 1–5.

The equilibrium is solved numerically by combining endogenous-grid backward induction on the household problem (Carroll, 2006) with the life-cycle stationary-distribution method of Young (2010), then iterating on (r, w, θ) until the three market-clearing conditions are satisfied within tolerance. The IR equilibrium is defined analogously with $\rho_\varphi = \sigma_\varphi = 0$, eliminating the φ state dimension from $\mathcal{S}_j$ and reducing the intratemporal allocation to a fixed share determined by φ_0 .

E.3 Calibration tables and figures

Table E.1 collects the parameters and demographic objects fixed outside the calibration loop and held common across the IR and ER economies. Preferences and technology

follow standard conventions: CRRA curvature $\gamma = 2$, Cobb-Douglas capital share $\alpha = 0.36$ (Cooley and Prescott, 1995), and a biennial depreciation rate compounding an 8% annual rate. The biennial income-process variances (σ_η^2 , σ_ε^2 , σ_τ^2) are own estimates from the income-components identification of Appendix B on the same PSID sample as the empirical analysis, with the persistent component imposed as a random walk ($\rho_\eta = 1$) following Blundell et al. (2008) and Kaplan and Violante (2010). The demographic block is drawn from external U.S. data: age-conditional survival from Social Security Administration male life tables, the deterministic age-income profile $\{\mu_j\}$ from PSID, and the rank-progressive replacement-rate schedule from Purcell (2012); the borrowing-limit pledgeable share is set to 0.20, following Kaplan et al. (2014) and Aguiar et al. (2025). Holding this block fixed across regimes is what lets the dynamic comparison of Section 6 load entirely on the internally calibrated parameters of Table 6.

Table E.1: Externally-set parameters and demographic objects.

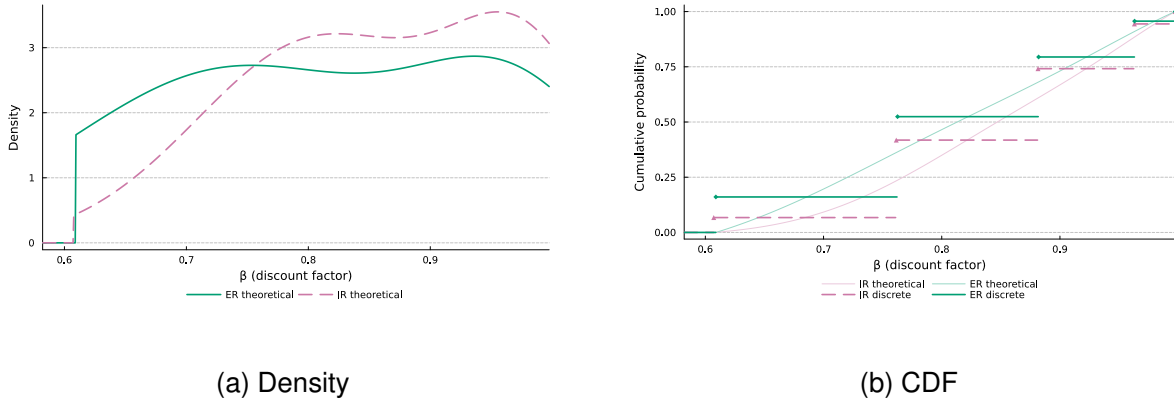
Parameter(s)	Value(s)	Source
<i>Preferences and technology</i>		
γ	2.0	Standard CRRA
α	0.36	Cooley and Prescott (1995)
δ (biennial)	0.1536	8% annual depreciation
A	1.0	Normalization
<i>Demographics</i>		
Period length	2 years	PSID biennial waves
J^W (working life)	20 periods (age 64)	—
J^D (death)	35 periods (age 94)	Terminal-age convention
n (pop. growth, biennial)	0.012	U.S. average, 2010s
$\{\mathfrak{s}_j\}$	compounded annual rates	SSA male life tables
$\{\mu_j\}$	lifecycle profile	PSID 1999–2023
$\{\text{rr}_q\}$ (replacement ratios)	[0.778, 0.591, 0.484, 0.422, 0.333]	Purcell (2012), SSA Bulletin
<i>Income risk (biennial frequencies)</i>		
ρ_η	1.0	BPP (2008); Kaplan and Violante (2010)
$\{\sigma_\eta^2, \sigma_\varepsilon^2, \sigma_\tau^2\}$	{0.0452, 0.1072, 0.1864}	own estimates; see B
<i>Borrowing limit</i>		
repay	0.20	Kaplan et al. (2014); Aguiar et al. (2025)

Notes: Externally-set parameters are held fixed across both the IR and ER calibrations.

Figure E.1 plots the calibrated discount-factor distributions the two economies settle on, as the continuous density (panel a) and the CDF together with the $n_\beta = 5$ discretization actually used in the model (panel b), both sharing the common upper truncation $\beta_{\max} = 0.9975$. The two mixtures are close; the one systematic difference is ER's lower and more dispersed impatient component, and the economic reading, namely that the marginal-utility-shock channel already generates precautionary saving that income risk alone does not, is given in Section 5.3.

Figure E.2 reports the hand-to-mouth share by age for PSID and the two calibrated

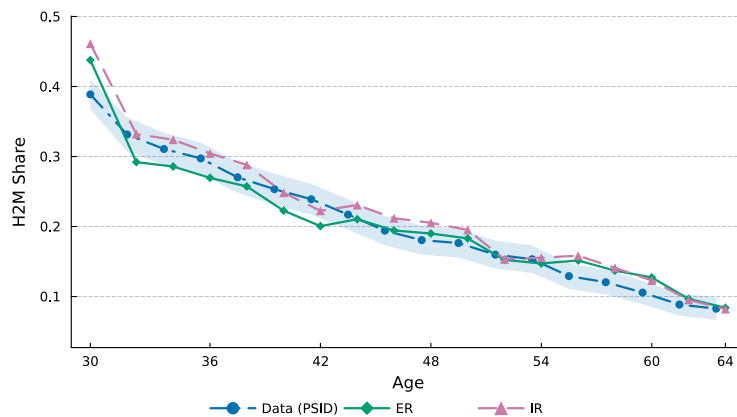
Figure E.1: Calibrated cross-household discount-factor distribution: IR vs ER.



Notes: Calibrated truncated two-component Gaussian mixtures of β for each economy, evaluated at the parameter vectors of Table 6. Panel (a): theoretical density (continuous). Panel (b): theoretical CDF (continuous) and its $n_\beta = 5$ discretization (step) actually used in the model. Both mixtures share the common upper truncation $\beta_{\max} = 0.9975$.

economies, the visual counterpart to the H2M-share level-benchmark fit of Section 5.4. Because the youth, midlife, and retirement H2M shares are themselves calibration targets, the figure doubles as a target-fit check: both economies track the PSID profile across the working life and into retirement, with all three targeted stages matched within 2.2 percentage points of the data (Section 5.4).

Figure E.2: Hand-to-mouth share by age: data vs calibrated economies.



Notes: Household-level H2M share (net worth ≤ 0) by age, for PSID (shaded 95% confidence band) and the two calibrated economies. The youth, midlife, and retirement H2M shares are internal-calibration targets (Table 6).

F Quantitative model: corollary results

This appendix collects corollary wealth-dynamics artifacts and a derivation cited in compressed form in Section 6. Subsection F.1 reports the full transition objects behind the main-text comparators: the full 2×2 hand-to-mouth (H2M) transition matrices underlying main-text Table 8, the age-conditional rank-rank wealth autocorrelation cited inline in Section 6.3, and the full IR / ER / PSID side-by-side 5×5 wealth-quintile heatmap underlying main-text Figure 7. Subsection F.2 walks through the algebraic steps from the c_2 -form Euler equation to the covariance wedge of main-text equation (10). Subsection F.3 reports the percent-change and absolute-unit dose-response figures of the σ_φ -scaling exercise of Section 6.5, together with the H2M transition rates across the same grid recomputed in the main-text convention and the implied wedge-versus- β -composition split of the IR-to-ER flow differences. Subsection F.6 collects the per-quintile consumption-growth cuts and the raw and log-tail density variants behind Section 6.2. Subsection F.4 reports the MPC-persistence dose-response behind the MPC-transience corollary of Section 6.4. Subsection F.5 adds the corollary's cross-sectional companion: MPC dispersion and its observable predictability across four economies, including the reverse-path cell (the IR β mixture with the φ process on) that completes the wedge-versus- β -composition split for this statistic.

F.1 Wealth transitions across PSID and the calibrated economies

This subsection reports in full the transition objects that the main-text comparators of Section 6.3 summarize. Table F.1 gives the complete 2×2 H2M transition matrices for the IR baseline, the ER economy, and PSID; Table F.2 the age-conditional rank-rank wealth autocorrelation for the same three economies; and Figure F.1 the side-by-side 5×5 wealth-quintile transition matrices. Each isolates the same gap-closure pattern, the ER economy moving the IR baseline toward the PSID anchor. Figure F.2 adds the post-escape dimension, plotting the share of escaped households that re-enter H2M by each horizon: the ER economy tracks the PSID recurrence curve more closely than the income-risk baseline does.

F.2 Covariance-wedge derivation

The intertemporal optimality condition for the risky-good consumption c_2 at interior optima, derived from the household problem of Section 5.1 and Appendix E.1, reads

$$\exp(\varphi_j) \cdot c_{2,j}^{-\gamma} = \beta \mathfrak{s}_j R \mathbb{E}_j [\exp(\varphi_{j+1}) \cdot c_{2,j+1}^{-\gamma}], \quad (\text{F.1})$$

with $R \equiv 1 + r$ the gross interest rate, $\mathfrak{s}_j$ the age- j conditional survival probability, and the household subscript suppressed throughout. Dividing both sides by $c_{2,j}^{-\gamma}$ rearranges

Table F.1: Full 2×2 hand-to-mouth (H2M) transition matrix: PSID, IR baseline, and ER economy.

State at $t - 2$	PSID		IR		ER	
	Not H2M	H2M	Not H2M	H2M	Not H2M	H2M
Not H2M	0.915	0.085	0.955	0.045	0.940	0.060
H2M	0.389	0.611	0.287	0.713	0.374	0.626

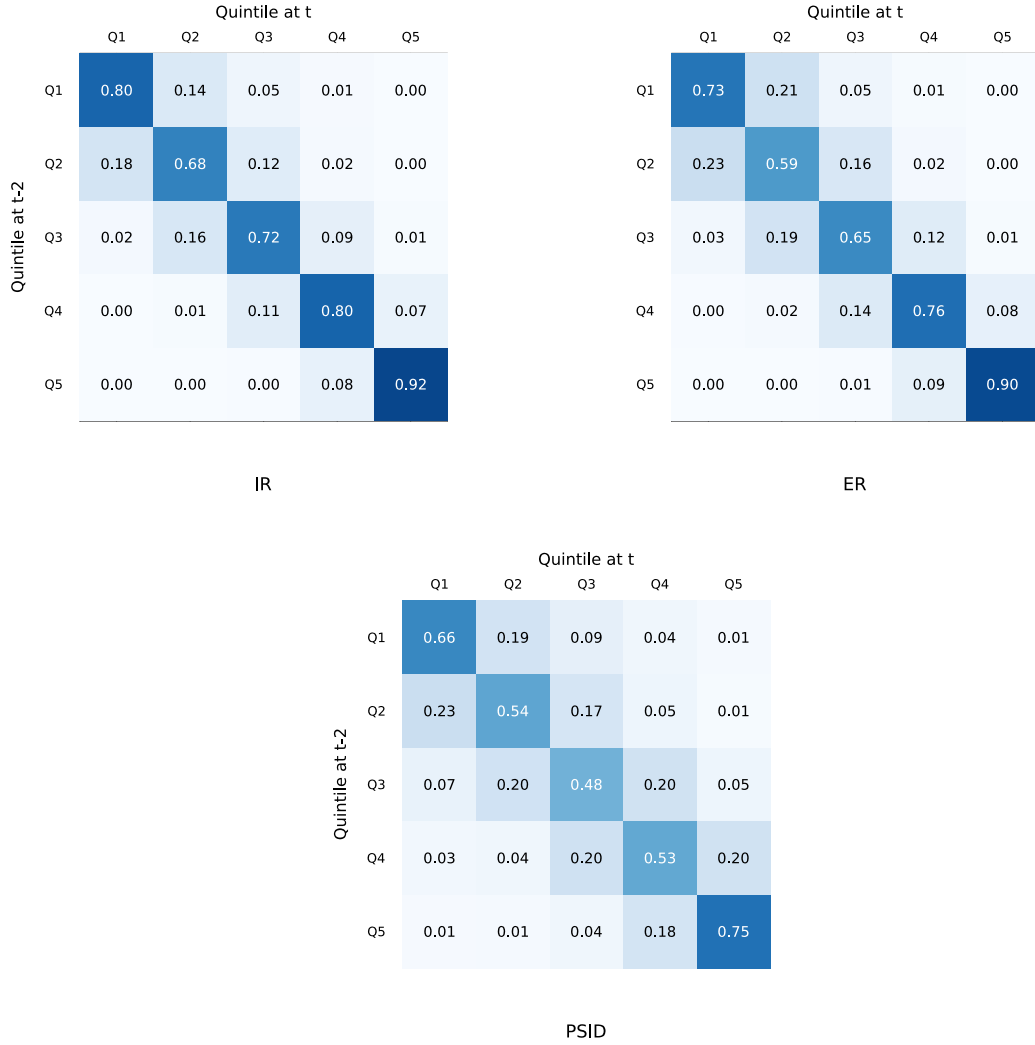
Notes: Rows index the H2M state at $t - 2$, columns the H2M state at t ; each row of each 2×2 panel sums to one. H2M defined as non-positive net worth, as in Section 2.2; rates are age-averaged over the working life (ages 26–64), with each transition dated to the wave at which the destination status is observed, matching the PSID construction. The off-diagonals are the biennial fall (Not H2M $\rightarrow$ H2M, top-row off-diagonal) and escape (H2M $\rightarrow$ Not H2M, bottom-row off-diagonal) rates reported in main-text Table 8; the diagonals are the implied stay probabilities.

Table F.2: Age-conditional rank-rank wealth autocorrelation: PSID, IR baseline, and ER economy.

Age tercile	PSID	IR	ER	Gap closure
Young (age 25–35)	0.740	0.890	0.861	20%
Middle (37–47)	0.848	0.958	0.942	15%
Old (49–63)	0.887	0.980	0.971	10%
All working ages	0.833	0.945	0.927	16%

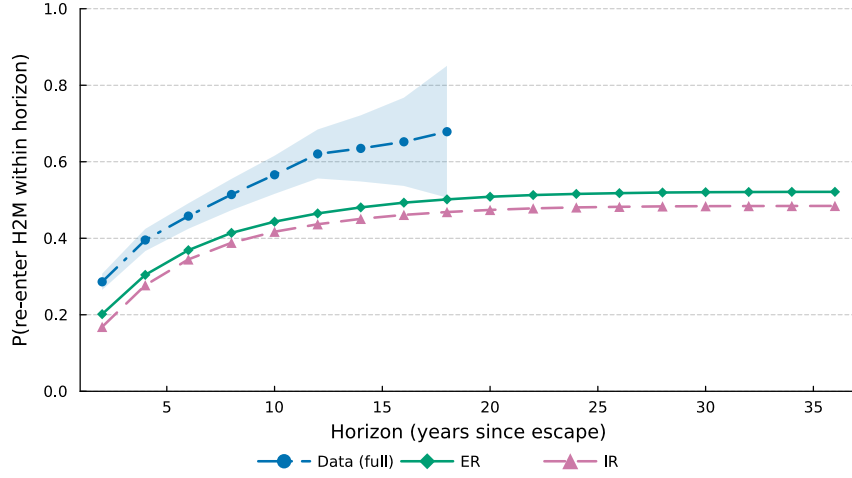
Notes: Each cell reports the OLS slope from a within-tercile regression of household wealth rank at t on wealth rank at $t - 2$. Working-age sample partitioned into three age terciles using the model’s biennial age periods j : young is $j \leq 6$ (ages 25–35), middle is $7 \leq j \leq 12$ (37–47), old is $j > 12$ (49–63); the pooled row uses all working-age periods $j \leq 20$. *Gap closure* is $(b^{IR} - b^{ER}) / (b^{IR} - b^{PSID})$, the fraction of the IR-Data persistence gap closed under ER. The closure tapers with age, mirroring the wealth-quintile tapering of Figure F.1.

Figure F.1: Biennial 5×5 wealth-quintile transition matrix: IR baseline, ER economy, and PSID.



Notes: Three panels reporting the biennial transition probabilities across wealth quintiles, with the two calibrated economies on the top row (IR left, ER right) and the PSID anchor on the bottom; rows index the quintile at $t-2$, columns the quintile at t , and each row sums to one. The progression from IR (most diagonal-dominant) through ER (intermediate, with thicker off-diagonal mass) to PSID (least diagonal-dominant) traces the gap-closure direction quantified in main-text Section 6.3: 48% of the IR-Data persistence gap closed at Q_1 , 64% at Q_2 , tapering to 8% at Q_5 , where the top-tail persistence puzzle of Section 5.4 carries through.

Figure F.2: Post-escape H2M recurrence across PSID and the calibrated economies.



Notes: For households observed escaping H2M, the share that have re-entered H2M by each horizon, in the IR baseline, the ER economy, and the PSID data. H2M is non-positive net worth, as in Section 2.2. The PSID curve carries a 95% confidence band and ends at the panel's last observable horizon, while the simulated economies run to the end of the working life. The ER economy tracks the PSID recurrence curve more closely than the income-risk baseline, the same gap-closure direction documented for the transition objects above.

(F.1) as $\exp(\varphi_j) = R\beta \mathfrak{s}_j \cdot \mathbb{E}_j[\exp(\varphi_{j+1}) \cdot (c_{2,j+1}/c_{2,j})^{-\gamma}]$. Applying the covariance identity $\mathbb{E}[XY] = \mathbb{E}[X] \mathbb{E}[Y] + \text{Cov}(X, Y)$ to the right-hand-side conditional expectation, with $X = \exp(\varphi_{j+1})$ and $Y = (c_{2,j+1}/c_{2,j})^{-\gamma}$, separates the joint expectation into a product term and a covariance term:

$$\exp(\varphi_j) = R\beta \mathfrak{s}_j \cdot \left\{ \mathbb{E}_j[\exp(\varphi_{j+1})] \cdot \mathbb{E}_j[(c_{2,j+1}/c_{2,j})^{-\gamma}] + \text{Cov}_j(\exp(\varphi_{j+1}), (c_{2,j+1}/c_{2,j})^{-\gamma}) \right\}. \quad (\text{F.2})$$

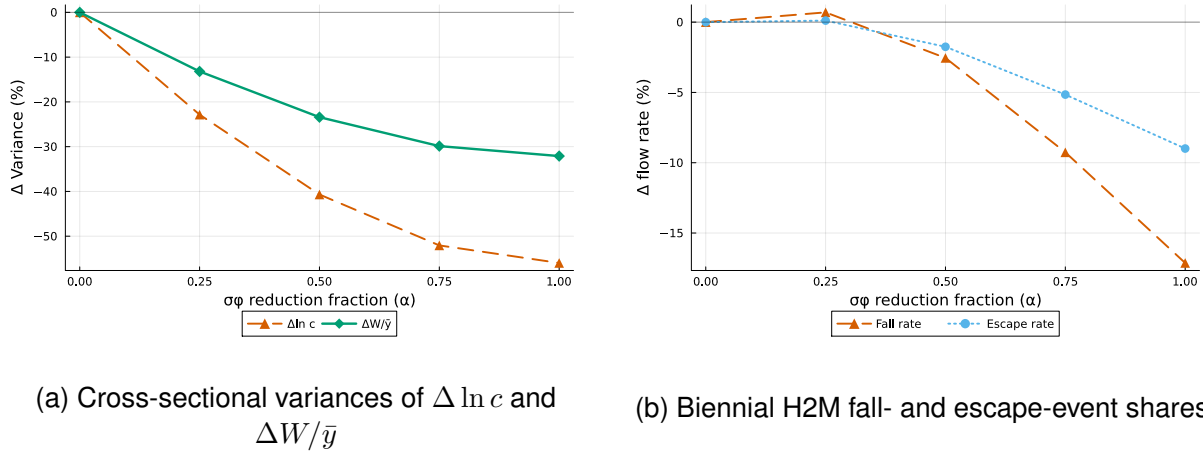
Solving (F.2) for $\mathbb{E}_j[(c_{2,j+1}/c_{2,j})^{-\gamma}]$ and dividing through by $\mathbb{E}_j[\exp(\varphi_{j+1})]$ yields the covariance-wedge form of main-text equation (10); the sign restriction ≥ 0 in (10) is established in the body of Section 6.1.

F.3 Dose-response of the σ_φ -scaling exercise

Figure F.3 reports the percent-change dose-response curves of the σ_φ -scaling exercise of Section 6.5, and Figure F.4 its absolute-unit companion; both plot variance changes (panel a) and H2M event-share changes (panel b) from the baseline ER calibration as σ_φ is scaled through $\sigma_\varphi^{\text{cf}} = (1 - \alpha) \sigma_\varphi^{\text{base}}$. Table F.3 then reports the H2M transitions across the same grid in the main-text convention of Table 8, whose $\sigma_\varphi \rightarrow 0$ corner row supports a decomposition of the IR-to-ER flow differences into a β -composition step and a wedge step.

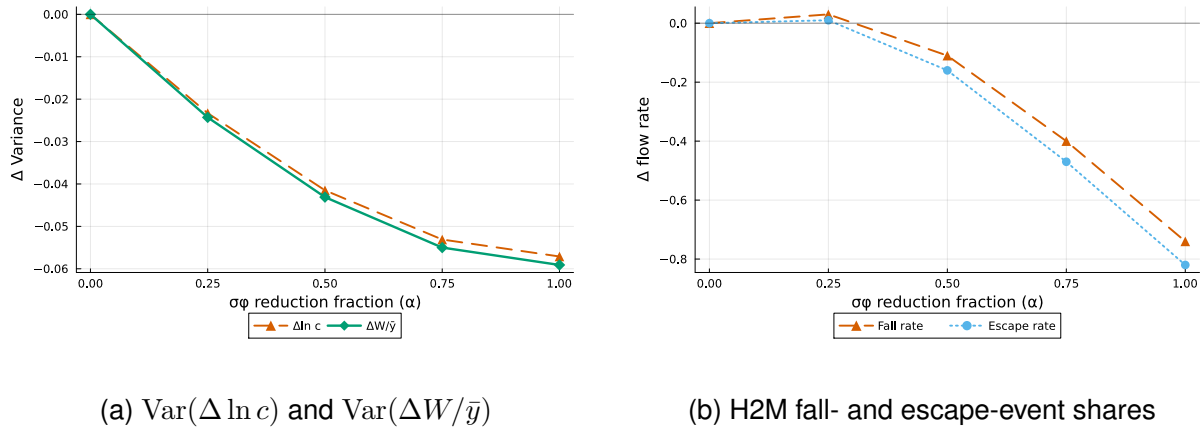
The flow panels of Figures F.3 and F.4 and the transition rates of Table F.3 measure different objects, and the difference matters for reading the wedge's strength. The figures plot

Figure F.3: Dose-response of consumption- and wealth-change variances and H2M event shares to σ_φ scaling.



Notes: Percent-change responses to scaling the marginal-utility-shock variance, $\sigma_\varphi^{\text{cf}} = (1 - \alpha) \sigma_\varphi^{\text{base}}$; $\alpha = 0$ is baseline ER and $\alpha = 1$ is the $\sigma_\varphi \rightarrow 0$ corner that turns the wedge of equation (10) off; α is a mechanism-isolating scaling parameter, not a policy instrument. Panel (a): the cross-sectional variance of consumption growth ($\Delta \ln c$) and the variance of biennial wealth changes ($\Delta W/\bar{y}$), in percent change relative to $\alpha = 0$. Panel (b): the biennial H2M fall- and escape-event shares of the simulated working-life panel (the share of household-waves at which each event occurs), in percent change relative to $\alpha = 0$; the conditional transition rates in the main-text convention are reported across the same grid in Table F.3.

Figure F.4: Dose-response in absolute units: σ_φ scaling on consumption- and wealth-change variances (panel a) and H2M event shares (panel b).



Notes: Both panels report changes from baseline ($\alpha = 0$) in original units. Panel (a): the cross-sectional variance of biennial consumption growth ($\Delta \ln c$) and the variance of biennial income-scaled wealth changes ($\Delta W/\bar{y}$); the two profiles are visually indistinguishable and converge on ≈ -0.06 at $\alpha = 1$, indicating that σ_φ contributes the same absolute amount of variance to both outcomes. Panel (b): the biennial H2M fall- and escape-event shares, in percentage points; the two shares decline by 0.7 and 0.8 percentage points across the grid, a pooled movement that blends the rising H2M stock with the falling conditional escape rate (Table F.3). The percent-change versions of these profiles are reported in Figure F.3.

the pooled share of working-life household-waves at which a fall or escape event occurs, the simulated panel's summary statistic; that share is the product of the at-risk stock and the conditional transition rate, and the two move in opposite directions as the wedge turns off, the panel-level H2M share rising by 8.0 pp (7.8 pp in the table's age-averaged convention) while the conditional escape rate falls. The pooled escape-event share therefore moves by only 0.8 pp across the grid even as the conditional escape rate falls from 0.374 to 0.256; the table's destination-indexed, age-averaged working-age rates are the objects comparable to main-text Table 8 and to the PSID construction.

Table F.3: H2M fall and escape rates across the σ_φ -scaling grid, main-text convention.

	Fall rate	Escape rate	H2M share	Fall share	Escape 1 – share
ER, $\alpha = 0$ (baseline)	0.060	0.374	0.181	0.333	0.457
$\alpha = 0.25$	0.063	0.334	0.209	0.303	0.422
$\alpha = 0.50$	0.064	0.298	0.233	0.273	0.389
$\alpha = 0.75$	0.061	0.270	0.251	0.243	0.361
$\alpha = 1$ ($\sigma_\varphi \rightarrow 0$ corner)	0.056	0.256	0.259	0.217	0.346
IR (no φ -process)	0.045	0.287	0.210	0.213	0.363
PSID	0.085	0.389	0.205	0.412	0.490

Notes: Biennial transition probabilities across the H2M margin in the convention of main-text Table 8: H2M status is non-positive net worth, each transition is dated to the wave at which the destination status is observed, and rates are age-averaged over the working life (ages 26–64). The α rows scale the marginal-utility-shock volatility within the ER calibration, $\sigma_\varphi^{\text{cf}} = (1 - \alpha) \sigma_\varphi^{\text{base}}$, holding the calibrated β distribution and all other parameters fixed; $\alpha = 1$ is the $\sigma_\varphi \rightarrow 0$ corner, an economy with the ER discount-factor mixture and no expenditure risk. IR and PSID reproduce main-text Table 8 for reference. *H2M share* is the age-averaged $a \leq 0$ stock in the same bin convention; the last two columns scale each flow by its destination stock, the fall rate by the H2M share and the escape rate by its complement, because the two calibrated economies are approximately stock-matched by calibration while the corner's stock rises as the wedge's precautionary saving is removed. Scaling each flow by its destination stock is one diagnostic normalization for holding stocks comparable across rows, not a derived flow decomposition; the unscaled rates it adjusts are reported alongside in the same table. An exact within-cohort accounting corroborates the split without any normalization: writing the fall profile as implied by the share and escape profiles through the identity $s_j = s_{j-2}(1 - e_j) + (1 - s_{j-2})f_j$ and swapping profiles across economies attributes the IR-to-corner fall difference to the corner's higher hand-to-mouth share path and the corner-to-ER difference to ER's uniformly higher escape rates.

Read against the two calibrated economies, the corner row splits the IR-to-ER flow differences of main-text Table 8 into two steps: a β -composition step from IR to the corner, across which neither economy carries expenditure risk and only the separately calibrated discount-factor mixtures and bequest tilts differ, and a wedge step from the corner to ER, across which the calibration is identical and only σ_φ switches on. On the escape rate the corner sits below IR (0.256 against 0.287), so the β -composition step is negative (−3.1 pp) and the wedge step (+11.8 pp) exceeds the full IR-to-ER difference (+8.7 pp): the escape closure of Section 6.3 is carried entirely by the wedge, with the ER calibration's more impatient lower tail working against it. On the raw fall rate the split reads the other way, the β

step carrying +1.2 pp of the +1.6 pp difference and the wedge step +0.4 pp; that comparison, however, is not stock-matched, because the corner's age-averaged H2M share rises to 0.259 against 0.181 in ER and 0.210 in IR, and a larger low-net-worth stock surrounds the boundary with more marginal households, mechanically raising falls. The destination-scaled columns hold stocks comparable: per unit of the H2M share, the fall split becomes 4% β -composition and 96% wedge (+0.004 and +0.116 of a +0.120 gap), and per unit of the non-H2M share the escape split reads -18% and $+118\%$, so on stock-comparable terms the wedge carries both flow closures. Because only the $(\beta^{\text{ER}}, \sigma_\varphi \rightarrow 0)$ corner is computed, the split is read along the $\text{IR} \rightarrow \text{corner} \rightarrow \text{ER}$ path; the reverse path, the IR β mixture with the φ process switched on, is not computed for the flow objects, so any interaction between the β composition and the wedge is folded into the wedge step. Subsection F.5 solves that economy for the MPC statistics, where the wedge step proves path-independent.

F.4 MPC persistence across the σ_φ -scaling exercise

This subsection reports the dose-response behind the MPC-transience corollary opening Section 6.4. At each node of the σ_φ -scaling grid the economy is re-solved in general equilibrium and the per-state marginal propensity to consume is attached to the simulated working-life panel, computed as the response of consumption to a small resource windfall from the saving policy function ($\text{MPC} = 1 - [g^{a'}(\tilde{x} + \varepsilon) - g^{a'}(\tilde{x})]/\varepsilon$, with ε one percent of mean income). Three statistics summarize the household-level MPC time series: its within-household lag-one autocorrelation, and, for a fixed high-MPC threshold, the mean length of a high-MPC spell (a run of consecutive waves above the threshold) and the share of those spells lasting a single wave. Table F.4 reports these across the grid at both thresholds; the re-solved mean MPC reproduces the value in the σ_φ -robustness experiment of Section 6.5 at every node.

Table F.4: MPC level and persistence across the σ_φ -scaling grid.

α	σ_φ	mean MPC	autocorr	spell _{0.5} (yr)	transient _{0.5}	spell _{0.3} (yr)	transient _{0.3}
0.00	1.199	0.269	0.755	4.30	0.501	8.19	0.393
0.25	0.899	0.281	0.779	4.77	0.468	9.64	0.354
0.50	0.599	0.292	0.802	5.39	0.448	11.58	0.307
0.75	0.300	0.298	0.819	6.06	0.441	13.78	0.256
1.00	0.000	0.301	0.825	6.26	0.429	15.18	0.207

Notes: Marginal propensity to consume and its persistence across the σ_φ -scaling grid of Section 6.5, where $\sigma_\varphi^{\text{cf}} = (1 - \alpha) \sigma_\varphi^{\text{base}}$ and $\alpha = 1$ is the $\sigma_\varphi \rightarrow 0$ corner. *autocorr*: within-household lag-one autocorrelation of the MPC. *spell_m*: mean length in years of a high-MPC spell at threshold $\text{MPC} \geq m$; *transient_m*: share of those spells lasting a single biennial wave. As σ_φ rises (moving up the table), the mean MPC falls, its autocorrelation falls, high-MPC spells shorten, and the transient share rises, all monotonically.

F.5 MPC dispersion across wealth and the marginal-utility state

This subsection reports the cross-sectional companion to Table F.4: how much of the dispersion in household MPCs the observable states price, and how much rides on the latent marginal-utility state. Household-period MPCs are computed on the simulated working-life panels with the same policy-function perturbation as in Subsection F.4, in four economies: the calibrated IR and ER baselines, the $\sigma_\varphi \rightarrow 0$ corner of the scaling grid (the ER β mixture with the wedge off), and the reverse-path economy left uncomputed in Subsection F.3, the IR β mixture with the ER φ process switched on, solved in general equilibrium and simulated at the same panel size and seed. The reverse-path economy is a diagnostic, calibrated to nothing; its general equilibrium is nonetheless informative: the interest rate falls to 0.091 against IR's 0.107, the patient IR mixture over-accumulating once expenditure risk adds its precautionary-saving motive, which is the general-equilibrium reason the separately calibrated ER economy pairs the wedge with a less patient mixture at the same wealth targets. Table F.5 reports the level, dispersion, and predictability statistics; Table F.6 the mean MPC across the joint distribution of wealth and φ in the ER economy.

Table F.5: MPC dispersion and its observable predictability across four economies.

	IR	$\sigma_\varphi \rightarrow 0$ corner (β^{ER} , wedge off)	ER	reverse path (β^{IR} , wedge on)
Mean MPC	0.259	0.301	0.269	0.234
Std. dev. of MPC	0.217	0.238	0.226	0.209
R^2 , wealth	0.553	0.594	0.487	0.446
R^2 , wealth + age + income	0.592	0.632	0.514	0.477
Within-cell variance share	0.386	0.354	0.473	0.507
Between-household share	0.688	0.716	0.581	0.550
General-equilibrium r	0.107	0.124	0.106	0.091

Notes: MPCs computed at every simulated household-wave as in Subsection F.4 (ε one percent of mean biennial income; values clamped to $[0, 1]$), on the working-life panel from the second wave on (net worth is degenerate at labor-market entry); all statistics are stable across windfall sizes of one-half, one, and five percent of mean income. R^2 , *wealth*: OLS of the MPC on within-age net-worth-quintile indicators plus a linear net-worth term. R^2 , *wealth + age + income* adds age indicators and log income. *Within-cell variance share*: share of MPC variance remaining within (age $\times$ wealth-quintile) cells. *Between-household share*: between-household share of age-adjusted MPC variance. The corner column reuses the $\alpha = 1$ node of the Table F.3 grid; the reverse-path column solves the IR β mixture (including its bequest tilt) with the calibrated ER φ process and simulates it at the headline panel size and seed. The R^2 entries are linear predictability in the specifications the empirical MPC literature runs, not structural variance shares; a saturated specification would fit exactly.

Read across columns, the wedge's effect on the wealth-MPC relation is path-independent where the flow decomposition of Subsection F.3 could not be: wealth's R^2 falls by 0.11 whether the wedge switches on at the ER mixture (0.594 to 0.487) or at the IR mixture (0.553 to 0.446), an interaction of 0.001, so no order caveat applies to this statistic. The

β -composition step runs the other way (+0.041 on both paths), and the two forces nearly cancel in the raw ER-to-IR comparison; the same cancellation shapes the mean, where the β step (+0.042) and the prudence-direction wedge step (−0.033; Table F.4) net to a raw gap of +0.010. An order-free Shapley decomposition of the full-state regression ($R^2 = 0.79$) attributes 0.29 of explained MPC variance to the discount-factor type, 0.28 to wealth, and 0.16 to φ , with age, income, and the income states carrying the remainder; adding φ to the observable set raises the R^2 by 0.13, and by 0.15 when the discount-factor type and the income states are also conditioned on.

Table F.6: Mean MPC by wealth quintile and φ tercile, ER economy.

Wealth quintile	Bottom φ tercile	Middle φ tercile	Top φ tercile
Q_1 (poorest)	0.37	0.50	0.73
Q_2	0.23	0.31	0.48
Q_3	0.16	0.21	0.31
Q_4	0.11	0.14	0.19
Q_5 (richest)	0.08	0.10	0.14

Notes: Mean MPC by within-age quintile of beginning-of-period net worth and pooled tercile of the marginal-utility state φ , ER economy, working-life panel from the second wave on. Moving from the middle to the top φ tercile raises the mean MPC by roughly one wealth quintile at every point of the distribution: the top-quintile top-tercile mean (0.14) matches the fourth quintile’s unconditional mean, and the policy-function slices confirm the MPC rises in φ over the whole cash-on-hand range at fixed age, income state, and discount factor.

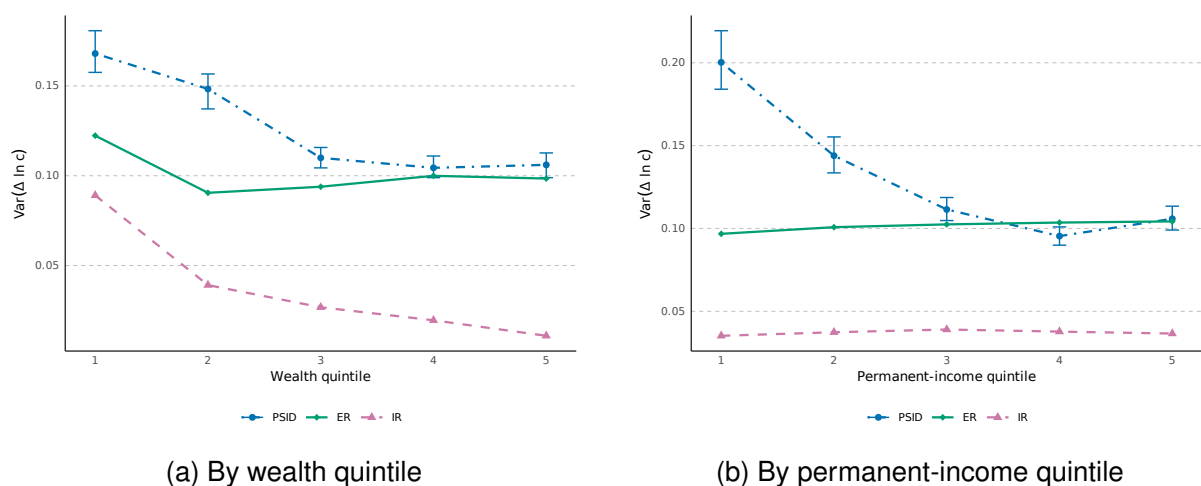
Three readings complete the picture. First, the state-like character of the dispersion: the between-household share of age-adjusted MPC variance falls from 0.69 under income risk to 0.58 under expenditure risk, and within households, movements in φ account for a third of the remaining variation while the persistent income state accounts for under one percent of it in either economy. Second, visibility: conditional on age, log income, and current wealth, adding next-wave wealth to the regression closes 72% of the R^2 gap that observing φ itself would close, while adding lagged wealth closes 5%; end-of-period wealth embeds the household’s consumption choice through the budget identity, so the number locates the state in wealth data rather than predicting it. Third, the latent-heterogeneity benchmark: a regression of simulated MPCs on income terciles, the average propensity to consume, and a quadratic in age, the covariate set of [Lewis et al. \(2026\)](#), explains 7% of MPC variance at the corner, 6% under IR, and 4% under ER, every economy inside their at-most-8% empirical ceiling; switching the wedge on pushes the observable share further down, the direction of their latent-heterogeneity reading, though levels are not comparable, simulated propensities and covariates carrying no measurement error and the propensity being biennial, so the direction is the claim. Two caveats bound all of the above: the R^2 statements are linear predictability, not structural variance shares, and the one-asset

economy speaks about net worth where the empirical weak-wealth findings concern liquid wealth (Fuster et al., 2021; Boehm et al., 2025), so the marginal-utility channel and the illiquid-balance-sheet channel of Kaplan and Violante (2014) remain complementary readings the model does not separate.

F.6 Consumption-growth dynamics: additional cuts

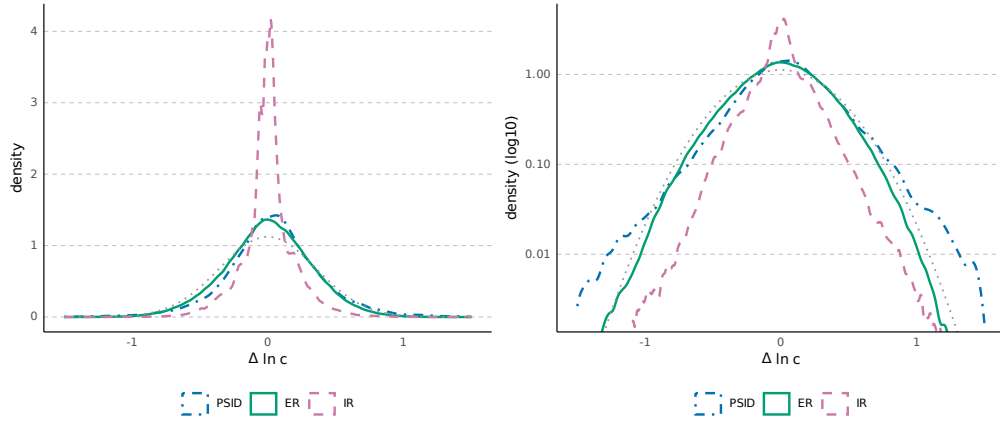
This subsection collects the supporting cuts behind the consumption-growth-risk results of Section 6.2. Figure F.5 reports $\text{Var}(\Delta \ln c)$ by wealth quintile (panel a) and by permanent-income quintile (panel b), the finer cuts behind the wealth-position exposure profile of main-text Figure 6: ER tracks the elevated PSID volatility over the upper three wealth quintiles and undershoots it in the bottom two, while IR collapses off the constraint, and both economies miss the declining permanent-income gradient in the data. Figure F.6 reproduces the pooled $\Delta \ln c$ density without the age and family-size residualization used in the body, on both a linear and a log scale; the raw and residualized densities tell the same dispersion story, confirming the residualization only strips deterministic life-cycle and demographic variation rather than manufacturing the match. Figure F.7 plots the residualized density on a log scale, where ER's near-Gaussian tails fall short of the PSID tail mass, the unmatched dimension noted in Section 6.2.

Figure F.5: Consumption-growth variance by wealth and permanent-income quintile: PSID, IR, and ER.



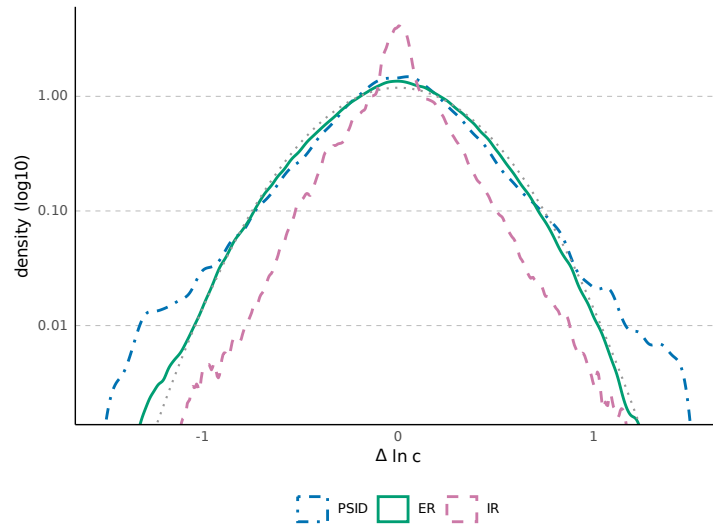
Notes: Finer-grid version of main-text Figure 6, panel b. Panel (a): $\text{Var}(\Delta \ln c)$ by wealth quintile; ER stays near the elevated PSID level over the upper three quintiles (the richest retains ≈ 0.10 in PSID and ER) and undershoots in the bottom two, while IR collapses toward 0.01 at the top. Panel (b): $\text{Var}(\Delta \ln c)$ by permanent-income quintile (within-household mean income, quintiled within economy); PSID shows a declining resource gradient (from ≈ 0.20 in the lowest quintile to ≈ 0.11 in the highest) that ER (flat) and IR (flat and far lower) both miss.

Figure F.6: Pooled density of consumption growth without residualization: PSID, IR, and ER.



Notes: Pooled density of raw consumption growth $\Delta \ln c$ (no age or family-size residualization), on a linear scale (left) and a log scale (right). The dispersion ordering and the ER-overlays-PSID / IR-spike pattern match the residualized body figure (main-text Figure 6, panel a), confirming the residualization is not doing the work.

Figure F.7: Residualized consumption-growth density, log scale: PSID, IR, and ER.



Notes: Log-scale density of residualized $\Delta \ln c$, the tail companion to main-text Figure 6, panel a. ER's Gaussian-innovation φ process produces near-Normal tails that fall short of the PSID tail mass (robust kurtosis ER 3.13 vs PSID 3.74); IR's apparent fat tails are relative to its tiny variance, a spike rather than fat absolute tails.